

Advanced Standard Mode: Safe-to-Use Technique Combinations for Transferring Solver State Across Related Verifications

VHAGaR / neta-s-paper — Internal technical note

Abstract

Advanced standard (`advstd`) mode is an execution mode of the VHAGaR global-robustness verifier that computes the global robustness bound $\delta_{N_2}^*$ for a target network N_2 by *reusing solver state* from a prior solve of a structurally similar network N_1 .

This note documents only the **safe-to-use** technique combinations — those empirically validated to produce average speedups with no severe slowdowns on any cell (geometric-mean time ratio $< 0.952\times$ and worst-cell time ratio $\leq 1.333\times$; i.e. `perf_tier` $\in \{\text{dominant}, \text{avg-win}\}$). Out of the full combinatorial space, 25 combinations satisfy this criterion. They use three independent techniques (Sections 4.1–4.3), all of which preserve the optimal global-robustness bound δ_{exact} . Section 5 gives the complete algorithm for every safe combination.

1 Motivation and Setting

Let N_1 and N_2 be two neural networks with the same architecture — same layer sizes, same ReLU structure, same input domain — but different parameter values. A typical instance is $N_2 = N_1$ after one additional SGD epoch.

The global robustness verification problem, as formulated in VHAGaR, asks: for a given source class c_s , target class c_t , perturbation type P , and bound ε , what is the smallest δ such that no input classified with confidence at least δ as c_s can be perturbed within $\mathcal{I}_\varepsilon(\cdot)$ to be classified as c_t ? This is encoded as a mixed-integer program (MIP) and solved with Gurobi.

When N_1 and N_2 are structurally identical but numerically close, the two MIPs have identical variable names, identical constraint structure (only coefficients differ), and similar solution geometry. Advanced standard mode exploits this similarity through a set of concrete, optional techniques.

2 Two-Phase Execution

Advanced standard mode runs in two phases, possibly in separate processes:

Phase 1 — N_1 solve

For each (c_s, c_t) class pair, solve the standard-mode MIP for N_1 to either optimality or a user-specified timeout. Extract:

- The full primal solution $(\text{var}_i, \text{val}_i)_{i=1}^{|V|}$.
- The per-neuron interval bounds $(l_{m,k}, u_{m,k})$ on all split ReLUs (the `layers_info` dictionary).

Serialise to disk. The resulting `n1_state` directory is a self-contained snapshot of “everything we learned while solving N_1 ”.

Phase 2 — N_2 sweep

For each (c_s, c_t) pair, build the standard-mode MIP for N_2 , load the N_1 state, apply each enabled technique, and call `optimize!`. Phase 2 never re-solves N_1 .

3 Background: ReLU Encoding in the MIP

The VHAGaR MIP encodes each ReLU neuron $z^+ = \max(0, \hat{z})$ using a *big-M formulation* with a binary variable $a \in \{0, 1\}$. Given pre-activation bounds $\hat{z} \in [\underline{z}, \bar{z}]$ (where $\underline{z} < 0 < \bar{z}$ for a “split” neuron), the encoding introduces:

$$z^+ \geq 0, \tag{1}$$

$$z^+ \geq \hat{z}, \tag{2}$$

$$z^+ \leq \hat{z} - \underline{z}(1 - a), \tag{3}$$

$$z^+ \leq \bar{z}a. \tag{4}$$

This is *exact*: when $a = 1$ (neuron active), constraints force $z^+ = \hat{z}$; when $a = 0$ (neuron inactive), they force $z^+ = 0$. The binary a is what makes the problem a *mixed-integer* program — solving over all possible $\{0, 1\}$ assignments is what makes verification NP-hard.

The LP relaxation of ReLU. If we *drop* the binary a and keep only the three constraints that do not involve a , we get the *triangle LP relaxation*:

$$z^+ \geq 0, \tag{5}$$

$$z^+ \geq \hat{z}, \tag{6}$$

$$z^+ \leq \frac{\bar{z}}{\bar{z} - \underline{z}} (\hat{z} - \underline{z}). \tag{7}$$

This is a *convex relaxation*: it allows z^+ to take any value inside a triangle in the $(\hat{z}, z^+)$ plane, not just the two exact ReLU branches. There is no binary variable, so the problem becomes a linear program (LP) — much faster to solve, but the feasible set is *larger* than the exact one. Any solution to the exact encoding also satisfies the LP relaxation, but not vice versa.

The LP relaxation is used in Technique 3: selected binary variables in the final N_2 MIP are replaced with triangle relaxations to reduce the integer search space.

4 Techniques Used in Safe Combinations

The safe combinations use exactly three techniques. Three techniques from the original advstcl toolchain — MIP Start, LP Basis, and Branch Priorities — are **not** used in any safe combination and are therefore omitted from this document.

Correctness invariant

Let $\mathcal{F}_2$ denote the feasible set of N_2 ’s standard-mode MIP and let

$$\delta_{\text{exact}} \triangleq \delta_{N_2}^* = \max_{x \in \mathcal{F}_2} \text{obj}_{N_2}(x).$$

For every subset of techniques, the optimal value reported by the solver when it terminates at OPTIMAL is exactly δ_{exact} . The techniques only change the *path* Gurobi takes through the branch-and-bound tree, not the set of feasible solutions or the objective function.

4.1 Technique 1 — Bound Tightening (**boundTight**)

Flag: `--adv_std_bound_tightening` (always true in safe combos). **Model modification:** tightens neuron bounds before the N_2 MIP is built.

This technique runs *before* the MIP is constructed. Its output — tighter per-neuron intervals — feeds into the MIP encoding itself (smaller big- M constants, fewer binary variables). The other techniques (branch priorities, variable hints, triangle relaxation) operate *on* the already-built MIP.

Technique 1 is independent of the *perturbed interval constraints* that are added to the MIP after construction (see the note at the end of this subsection).

Goal. For each ReLU neuron (m, k) in N_2 , compute a sound interval $[d_{m,k}^\downarrow, d_{m,k}^\uparrow]$ on the *weight-drift difference* $\hat{z}_{m,k}^{N_2} - \hat{z}_{m,k}^{N_1}$ (how much the pre-activation changed due to N_2 having different weights than N_1). Once we have this interval, we combine it with N_1 ’s known bounds from Phase 1 to get N_2 bounds:

$$l_{m,k}^{N_2} \geq l_{m,k}^{N_1} + d_{m,k}^\downarrow, \quad u_{m,k}^{N_2} \leq u_{m,k}^{N_1} + d_{m,k}^\uparrow.$$

When a neuron’s tightened interval lies strictly in $\mathbb{R}_{<0}$ or $\mathbb{R}_{\geq 0}$, the binary $a_{m,k}$ is fixed and removed from the MIP.

There are two methods to compute $[d_{m,k}^\downarrow, d_{m,k}^\uparrow]$. The safe combinations use one of these two, controlled by the `zono` flag.

4.1.1 Method 1: Interval arithmetic (baseline)

Used when: `zono=false`.

This is the default method. From Phase 1 we have N_1 ’s per-neuron bounds $(l_{m,k}^{N_1}, u_{m,k}^{N_1})$, so the post-ReLU range is $z_{m,k}^{N_1} \in [\max(0, l_{m,k}^{N_1}), \max(0, u_{m,k}^{N_1})]$.

Weight-drift difference. Because N_1 and N_2 share the same architecture, every neuron (m, k) in N_2 has a corresponding neuron in N_1 . The *difference* in their pre-activations, $z_{m,k}^{N_2} - z_{m,k}^{N_1}$, depends only on the weight difference $\Delta W = W_2 - W_1$ (the “weight drift” from training). Sound intervals on this difference are computed layer by layer using interval arithmetic.

Affine layers. Let $\Delta w_{m,k} = w_{m,k}^2 - w_{m,k}^1$ and $\Delta b_{m,k} = b_{m,k}^2 - b_{m,k}^1$ be the weight and bias differences at layer m . From Phase 1 we have N_1 ’s per-neuron bounds $(l_{m,k}^{N_1}, u_{m,k}^{N_1})$ for every ReLU neuron. Since the post-ReLU output is $z_{m,k} = \max(0, \hat{z}_{m,k})$, the post-ReLU range for each neuron k in layer $m-1$ is $z_{m-1,k}^{N_1} \in [\max(0, l_{m-1,k}^{N_1}), \max(0, u_{m-1,k}^{N_1})]$.

The pre-activation difference at neuron k of layer m decomposes as:

$$\hat{z}_{m,k}^{N_2} - \hat{z}_{m,k}^{N_1} = \underbrace{\sum_{k'} w_{m,k,k'}^2 (z_{m-1,k'}^{N_2} - z_{m-1,k'}^{N_1})}_{\text{propagated diff from previous layer}} + \underbrace{\sum_{k'} \Delta w_{m,k,k'} \cdot z_{m-1,k'}^{N_1} + \Delta b_{m,k}}_{\text{direct weight-drift contribution}}.$$

Interval arithmetic bounds each term per output neuron k :

$$[d_{m,k}^\downarrow, d_{m,k}^\uparrow] = \sum_{k'} w_{m,k,k'}^2 \cdot [d_{m-1,k'}^\downarrow, d_{m-1,k'}^\uparrow] + \sum_{k'} \Delta w_{m,k,k'} \cdot [\max(0, l_{m-1,k'}^{N_1}), \max(0, u_{m-1,k'}^{N_1})] + \Delta b_{m,k},$$

ReLU layers. The ReLU difference satisfies $\sigma(z+d) - \sigma(z) \in [\min(0, d^\downarrow), \max(0, d^\uparrow)]$, so the diff interval is clipped: $d^\uparrow \leftarrow \max(0, d^\uparrow)$ and $d^\downarrow \leftarrow -\max(0, -d^\downarrow)$.

The final output at each layer is a sound interval:

$$d_{m,k}^\downarrow \leq z_{m,k}^{N_2} - z_{m,k}^{N_1} \leq d_{m,k}^\uparrow.$$

These are then combined with N_1 's known bounds to get N_2 bounds:

$$l_{m,k}^{N_2} \geq l_{m,k}^{N_1} + d_{m,k}^\downarrow, \quad u_{m,k}^{N_2} \leq u_{m,k}^{N_1} + d_{m,k}^\uparrow.$$

This method is fast but loose: it treats every neuron as independent, losing all correlations between neurons in the same layer.

4.1.2 Method 2: Zonotope arithmetic (Source A + B)

Used when: `zono=true`. **Flag:** `--adv_std_zono_bounds`.

This method computes the same $[d_{m,k}^\downarrow, d_{m,k}^\uparrow]$ as Method 1, but using *zonotope arithmetic* instead of interval arithmetic. A zonotope tracks correlations between neurons by representing each neuron's value as a linear combination of shared noise variables, producing tighter bounds. Additionally, a second pass (Source B) independently bounds N_2 's absolute activations and intersects the result with the diff-based bounds.

Source A — Zonotope diff bounds. Each neuron's difference is represented as a zonotope — a centre plus generators sharing noise variables $\epsilon_j \in [-1, 1]$ across all neurons in the layer. We store a centre vector $\mathbf{c}_m \in \mathbb{R}^K$ and a generator matrix $G_m \in \mathbb{R}^{K \times p}$ (rows = K neurons, columns = p generators, where p is the number of columns accumulated so far — initially $p = K_0$ at the input layer, and each affine layer adds K' columns while each ReLU adds one column per split neuron):

$$d_{m,k} = c_{m,k} + \sum_j g_{m,k,j} \epsilon_j, \quad d_{m,k}^\downarrow = c_{m,k} - \sum_j |g_{m,k,j}|, \quad d_{m,k}^\uparrow = c_{m,k} + \sum_j |g_{m,k,j}|.$$

Shared ϵ_j variables encode correlations: interval arithmetic is the special case where G_m is diagonal (no correlations).

Initialisation. At the input layer, N_1 and N_2 receive the same input, so the diff is zero: $\mathbf{c}_0 = \mathbf{0}$, $G_0 = \mathbf{0}_{K_0 \times 0}$ (no generators yet, $p = 0$). The first affine update creates the initial generators via G'_{indep} .

Affine layer update. Let $K' =$ neurons in layer $m-1$, $K =$ neurons in layer m :

$$c_{m,k} = \sum_{k'} w_{m,k,k'}^2 c_{m-1,k'} + \sum_{k'} \Delta w_{m,k,k'} \cdot \text{mid}_{k'} + \Delta b_{m,k} \quad (8)$$

$$\underbrace{G'_{\text{corr}}}_{K \times p} = \underbrace{W_2}_{K \times K'} \underbrace{G_{m-1}}_{K' \times p} \quad (9)$$

$$\underbrace{G'_{\text{indep}}}_{K \times K'} = \underbrace{\Delta W}_{K \times K'} \cdot \underbrace{\text{diag}(\text{rad}_1, \dots, \text{rad}_{K'})}_{K' \times K'} \quad (10)$$

$$\underbrace{G_m}_{K \times (p+K')} = [G'_{\text{corr}} \mid G'_{\text{indep}}] \quad (11)$$

where $\text{mid}_{k'}$ and $\text{rad}_{k'}$ are the midpoint and half-width of N_1 's post-ReLU interval at neuron k' : $\text{mid}_{k'} = \frac{1}{2}(\max(0, u_{m-1,k'}^{N_1}) + \max(0, l_{m-1,k'}^{N_1}))$, $\text{rad}_{k'} = \frac{1}{2}(\max(0, u_{m-1,k'}^{N_1}) - \max(0, l_{m-1,k'}^{N_1}))$.

After the affine update, the column count grows from p to $p + K'$.

ReLU layer update (DeepZ). Compute hull per neuron: $l_k = c_{m,k} - \sum_j |g_{m,k,j}|$, $u_k = c_{m,k} + \sum_j |g_{m,k,j}|$.

- $u_k \leq 0$: $c_{m,k} \leftarrow 0$, $g_{m,k,\cdot} \leftarrow 0$.
- $l_k \geq 0$: unchanged.
- Split ($l_k < 0 < u_k$): $\lambda_k = \frac{u_k}{u_k - l_k}$, $\mu_k = -\frac{u_k l_k}{2(u_k - l_k)}$,

$$c_{m,k} \leftarrow \lambda_k c_{m,k} + \mu_k, \quad g_{m,k,\cdot} \leftarrow \lambda_k g_{m,k,\cdot},$$

append column to G_m : value μ_k at row k , zero elsewhere.

After the ReLU update, $p \leftarrow p + |\{k : l_k < 0 < u_k\}|$ (one new column per split neuron).

So for a network with L affine layers of widths $K_1, \dots, K_L$ and s_m split neurons at ReLU layer m :

$$p_{\text{final}} = K_0 + \sum_{m=1}^L (K_{m-1} + s_m).$$

Source B — N1-tightened absolute N2 zonotope. A separate zonotope propagates N_2 's *absolute* pre-activations (not the diff) through N_2 alone, using the same zonotope machinery as Source A but applied to N_2 's own values. Initialise with a single zonotope whose hull covers both the clean input $v_{\text{in}} \in [0, 1]^n$ and the perturbed input $v_{x_0} = v_{\text{in}} + v_e \in [-\varepsilon, 1 + \varepsilon]^n$ simultaneously:

$$\mathbf{c}_0^B = \frac{1}{2}\mathbf{1}, \quad G_0^B = \text{diag}(\frac{1}{2} + \varepsilon, \dots, \frac{1}{2} + \varepsilon) \in \mathbb{R}^{n \times n},$$

so the initial hull is $[-\varepsilon, 1 + \varepsilon]^n$ rather than $[0, 1]^n$. This widens the input box by ε on each side relative to the clean-only range, which is the price of using one zonotope to bound N_2 's pre-activations under both the clean and perturbed inputs (rather than propagating two separate zonotopes and taking their union). Propagate through N_2 's affine layers ($c \leftarrow W_2 c + b_2$, $G \leftarrow W_2 G$) and DeepZ ReLU relaxations exactly as in Source A. At each ReLU layer, the zonotope hull gives per-neuron bounds:

$$u_{m,k}^{\text{hull}} = c_{m,k}^B + \sum_j |g_{m,k,j}^B|, \quad l_{m,k}^{\text{hull}} = c_{m,k}^B - \sum_j |g_{m,k,j}^B|.$$

These are then *intersected* with Source A's diff-based bounds:

$$u_{m,k}^{N_2} = \min(u_{m,k}^{\text{hull}}, u_{m,k}^{N_1} + d_{m,k}^{\uparrow}), \quad l_{m,k}^{N_2} = \max(l_{m,k}^{\text{hull}}, l_{m,k}^{N_1} + d_{m,k}^{\downarrow}). \quad (12)$$

The first term in each min/max comes from Source B's own zonotope pass on N_2 ; the second term comes from Source A's diff bounds applied to N_1 's known bounds (the same formula as in the Goal paragraph above). Taking the tighter of the two at each neuron is sound because both are valid over-approximations.

The DeepZ ReLU relaxation is then applied using these tightened bounds, so Source B's zonotope passes through a narrower "tube" at each layer.

4.1.3 Precision guarantees

The two methods are ordered in precision, and Source A and Source B within Method 2 are incomparable.

Theorem 1 (Zonotope diff dominates interval diff). *Let $[d_{m,k}^{\downarrow,I}, d_{m,k}^{\uparrow,I}]$ denote the diff bound produced by Method 1 (interval arithmetic, Section 4.1.1) and $[d_{m,k}^{\downarrow,Z}, d_{m,k}^{\uparrow,Z}]$ the diff bound produced by Source A of Method 2 (zonotope, Section 4.1.2) when both start from the same N_1 state. Then for every ReLU neuron (m, k) ,*

$$d_{m,k}^{\downarrow,I} \leq d_{m,k}^{\downarrow,Z} \leq \hat{z}_{m,k}^{N_2} - \hat{z}_{m,k}^{N_1} \leq d_{m,k}^{\uparrow,Z} \leq d_{m,k}^{\uparrow,I}.$$

The inequality is strict whenever some intermediate affine layer combines two generators whose signs disagree after multiplication by the layer's weights.

Proof sketch. Interval arithmetic is the special case of zonotope arithmetic in which the generator matrix G_m is restricted to be diagonal (each neuron has its own independent noise variable). Zonotope preserves correlations: when a subsequent affine layer computes $\sum_{k'} w_{k'} z_{m-1,k'}$, the shared noise ϵ_j combines as $(\sum_{k'} w_{k'} g_{m-1,k',j}) \epsilon_j$, contributing $|\sum_{k'} w_{k'} g_{m-1,k',j}|$ to the half-width. Interval arithmetic discards the correlation and contributes $\sum_{k'} |w_{k'} g_{m-1,k',j}|$ instead. The triangle inequality gives the former $\leq$ the latter, with equality iff all $w_{k'} g_{m-1,k',j}$ share a sign. Inducting on depth and applying the sound DeepZ ReLU relaxation at each layer yields the claim. The ReLU relaxation $\lambda_k z + \mu_k$ is affine with $\lambda_k \in [0, 1]$ and therefore preserves width ordering. $\square$

Corollary 2 (Precision gap grows with depth and drift). *Let $w_{m,k}^\bullet \triangleq d_{m,k}^{\uparrow,\bullet} - d_{m,k}^{\downarrow,\bullet}$ denote the width of the diff bound at neuron (m, k) under Method $\bullet \in \{I, Z\}$. The width gap $w_{m,k}^I - w_{m,k}^Z$ is non-decreasing in m and grows monotonically with $\|\Delta W\|_F$.*

Theorem 3 (Source A and Source B are incomparable). *Let $[l_{m,k}^A, u_{m,k}^A] \triangleq [l_{m,k}^{N_1} + d_{m,k}^{\downarrow}, u_{m,k}^{N_1} + d_{m,k}^{\uparrow}]$ be the Source-A N_2 bound and $[l_{m,k}^B, u_{m,k}^B] \triangleq [l_{m,k}^{\text{hull}}, u_{m,k}^{\text{hull}}]$ the Source-B bound.*

- (a) *There exist instances on which $u^A < u^B$ (equivalently $l^A > l^B$) strictly, and instances on which the opposite strict inequality holds.*
- (b) *The intersection bound $[\max(l^A, l^B), \min(u^A, u^B)]$ (Eq. 12) has width strictly less than $\min(w^A, w^B)$ on any neuron where the two sources disagree in orientation (i.e. $u^A < u^B$ at the upper end and $l^A > l^B$ at the lower end, or vice-versa).*

Proof sketch. (a) *A tighter than B.* Take $N_1 = N_2$ and a generic $\varepsilon > 0$. Then $\Delta W = 0$, so every Source-A generator column $G'_{\text{indep}} = 0$ and the diff zonotope remains zero, giving $[d^{\downarrow}, d^{\uparrow}] = 0$ and $l^A = l^{N_1}$, $u^A = u^{N_1}$ (i.e. N_1 's MIP-level bounds). Source B propagates the wider hull $[-\varepsilon, 1 + \varepsilon]^n$ through DeepZ relaxations, which discard ReLU correlations and accumulate error, so $l^B \leq l^A - c_\varepsilon$ and $u^B \geq u^A + c_\varepsilon$ for some $c_\varepsilon > 0$ on any neuron with a split ancestor.

B tighter than A. Take an N_1 whose Phase-1 solve timed out before closing the root relaxation, so its MIP bounds are vacuous: $l_{m,k}^{N_1} = -\infty$ (in practice, the big- M input bound). Source A inherits these, giving l^A, u^A effectively vacuous. Source B is independent of N_1 's MIP state and produces finite bounds from N_2 's weights alone.

(b) Immediate from the min/max definition: when $u^A < u^B$ and $l^A > l^B$ the intersection keeps u^A and l^A (width w^A); when the sources disagree in orientation, say $u^A < u^B$ and $l^B > l^A$, the intersection is $[l^B, u^A]$ with width strictly less than both w^A and w^B . $\square$

Implication for the safe combinations. Theorem 1 explains the empirical pattern that `zono=true` never *loses* precision relative to `zono=false` but may cost more to compute. The incomparability in Theorem 3 explains why the Source A + Source B intersection is always at least as useful as either alone, and why on benchmarks where N_1 is solved to optimality the benefit of Source B over Source A alone is small, whereas on timed-out N_1 instances it is large.

4.1.4 Final bound integration

When N_2 's MIP is built, each ReLU neuron's bounds are the tightest (intersection) of all available sources:

$$u_{m,k}^{N_2} \leftarrow \min(u_{m,k}^{N_1} + d_{m,k}^{\uparrow}, u_{m,k}^B), \quad (13)$$

$$l_{m,k}^{N_2} \leftarrow \max(l_{m,k}^{N_1} + d_{m,k}^{\downarrow}, l_{m,k}^B), \quad (14)$$

Terms from disabled methods are simply omitted from the min/max. When the final interval is single-signed ($u_{m,k}^{N_2} \leq 0$ or $l_{m,k}^{N_2} \geq 0$), the associated binary is eliminated from the MIP entirely.

Soundness. Both methods produce sound over-approximations:

- Method 1 (interval arithmetic): standard interval arithmetic rules over-approximate the true range at each layer.
- Method 2 (zonotope): zonotope arithmetic is a sound abstract interpretation; Source B’s intersection of two sound over-approximations is itself sound.

Every resulting bound is satisfied by every $x \in \mathcal{F}_2$, so x^* remains feasible and δ_{exact} is preserved.

Summary of what operates where:

- **Technique 1** (bound tightening, Sources A/B): runs *before* MIP construction. Output: tighter per-neuron intervals $\rightarrow$ smaller big- M constants, fewer binaries.
- **Perturbed interval constraints:** runs *after* MIP construction. Output: additional linear constraints linking the org and perturbation copies of N_2 .
- Both are always active in the safe combinations and complement each other.

4.2 Technique 2 — Variable Hints (varHint=prev/direct/direct_pgd/prev_pgd)

Flag: `--adv_std_var_hint` $\in$ {off, prev, direct, direct_pgd, prev_pgd} (five-valued; legacy true/false still accepted, mapped to prev/off with a deprecation warning). **Gurobi attributes:** `VarHintVal`, `VarHintPri` (prev/direct) or `Start` via `set_start_value` (prev_pgd/direct_pgd). **Filename tags:** `_varHintFixed` (prev), `_varHintDirect` (direct), `_varHintDirectPGD` (direct_pgd), `_varHintPrevPGD` (prev_pgd). **Retired:** the previously opt-in `--adv_std_var_hint_fix` flag has been merged into `--adv_std_var_hint`; the “fix” (filtering N_1 pseudocosts down to N_2 -surviving binaries before ranking) is now the default and only behavior. Pre-merger result files tagged `_varHint` or `_varHint_varHintFix` are reported as `vh_legacy` by the post-processing extractor and excluded from the auto-tex tables (they are not directly comparable to the merged-behavior runs).

Background: how Gurobi solves a MIP. When Gurobi solves N_2 ’s MIP, it uses *branch-and-bound*: it maintains a search tree where each node is a sub-problem with some binaries fixed and the rest relaxed to continuous $[0, 1]$. At each node, Gurobi solves an LP relaxation of that sub-problem. The LP solution gives fractional values for the unfixed binaries (e.g. $a_{m,k} = 0.37$). Gurobi then needs to decide:

1. *Which* fractional binary to branch on next.
2. *Which direction* to try first ($a_i = 0$ or $a_i = 1$).
3. Whether to run *heuristics* (RINS, diving, rounding) that try to round the fractional LP solution into a feasible integer solution (an “incumbent”).

Variable hints influence all three decisions as soft suggestions.

What the two hint attributes do. For each binary a_i in N_2 ’s MIP, we set:

- `VarHintVal( $a_i$ )`: “what value should a_i take?” — Gurobi’s heuristics prefer rounding a_i toward this value.
- `VarHintPri( $a_i$ )`: “how confident are we?” — higher priority means Gurobi follows the hint more strongly. Priority 0 means “ignore unless nothing else is available.”

Both are *soft*: they do not add constraints or change the MIP. If the LP relaxation at some node clearly disagrees with the hint, Gurobi overrides it. A wrong hint wastes some effort but cannot change the final answer.

Transfer-probability signal. The right question for the hint is not “how useful was this binary during N_1 ’s branch-and-bound” — it is “*what is the chance that N_1 ’s binary value for this neuron is still correct for N_2 ?*” We build a continuous signal $p_i \in [0, 1]$ that answers this question directly, using only state that N_1 ’s solve and Technique 1 already produce.

Flag values. The `--adv_std_var_hint` flag takes five values. `off` sets no hint on any binary. The remaining four are the Cartesian product of two independent choices — (i) how the transfer probability p_i is computed (*prev-rule* vs. *direct-rule*), and (ii) which Gurobi channel carries the result (*soft VarHint* vs. *PGD-consensus Start*):

- `prev` — p_i from *prev-rule*, emitted via `VarHintVal/VarHintPri`.
- `direct` — p_i from *direct-rule*, emitted via `VarHintVal/VarHintPri`.
- `prev_pgd` — p_i from *prev-rule*, emitted via `Start` with PGD-consensus filtering (see below). No `VarHintVal/VarHintPri` is written.
- `direct_pgd` — p_i from *direct-rule*, emitted via `Start` with PGD-consensus filtering. No `VarHintVal/VarHintPri` is written.

The two rules for p_i :

- *prev-rule*: build a heuristic interval $\tilde{I}_i$ by shifting N_1 ’s pre-activation $\hat{z}_i^{N_1}$ by the diff bound $[d_i^\downarrow, d_i^\uparrow]$ and clipping to $[l_i^{N_2}, u_i^{N_2}]$, then derive p_i from interval-length ratios.
- *direct-rule*: derive p_i directly from Technique 1’s tightened bounds $[l_i^{N_2}, u_i^{N_2}]$, without constructing $\tilde{I}_i$ or using $\hat{z}_i^{N_1}$.

All four non-off modes share the same hint-value formula (agree with $v_i^{N_1}$ when $p_i \geq 0.5$, flip otherwise); they differ only in how p_i is computed and how the resulting `hint_val` is delivered to Gurobi.

Ingredients. For each ReLU neuron $i = (m, k)$ that survives Technique 1’s elimination (i.e., $l_{m,k}^{N_2} < 0 < u_{m,k}^{N_2}$):

- $v_i^{N_1} \in \{0, 1\}$ — N_1 ’s optimal binary (from N_1 ’s primal). Used by both modes to select N_1 ’s supporting side of zero.
- $[l_i^{N_2}, u_i^{N_2}]$ — N_2 ’s final tightened bounds after Technique 1’s integration step (Section 4.1.4). Used by both modes.
- (*prev only*) $\hat{z}_i^{N_1}$ — N_1 ’s pre-activation scalar. When $v_i^{N_1} = 1$ this equals N_1 ’s post-ReLU value z_i^{+, N_1} and is read directly from the primal. When $v_i^{N_1} = 0$ the big- M encoding only constrains $\hat{z}_i^{N_1} \in [l_i^{N_1}, 0]$; the *prev* rule takes its midpoint $l_i^{N_1}/2$ as a proxy.
- (*prev only*) $[d_i^\downarrow, d_i^\uparrow]$ — the weight-drift diff bound from Technique 1.

Mode *prev* — previous rule. Shift N_1 ’s achieved pre-activation by the diff bound, clip to N_2 ’s sound bounds, and take p_i as the fraction of the clipped interval on N_1 ’s supporting side of zero:

$$I_i = [\hat{z}_i^{N_1} + d_i^\downarrow, \hat{z}_i^{N_1} + d_i^\uparrow], \quad \tilde{I}_i = I_i \cap [l_i^{N_2}, u_i^{N_2}],$$

$$p_i = \begin{cases} \frac{\text{length}(\tilde{I}_i \cap (-\infty, 0])}{\text{length}(\tilde{I}_i)} & \text{if } v_i^{N_1} = 0, \\ \frac{\text{length}(\tilde{I}_i \cap [0, +\infty))}{\text{length}(\tilde{I}_i)} & \text{if } v_i^{N_1} = 1. \end{cases}$$

$\tilde{I}_i$ is a heuristic prior on where $\hat{z}_i^{N_2}$ lands under the assumption that N_2 's optimal input is close to N_1 's. It is not a sound over-approximation of $\hat{z}_i^{N_2}$ at N_2 's own optimum (which is allowed to use a different input), but for an advisory hint this is the right quantity.

Mode direct — new rule. Skip the $\tilde{I}_i$ construction entirely. p_i is the fraction of N_2 's tightened pre-activation range on N_1 's supporting side of zero:

$$p_i = \begin{cases} \frac{-l_i^{N_2}}{u_i^{N_2} - l_i^{N_2}} & \text{if } v_i^{N_1} = 0, \\ \frac{u_i^{N_2}}{u_i^{N_2} - l_i^{N_2}} & \text{if } v_i^{N_1} = 1. \end{cases}$$

Technique 1's integration step already folds N_1 's ReLU bounds and the weight drift into $[l_i^{N_2}, u_i^{N_2}]$, so re-deriving an interval by shifting $\hat{z}_i^{N_1}$ only re-introduces the midpoint-proxy error (when $v_i^{N_1} = 0$) and the diff-width noise (when $[d_i^\downarrow, d_i^\uparrow]$ is wide). Using $[l_i^{N_2}, u_i^{N_2}]$ directly self-calibrates: a wider diff bound flows through Technique 1 to produce a wider $[l_i^{N_2}, u_i^{N_2}]$, which pulls p_i toward 0.5 and lowers the priority automatically.

Three regimes of p_i . Under either mode:

- $p_i = 1$: the source range lies entirely on N_1 's side — N_1 's choice almost certainly transfers.
- $0 < p_i < 1$: the source range straddles zero; p_i measures how much of its length still backs N_1 .
- $p_i = 0$: the source range lies entirely on the opposite side — the drift has likely flipped the neuron; N_1 's raw choice is wrong for N_2 .

The “source range” is $\tilde{I}_i$ under **prev** and $[l_i^{N_2}, u_i^{N_2}]$ under **direct**.

Continuous hint rule (all four non-off modes). Agree with N_1 when $p_i \geq 0.5$, flip otherwise:

$$\text{hint_val}(a_i) \leftarrow \begin{cases} v_i^{N_1} & \text{if } p_i \geq 0.5, \\ 1 - v_i^{N_1} & \text{if } p_i < 0.5. \end{cases}$$

Under **prev/direct** this is paired with a V-shaped priority, maximal at both extremes and minimal at the 50/50 line:

$$\text{VarHintVal}(a_i) \leftarrow \text{hint_val}(a_i), \quad \text{VarHintPri}(a_i) \leftarrow \max(1, \text{round}(100 \cdot |2p_i - 1|)).$$

Under **prev_pgd/direct_pgd** the same **hint_val** is instead routed through **Start** with a consensus filter against the PGD warm-start (see below); no **VarHintVal/VarHintPri** is written.

Anchor points, for intuition:

p_i	VarHintVal	VarHintPri	Interpretation
1.00	$v_i^{N_1}$	100	Source range fully on N_1 's side
0.75	$v_i^{N_1}$	50	Majority still backs N_1
0.50	either	1	No usable signal
0.25	$1 - v_i^{N_1}$	50	Majority opposes N_1
0.00	$1 - v_i^{N_1}$	100	Source range fully opposite to N_1 ; drift flipped the neuron

The hint value flips exactly at the $p_i = 0.5$ crossing. The priority is continuous in p_i and never requires a hand-tuned threshold beyond the floor at 1 (priority 0 is reserved by Gurobi for “ignore this hint”).

PGD-consensus Start channel (prev_pgd/direct_pgd). The “_pgd” modes use the same `hint_val( $a_i$ )` as their non-pgd siblings but deliver it via `Start` (`set_start_value` in JuMP) instead of `VarHintVal`/`VarHintPri`. Before Technique 2 runs, PGD has already populated `Start` on some binaries via `hyper_attack_hints()`; for each surviving binary a_i we read `JuMP.start_value( $a_i$ )` and apply a three-way consensus filter:

- **PGD silent** (`start_value = nothing`): N_1 's transfer hint is the only signal available — call `set_start_value( $a_i$ , hint_val)` (“*fill*”).
- **PGD agrees** (`start_value = hint_val`): PGD and the N_1 transfer point at the same branch; leave PGD's `Start` in place, no update (“*leave*”).
- **PGD disagrees** (`start_value  $\neq$  hint_val`): the two signals contradict — call `set_start_value( $a_i$ , nothing)` to withdraw PGD's value, leaving the binary unstated for Gurobi to decide (“*withdraw*”).

The intent is to keep only the warmstart bits both sources agree on (or that only one source filled in), instead of fighting between two incomplete MIP starts. If PGD failed entirely (all starts are `nothing`), the modes degrade to “variable hints fill every gap via `Start`” — equivalent to unconditional N_1 -transfer warmstart.

Relation to prev/direct. The p_i formulas, the flip rule, and the handling of eliminated binaries are unchanged; only the Gurobi channel differs. Priority information is discarded (Gurobi's `Start` API has no analogue of `VarHintPri`) — every `hint_val` that survives the consensus filter has equal weight.

Optimal-hint characterization of direct. The p_i formula for `direct` is not a heuristic — it is a posterior probability under a canonical prior derived from Technique 1's output. Specifically:

Theorem 4 (Bayes-optimal hint under max-entropy prior). *Fix a surviving binary a_i (so $l_i^{N_2} < 0 < u_i^{N_2}$). Let π_i be the maximum-entropy prior on $\hat{z}_i^{N_2}$ whose support is the Technique-2 bound, i.e. $\hat{z}_i^{N_2} \sim \text{Unif}([l_i^{N_2}, u_i^{N_2}])$, and let $v_i^{N_2,*} \in \{0, 1\}$ be the (unknown) value taken by a_i at N_2 's optimum. Then:*

- (a) *The transfer probability of the direct rule equals the posterior probability of agreement with N_1 :*

$$\Pr_{\pi_i}[v_i^{N_2,*} = v_i^{N_1} \mid l_i^{N_2}, u_i^{N_2}] = p_i^{\text{direct}}.$$

- (b) *The hint value $\text{VarHintVal}(a_i) = (p_i \geq 0.5) ? v_i^{N_1} : 1 - v_i^{N_1}$ is the maximum a posteriori estimate of $v_i^{N_2,*}$ under π_i .*
- (c) *The hint priority $\text{VarHintPri}(a_i) = \max(1, \text{round}(100 \cdot |2p_i - 1|))$ is monotone in the Bayes-risk reduction $\mathbb{E}_{\pi_i}[\mathbf{1}\{v_i^{N_2,*} \neq \hat{v}\}] - \mathbb{E}_{\pi_i}[\mathbf{1}\{v_i^{N_2,*} \neq \text{MAP}\}]$ over the uniform random hint $\hat{v}$.*

Proof. The big- M encoding ties $a_i = \mathbf{1}\{\hat{z}_i^{N_2} \geq 0\}$ (up to tie-breaking on the measure-zero set $\hat{z}_i^{N_2} = 0$). Under π_i ,

$$\Pr_{\pi_i}[\hat{z}_i^{N_2} > 0] = \frac{u_i^{N_2}}{u_i^{N_2} - l_i^{N_2}}, \quad \Pr_{\pi_i}[\hat{z}_i^{N_2} < 0] = \frac{-l_i^{N_2}}{u_i^{N_2} - l_i^{N_2}}.$$

Substituting into the **direct** case split gives (a). Part (b) follows because the MAP estimate under a Bernoulli posterior with parameter q is $\mathbf{1}\{q \geq 0.5\}$; the **direct** rule assigns $v_i^{N_1}$ when $p_i \geq 0.5$ and $1 - v_i^{N_1}$ otherwise, which is exactly the MAP value. Part (c) is immediate from $|2p_i - 1|$ being the total-variation distance between the posterior and the uniform distribution. $\square$

Why prev is biased. The **prev** rule computes p_i on the interval $\tilde{I}_i = [\hat{z}_i^{N_1} + d_i^\downarrow, \hat{z}_i^{N_1} + d_i^\uparrow] \cap [l_i^{N_2}, u_i^{N_2}]$, which is a strict subinterval of $[l_i^{N_2}, u_i^{N_2}]$ whenever the diff bound is narrower than the tightened bound. $\tilde{I}_i$ is not the maximum-entropy support of $\hat{z}_i^{N_2}$: it presumes that N_2 's optimum will evaluate $\hat{z}_i^{N_2}$ near $\hat{z}_i^{N_1}$, a commitment that the big- M encoding does not license (the two MIPs can pick different inputs). Consequently **prev** is a biased estimator of $\Pr[v_i^{N_2,*} = v_i^{N_1}]$, with bias proportional to the diff-bound width $d_i^\uparrow - d_i^\downarrow$. This explains the empirical self-calibration observation already noted above: on instances where the diff bound is wide relative to $[l_i^{N_2}, u_i^{N_2}]$, **direct**'s Bayes-optimal pull toward $p_i = 0.5$ correctly suppresses the priority, whereas **prev** emits a confidently-wrong hint.

Applicability. The rule runs only for binaries that survive Technique 1's elimination. When $l_i^{N_2} \geq 0$ or $u_i^{N_2} \leq 0$ the binary is already fixed and removed from the MIP, so there is nothing to hint.

Hints vs. candidate solution — and why we use both channels. Gurobi exposes two advisory channels for biasing the search. Soft *variable hints* (**VarHintVal/VarHintPri**) nudge each binary independently with a confidence weight; Gurobi is free to ignore any individual hint the LP relaxation disagrees with. A full *MIP Start* (**Start**) submits a candidate assignment that Gurobi's start heuristic tries to turn into a feasible incumbent. **Start** is a stronger commitment: if N_1 's solution is infeasible for N_2 , Gurobi spends effort repairing (or discarding) it. Blindly transferring the entire N_1 primal as a **Start** was harmful in all tested combinations (“MIP Start mode”, all LOSER) — which motivated first the soft-hint modes (**prev**, **direct**), and then the PGD-consensus **Start** modes (**prev_pgd**, **direct_pgd**) that only emit a **Start** bit when the N_1 -transfer hint and the PGD warmstart agree (or when only one of them has an opinion). The consensus filter replaces the “repair-everything” failure mode with a “warmstart only the bits both signals trust” policy.

Soundness: δ_{exact} **is preserved under all five modes.** We claim $\delta_{\text{varHint}} = \delta_{\text{exact}}$ for every mode $\in \{\text{off}, \text{prev}, \text{direct}, \text{prev_pgd}, \text{direct_pgd}\}$ whenever Gurobi terminates at **OPTIMAL**. The argument is uniform:

- **No constraint, coefficient, or bound is modified by any mode.** **prev/direct** only set the advisory Gurobi attributes **VarHintVal/VarHintPri**; **prev_pgd/direct_pgd** only call **set_start_value** (which maps to Gurobi's **Start** attribute) or withdraw it (**set_start_value(.,nothing)**). None of **VarHintVal**, **VarHintPri**, **Start** participates in the constraint system; they are inputs to Gurobi's heuristics and branching priority, not to feasibility. In particular, **Start** does *not* bound the variable — the feasible set is whatever the big- M /triangle constraints define, independent of any **Start** value.

- **Feasible-set invariance.** Consequently:

$$\mathcal{F}_2^{(\text{off})} = \mathcal{F}_2^{(\text{prev})} = \mathcal{F}_2^{(\text{direct})} = \mathcal{F}_2^{(\text{prev-pgd})} = \mathcal{F}_2^{(\text{direct-pgd})} = \mathcal{F}_2.$$

Since $\delta_{\text{exact}} \triangleq \max_{x \in \mathcal{F}_2} \text{obj}_{N_2}(x)$ is defined purely in terms of $\mathcal{F}_2$ and the unchanged objective, δ_{exact} is identical across all five modes.

- **Consequence at termination.** At OPTIMAL, Gurobi returns $\max_{x \in \mathcal{F}_2^{(\text{mode})}} \text{obj}_{N_2}(x) = \delta_{\text{exact}}$ in every mode. At TIME_LIMIT the reported incumbent is a sound lower bound on δ_{exact} whose numeric value may differ across modes (hints shift the branching path and which incumbents are found first), but the underlying δ_{exact} is fixed.

This covers both the soft-hint modes and the PGD-consensus Start modes in a single argument — neither channel writes a constraint, so the optimum is locked to δ_{exact} regardless of whether the hint is right, wrong, or contradictory. The only thing a bad hint costs is search time.

4.3 Technique 3 — Per-Copy Triangle Drop (**BoundTightPertRelax**)

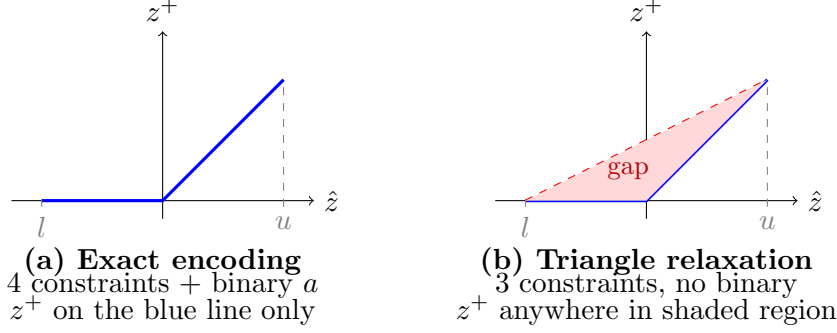
Flag: `--adv_std_n2_relax_threshold` $\in \{\text{off}, 0.01, 0.1, 0.5, 1.0\}$ (in safe combos; off means Technique 3 is disabled). **Model modification:** replaces selected N_2 binary variables with LP relaxations. **Filename tag:** runs with $\tau \geq 0$ emit `_BoundTightPertRelax $\{\tau\}$` in place of `_boundTight` (the tag implies bound-tightening is active). The retired `_relaxT $\{\tau\}$` tag is not accepted by the post-processing pipeline.

What is relaxation, and why would we want it? The MIP we hand to Gurobi contains two kinds of variables: *continuous* variables (real numbers, e.g. the pre-activation $\hat{z}$ of a neuron) and *binary* variables $a \in \{0, 1\}$ (one per split ReLU neuron, indicating whether the neuron is active or inactive). Binaries are what make the problem hard: Gurobi has to branch over every possible 0/1 assignment in the worst case, so each binary in principle doubles the search tree. Our networks have hundreds of split ReLUs, which is why verification is slow.

Relaxation means: replace the hard $\{0, 1\}$ choice on a binary with a looser, purely continuous set of constraints that still over-covers the exact ReLU behavior. The continuous version is a linear program (LP) — no branching needed. The price is that the feasible set becomes larger (it now includes some non-ReLU points), so the reported δ may come out slightly higher than the true optimum. If we relax only neurons that are “almost decided anyway,” the price is small and the speedup is large.

This subsection explains: (i) what exactly the relaxation replaces, (ii) how we measure “almost decided,” (iii) what $\tau = 0.5$ means concretely, and (iv) which binaries get dropped.

Exact vs. relaxed ReLU encoding. Recall from Section 3 that a split ReLU neuron (pre-activation bounds $l < 0 < u$) can be encoded in two ways:



- **Exact** (a): uses a binary variable $a \in \{0, 1\}$ and 4 constraints (Eqs. (1)–(4)). $a = 1$ means “neuron active, $z^+ = \hat{z}$ ”; $a = 0$ means “neuron inactive, $z^+ = 0$ ”. The post-ReLU value z^+ is forced onto the exact ReLU curve (blue line in diagram (a)). This is precise but expensive: having to branch on a is what makes verification NP-hard.
- **Relaxed** (b): drops the binary a and uses 3 linear constraints (Eqs. (5)–(7)). z^+ is now allowed to take any value inside the shaded triangle in diagram (b). Every point on the true ReLU curve is still inside this triangle (so the relaxation *over-covers* reality), but the triangle also admits some points that are not on the curve — these extra admissible points are the price we pay. The shaded region between the blue line and the red dashed envelope is the *gap*: the set of $(\hat{z}, z^+)$ pairs that the relaxation allows but the exact encoding would not.

The smaller the triangle, the less the relaxation loses. If l is very close to zero (the neuron “wants” to be active nearly always) or u is very close to zero (the neuron wants to be inactive nearly always), the triangle is a thin sliver and the relaxation is almost indistinguishable from the exact ReLU. Relaxing such a neuron costs almost nothing and saves one binary. When $|l| \approx u$ the triangle is fat — the neuron is genuinely ambiguous and we do not want to relax it.

Measuring “almost decided”: the triangle’s gap area. We quantify how thin the triangle is by its area. For a ReLU neuron with pre-activation bounds $[l, u]$:

$$\text{gap}(l, u) = \begin{cases} 0 & \text{if } l \geq 0 \text{ or } u \leq 0 \quad (\text{neuron is } \textit{stable} \text{ — not split at all}), \\ \frac{u \cdot |l|}{2(u - l)} & \text{otherwise (neuron is split; compute triangle area).} \end{cases} \quad (15)$$

The second case is just the base-times-height formula for the triangle in diagram (b) (base $u - l$ along the $\hat{z}$ -axis, peak offset $u \cdot |l|/(u - l)$ above the ReLU line at $\hat{z} = 0$). Some concrete numbers to build intuition:

l	u	$\text{gap}(l, u)$	Interpretation
−0.1	2.0	0.095	sliver: almost always active; cheap to relax
−2.0	0.1	0.095	sliver: almost always inactive; cheap to relax
−0.5	0.5	0.125	still fairly thin
−1.0	1.0	0.25	moderate
−2.0	2.0	0.5	big triangle; relaxation is noticeably loose
−5.0	5.0	1.25	very fat; ambiguous neuron, leave exact

A threshold τ drawn in the same units of (pre-activation area) lets us say: “relax any neuron whose triangle has area $\leq \tau$, keep the rest exact.” Concretely:

- $\tau = 0.01$: relax only extremely thin slivers ($|l| < 0.02$ or $u < 0.02$ in the table above).

- $\tau = 0.1$: relax most slivers but keep anything more than moderately split.
- $\tau = 0.5$: relax everything with gap ≤ 0.5 , including the $[-2, 2]$ case above; genuinely ambiguous neurons (gap > 0.5) still keep their binary. This is aggressive but still below the fully-symmetric case.
- $\tau = 1.0$: relax almost every split neuron; fat triangles get replaced too. Strongest speedup, loosest answer.
- $\tau = \text{off}$: relax nothing — every split neuron keeps its binary and its exact big- M encoding.

So the “0.5” in `--adv_std_n2_relax_threshold=0.5` is literally the maximum triangle-area any neuron can have and still be converted from exact-with-binary to relaxed-without-binary.

Which binaries actually get dropped? Because VHAGaR encodes the network twice (see the next paragraph), every split ReLU neuron (m, k) owns *two* binaries in the MIP: $a_{m,k}^{\text{org}} \in \{0, 1\}$ on the clean-input copy and $a_{m,k}^{\text{pert}} \in \{0, 1\}$ on the perturbed-input copy. The relaxation decision is made per copy: for each copy $c \in \{\text{org}, \text{pert}\}$ whose gap $g_{m,k}^c$ satisfies the tiered rule below, we remove $a_{m,k}^c$ together with the two big- M constraints that mention it (Eqs. (3)–(4)) and replace them with the three triangle inequalities (Eqs. (5)–(7)) on that copy’s interval $[l_{m,k}^{N_2,c}, u_{m,k}^{N_2,c}]$. Concretely, three outcomes are possible for a single neuron:

- drop *both* $a_{m,k}^{\text{org}}$ and $a_{m,k}^{\text{pert}}$ (relax both copies);
- drop exactly one of them (relax only that copy; the other copy keeps its binary and its four big- M constraints);
- keep both (no relaxation on this neuron).

Nothing else is touched in any of the three outcomes: the pre-activations $\hat{z}_{m,k}^{\text{org}}, \hat{z}_{m,k}^{\text{pert}}$, the post-ReLUs $z_{m,k}^{+, \text{org}}, z_{m,k}^{+, \text{pert}}$, the neurons in adjacent layers, the inter-layer linear constraints, and the perturbation-coupling constraints that link the two copies all remain exactly as they were.

Why there are *two* bounds per neuron (the org/pert twist). VHAGaR encodes each network twice inside one MIP: once for the clean input v_{in} (the “org” copy) and once for the perturbed input $v_{x_0} = v_{\text{in}} + v_e$ (the “pert” copy). Every ReLU neuron therefore has *two* instances in the MIP: one on the org copy with pre-activation bounds $[l_{m,k}^{N_2, \text{org}}, u_{m,k}^{N_2, \text{org}}]$, and one on the pert copy with (generally looser, because the pert input box is wider) bounds $[l_{m,k}^{N_2, \text{pert}}, u_{m,k}^{N_2, \text{pert}}]$. Each instance has its own binary and its own triangle gap. This lets us make the relaxation decision *independently* per copy: we can relax the org-copy binary, the pert-copy binary, both, or neither.

Decision rule (tiered, per-copy). Put everything together. For each ReLU neuron (m, k) we compute two gap numbers, one per copy:

$$g_{m,k}^{\text{org}} = \text{gap}(l_{m,k}^{N_2, \text{org}}, u_{m,k}^{N_2, \text{org}}), \quad g_{m,k}^{\text{pert}} = \text{gap}(l_{m,k}^{N_2, \text{pert}}, u_{m,k}^{N_2, \text{pert}}),$$

then compare each to the threshold τ . There are three cases (hence the word “tiered”):

1. **Both tiny** — if $\max(g_{m,k}^{\text{org}}, g_{m,k}^{\text{pert}}) \leq \tau$, both copies have thin triangles, so we drop *both* binaries $a_{m,k}^{\text{org}}$ and $a_{m,k}^{\text{pert}}$ and put the triangle inequalities on both copies instead.
2. **Only one tiny** — if exactly one of the two gaps is $\leq \tau$, we drop the binary for the smaller-gap copy only. The other copy keeps its binary and exact big- M encoding. This is the case that motivates the name **BoundTightPertRelax**: under a non-trivial perturbation the pert copy’s bounds are looser (its triangle is bigger), so a neuron can easily be tight on the org side and still split on the pert side.

3. **Neither tiny** — if both gaps exceed τ , the neuron is genuinely ambiguous on both sides. Keep the binary on both copies, do not relax.

The per-copy bounds $l_{m,k}^{N_2,\text{copy}}, u_{m,k}^{N_2,\text{copy}}$ that feed $\text{gap}(\cdot)$ are the ones Technique 1 already computed — specifically, they are the intersection of Source A (N_1 's per-copy bound plus the diff) and Source B (the shared absolute- N_2 zonotope hull):

$$u_{m,k}^{N_2,\text{copy}} = \min\left(\underbrace{u_{m,k}^{N_1,\text{copy}} + d_{m,k}^\uparrow}_{\text{Source A: } N_1 \text{ bound} + \text{diff}}, \underbrace{u_{m,k}^B}_{\text{Source B: } N_2 \text{ zonotope hull}}\right), \quad (16)$$

$$l_{m,k}^{N_2,\text{copy}} = \max\left(l_{m,k}^{N_1,\text{copy}} + d_{m,k}^\downarrow, l_{m,k}^B\right). \quad (17)$$

The org/pert asymmetry flows entirely through Source A: N_1 's Phase-1 MIP is itself dual-copy, so its stored bound table $\{(l_{m,k}^{N_1,\text{copy}}, u_{m,k}^{N_1,\text{copy}})\}$ distinguishes the two copies (the pert copy inherits a wider input box $[-\varepsilon, 1+\varepsilon]^n$, so its bounds are typically looser). The diff interval $[d_{m,k}^\downarrow, d_{m,k}^\uparrow]$ is shared between copies. Source B is also shared: its zonotope is initialised to cover both input boxes at once, so a single hull is valid for either copy. Therefore, the only reason $g_{m,k}^{\text{org}}$ and $g_{m,k}^{\text{pert}}$ ever disagree is N_1 's own per-copy bounds; when those happen to be identical at a neuron, both gaps collapse to the same value and Tier 2 degenerates into Tier 1 or Tier 3 there.

Safety net: the two copies are still linked. Even when we relax one copy and leave the other exact, the two are not free to drift apart. VHAGaR's perturbed-interval coupling constraints (always active in advstd safe combinations) link them via $z_{\text{pert}}^+ - z_{\text{org}}^+ \in [\Delta^\downarrow, \Delta^\uparrow]$, where $[\Delta^\downarrow, \Delta^\uparrow]$ is a sound interval on how much the pert post-ReLU value can differ from the org one under the perturbation budget. So relaxing a binary on one side still indirectly constrains the other side.

Why N_2 's per-copy bounds (not N_1 's)? We judge a neuron's gap using N_2 's tightened bounds, not N_1 's, for three reasons. (i) The triangle constraints we emit use exactly those N_2 bounds — deciding on the same interval we later write into the MIP is what makes the soundness argument work (see “Consistency requirement” and “Soundness” below). (ii) Under non-trivial perturbations, N_2 's org- and pert-copy bounds can differ substantially (the pert copy sees a wider input box), so a single-gap rule built on N_1 would miss the asymmetric opportunities Tier 2 exploits. (iii) The flag name itself — `BoundTightPertRelax` — advertises the dependency on Technique 1's per-copy bound tightening.

Consistency requirement. The decision-time gaps $g_{m,k}^{\text{org}}$ and $g_{m,k}^{\text{pert}}$ must be computed from the *same* intersected bounds that get baked into the emitted triangle constraints. In the code, both sites call the shared helper `intersect_per_copy_bounds` in `core_ops.jl`, so there is no drift between “the interval we decided was thin” and “the interval of the triangle we wrote into the MIP.” This invariant is what the soundness argument below rests on.

What does this cost, and what do we gain? Each neuron we relax removes one binary variable and two big- M constraints from the MIP (and adds three linear inequalities), so on a network with hundreds of split ReLUs we typically save dozens of binaries. Since each binary in principle doubles Gurobi's branch-and-bound tree, removing even a handful of binaries can cut solve time substantially.

The price is that the reported δ is no longer guaranteed equal to the exact optimum δ_{exact} ; it is now a *sound upper bound*:

$$\delta_{\text{relaxed}} \geq \delta_{\text{exact}}.$$

In other words, if the relaxed MIP certifies robustness at some δ , then the exact MIP would also certify robustness at (at worst) that same δ . The overshoot grows with τ , but since we only relax neurons with $\text{gap} \leq \tau$, each relaxation contributes at most a small slice of looseness. In practice the observed overshoot is often in the third decimal digit even at $\tau = 0.5$.

Soundness. Let $R \subseteq \{(m, k, \text{copy})\}$ be the set of (neuron, copy) pairs selected for relaxation by the tiered rule, and let $\mathcal{F}_2^{\{\{6\}\}}$ denote the feasible set of the resulting N2 MIP. We claim $\mathcal{F}_2 \subseteq \mathcal{F}_2^{\{\{6\}\}}$, hence $\delta_{\text{BTPR}} \geq \delta_{\text{exact}}$.

- For $(m, k, \text{copy}) \notin R$ the exact big- M encoding is emitted, identical to the baseline; every $x \in \mathcal{F}_2$ trivially satisfies the same constraints.
- For $(m, k, \text{copy}) \in R$ the binary $a_{m,k,\text{copy}}$ and its four big- M constraints are dropped; the three triangle inequalities are emitted on the *same* per-copy bound $[l_{m,k}^{N_2,\text{copy}}, u_{m,k}^{N_2,\text{copy}}]$ that would have gated the exact encoding (the consistency requirement above). By the triangle-relaxation lemma (Eqs. (5)–(7)), every $(\hat{z}, z^+)$ satisfying the exact ReLU encoding on $[l_{m,k}^{N_2,\text{copy}}, u_{m,k}^{N_2,\text{copy}}]$ lies inside the triangle on the same interval. So every $x \in \mathcal{F}_2$ satisfies the relaxed constraints.

All other constraints (perturbed-interval coupling, VHAGaR dependencies, inter-layer linking, warmstart hints as non-constraining hints) are unchanged. Therefore $\mathcal{F}_2 \subseteq \mathcal{F}_2^{\{\{6\}\}}$, and taking the max of the unchanged objective over a superset can only raise the value: $\delta_{\text{BTPR}} = \max_{x \in \mathcal{F}_2^{\{\{6\}\}}} \text{obj}_{N_2}(x) \geq \max_{x \in \mathcal{F}_2} \text{obj}_{N_2}(x) = \delta_{\text{exact}}$.

4.4 Technique 4 — Sibling-Gated Refinement (SibGate)

Flag: `--adv_std_n2_sibling_gate` $\in \{\text{false}, \text{true}\}$. **Requires:** `--adv_std_n2_relax_threshold` (this technique augments Technique 3’s tiered decision rule; it does not introduce one). **Model modification:** replaces the simple Wong–Kolter triangle with a conditional triangle gated on the surviving sibling binary in the “one thin” tier, and adds a pre-activation coupling constraint linking the two copies in the “both thin” tier. **Filename tag:** `_SibGate` appended to the existing `_BoundTightPertRelax $\{\tau\}$` tag.

What is changed. Technique 3’s tiered per-copy decision rule (Section 4.3) is inherited verbatim. Only the constraints emitted on each tier change:

Tier	Technique 3 (today)	Technique 4 (proposed)
both thin	simple triangle on each copy	simple triangle on each copy <i>plus</i> a single pre-activation coupling line linking them
one thin	simple triangle on the relaxed copy	conditional triangle on the relaxed copy, gated on the kept sibling’s binary, with int
neither	exact big- M on both	unchanged

Tier “one thin” — conditional triangle

Let $c \in \{\text{org}, \text{pert}\}$ be the relaxed copy and pre its sibling (the copy whose binary survives in the MIP): $\text{pre} = \text{pert}$ when $c = \text{org}$, and $\text{pre} = \text{org}$ when $c = \text{pert}$. Write $a_{m,k}^{\text{pre}}$ for the surviving sibling

binary. Instead of the three inequalities (5)–(7) the MIP receives four:

$$z_{m,k}^c \geq 0, \quad z_{m,k}^c \geq \hat{z}_{m,k}^c, \quad (18)$$

$$z_{m,k}^c \leq \frac{u_A}{u_A - l_A} (\hat{z}_{m,k}^c - l_A) + M (1 - a_{m,k}^{\text{pre}}), \quad (19)$$

$$z_{m,k}^c \leq \frac{u_I}{u_I - l_I} (\hat{z}_{m,k}^c - l_I) + M a_{m,k}^{\text{pre}}, \quad (20)$$

with $M = u_{m,k}^{N_2,c} + |l_{m,k}^{N_2,c}|$ chosen large enough to make the “off” upper bound vacuous over the feasible region. (For $c = \text{org}$ this is exactly the conditional triangle of the existing `--use_relaxations` encoding; the $c = \text{pert}$ case is the symmetric mirror, gated on $a_{m,k}^{\text{org}}$ instead.)

The four conditional bounds l_A, u_A, l_I, u_I are built from three sound interval inputs — two of them produced by Technique 1 (`--adv_std_bound_tightening`) and one of them produced by the perturbation-coupling pipeline that backs `--use_relaxations`:

- $[l_{m,k}^{N_2,c}, u_{m,k}^{N_2,c}]$ — the unconditional pre-activation bound on the *relaxed* copy c , produced by Technique 1’s `intersect_per_copy_bounds()` routine (Section 4.1.4).
- $[l_{m,k}^{N_2,\text{pre}}, u_{m,k}^{N_2,\text{pre}}]$ — the same routine applied to the *sibling* (kept-exact) copy *pre*.
- $[l_{\text{int}}, u_{\text{int}}]$ — the *intended* input is a sound interval bound on the perturbation-induced pre-activation difference $\hat{z}_{m,k}^c - \hat{z}_{m,k}^{\text{pre}}$ in N_2 . The formula derivation in (21)–(22) and the soundness proof below assume exactly that quantity, which is the pre-activation analogue of the post-activation coupling bound $[\Delta^\downarrow, \Delta^\uparrow]$ already used by the perturbed-interval constraints. **Caveat on today’s code:** the routines that populate the global `relu_comp_*_bounds` actually store $\hat{z}_{N_2,\text{pert}} - \hat{z}_{N_1,\text{org}}$, which bundles the org–pert perturbation diff with the N_1 – N_2 weight drift; the dispatch is gated on `--use_relaxations` or `--adv_std_zono_bounds`, *not* on `--adv_std_bound_tightening`, and the diff component (but not the pert component) becomes zonotope-tightened when `--adv_std_zono_bounds=true`. The next paragraph spells out the recurrence the code uses and what changes are needed to expose the pure pert bound that this derivation needs.

How $[l_{\text{int}}, u_{\text{int}}]$ is computed in the actual code. The quantity Technique 4 conceptually wants is the pre-activation *org*–*pert* diff $\hat{z}_{m,k}^{\text{pert}} - \hat{z}_{m,k}^{\text{org}}$ in N_2 . Inside both `compute_diff_and_comp_bounds()` and its zonotope sibling `compute_diff_bounds_zonotope()` (`utils/perturbation_intervals.jl`), this diff lives in two 4D arrays `pert_up`, `pert_down`. The recurrence that propagates them through N_2 alone is verbatim:

Initialisation (input layer). `pert_up` $\leftarrow +\varepsilon \cdot \mathbf{1}$, `pert_down` $\leftarrow -\varepsilon \cdot \mathbf{1}$.

Linear layer (with weights W_2). Plain interval arithmetic; the bias cancels in the difference:

$$[\text{pert_down}', \text{pert_up}'] = W_2 \cdot [\text{pert_down}, \text{pert_up}].$$

Conv layer (with filter F_2). Same idea, via `interval_conv2d_bounds( $F_2, F_2, \cdot$ )` with zero bias.

ReLU layer. Per-neuron clipping: `pert_up` $\leftarrow \max(0, \text{pert_up})$, `pert_down` $\leftarrow -\max(0, -\text{pert_down})$.

The two values stored at the input to each ReLU are then exactly the quantity Technique 4 wants, and they would be the natural source for $[l_{\text{int},m,k}, u_{\text{int},m,k}]$. The pert recurrence is *always interval arithmetic*, in both compute routines; the zono flag does not switch it. (What the zono flag switches is the *diff* state that runs alongside, propagating $\hat{z}_{m,k}^{N_2} - \hat{z}_{m,k}^{N_1}$ as a zonotope — a separate quantity.)

What the code actually exposes today (Important caveat). At each ReLU layer the code does not push `pert_up`, `pert_down` to a global. It only pushes the *composed* bound

$$\text{relu_comp_up}_{m,k} = \text{diff_up}_{m,k} + \text{pert_up}_{m,k}, \quad \text{relu_comp_down}_{m,k} = \text{diff_down}_{m,k} + \text{pert_down}_{m,k},$$

which bounds $\hat{z}_{m,k}^{N_2, \text{pert}} - \hat{z}_{m,k}^{N_1, \text{org}}$ (weight drift *and* input perturbation combined). These are the `relu_comp_*_bounds` globals that today's `--use_relaxations` code reads. Two consequences:

1. If Technique 4 reads `relu_comp_*_bounds` verbatim, its $[l_{\text{int}}, u_{\text{int}}]$ inherits the N_1 – N_2 diff component on top of the org–pert perturbation — that is *not* what the conditional triangle's derivation in (21)–(22) assumes, and the diff component does become zonotope-tightened when `--adv_std_zono_bounds=true`.
2. To match the derivation, the code change for Technique 4 needs to expose `pert_up`, `pert_down` at each ReLU as new globals (call them `relu_pert_*_bounds`) and have Technique 4 read those instead of `relu_comp_*`. With that change, $[l_{\text{int}}, u_{\text{int}}]$ would always be interval, regardless of `--adv_std_zono_bounds`, as the derivation requires.

Flags that today trigger the routines (and what `run_relaxation_sweep.py` sets). Both `compute_diff_and_comp_bounds()` and `compute_diff_bounds_zonotope()` are dispatched from `utils/perturbation.py`. The dispatch fires only when at least one of these flags is true:

- `--use_relaxations=true` (with N1 binaries kept and N1 encoded), *or*
- `--adv_std_zono_bounds=true` (`bound_n2_relu_using_zonotope`), *or*
- one of the niche flags `bound_n2_xp_using_composed`, `bound_by_zonotope_n2_hidden_neurons_which_are_not_no_n2_xp_encoding`, `no_n1_encoding_at_all`, or `n1_stability_relax_threshold ≥ 0`.

When the dispatch does fire, the choice between the two routines is `--adv_std_zono_bounds: false → compute_diff_and_comp_bounds()`, `true → compute_diff_bounds_zonotope()`.

The Phase-2 invocation built by `run_relaxation_sweep.py` (`run_relaxation_sweep.py:2486-2513`) does *not* pass `--use_relaxations`, and Technique 4's flag `--adv_std_n2_sibling_gate` is not yet wired up. So in the sweep the routines only fire on the rows where `--adv_std_zono_bounds=true`. Concretely:

- Sweep row with `--adv_std_zono_bounds=true`: `relu_comp_*_bounds` are populated by the zonotope variant. The diff side is zonotope-tightened, the pert side is interval — so the comp bound is partially zonotope.
- Sweep row with `--adv_std_zono_bounds=false`: `relu_comp_*_bounds` are not populated at all. A Technique 4 emission that reads them would see empty arrays and have nothing to gate on; it would degrade to Technique 3.

The minimum code change to make Technique 4 active in the sweep is to broaden the dispatch condition to also fire on `--adv_std_n2_sibling_gate=true`, and to expose the pert state as its own global so $[l_{\text{int}}, u_{\text{int}}]$ matches the formula derivation rather than carrying weight drift on top.

With those inputs in hand, the conditional bounds are obtained by intersecting Source-A-style conditional intervals (the first argument of each max/min) with the unconditional per-copy bound on the relaxed copy (the second argument):

$$l_A \triangleq \max(l_{\text{int}}, l_{m,k}^{N_2,c}), \quad u_A \triangleq \min(u_{m,k}^{N_2,\text{pre}} + u_{\text{int}}, u_{m,k}^{N_2,c}), \quad (21)$$

$$l_I \triangleq \max(l_{m,k}^{N_2,\text{pre}} + l_{\text{int}}, l_{m,k}^{N_2,c}), \quad u_I \triangleq \min(u_{\text{int}}, u_{m,k}^{N_2,c}). \quad (22)$$

The first argument of each max/min is what today’s `--use_relaxations` code uses on its own (computed by `compute_diff_and_comp_bounds()`, currently gated on `--use_relaxations`, *not* on `--adv_std_bound_tightening`). The second argument is the new contribution: the per-copy unconditional bound $[l_{m,k}^{N_2}, u_{m,k}^{N_2}]$ that Technique 1 (`--adv_std_bound_tightening`) produces via `intersect_per_copy_bounds()`. So the conditional bounds genuinely need *both* pipelines to be active: the comp-bound machinery to give the first argument, and Technique 1’s per-copy tightening to give the second. With either pipeline missing, the intersection collapses to whichever side is still finite and Technique 4 degrades — in the limit, all the way back to Technique 3.

Tier “both thin” — pre-activation coupling

Both binaries are dropped, both copies receive the simple triangle exactly as in Technique 3. We additionally emit *one* linear constraint per neuron:

$$l_{\text{int}} \leq \hat{z}_{m,k}^{\text{pert}} - \hat{z}_{m,k}^{\text{org}} \leq u_{\text{int}}. \quad (23)$$

This constraint involves only existing pre-activation variables; no binary, no big- M . Its purpose is to prevent the LP relaxation from choosing the two copies’ pre-activations independently. (At the post-activation level the analogous coupling $z^{+, \text{pert}} - z^{+, \text{org}} \in [\Delta^\downarrow, \Delta^\uparrow]$ is already contributed by the perturbed-interval coupling described after Section 4.1.4; the new line is its missing pre-activation analogue.)

Soundness — “one thin”. Let $\mathcal{F}_2$ be the feasible set of N_2 ’s exact MIP and let $\mathcal{F}_2^{(4)}$ be the feasible set after Technique 4 emission. We claim $\mathcal{F}_2 \subseteq \mathcal{F}_2^{(4)}$.

Take any $x \in \mathcal{F}_2$ and any neuron (m, k) in tier “one thin”; let $b = a_{m,k}^{\text{pre}}(x) \in \{0, 1\}$ be the realised value of the kept sibling binary at x . We verify each of (18)–(20) at x .

The two lower bounds in (18) appear in the exact ReLU encoding too, so x satisfies them.

For the upper bounds we split on b .

(i) $b = 1$. The exact big- M on the sibling enforces $\hat{z}^{\text{pre}}(x) \in [0, u_{m,k}^{N_2, \text{pre}}]$. Technique 1’s diff bound gives $\hat{z}^c(x) - \hat{z}^{\text{pre}}(x) \in [l_{\text{int}}, u_{\text{int}}]$, so

$$\hat{z}^c(x) \in [l_{\text{int}}, u_{m,k}^{N_2, \text{pre}} + u_{\text{int}}].$$

Combining with Technique 1’s unconditional bound $\hat{z}^c(x) \in [l_{m,k}^{N_2, c}, u_{m,k}^{N_2, c}]$ and the definition in (21) yields $\hat{z}^c(x) \in [l_A, u_A]$. The Wong–Kolter envelope on $[l_A, u_A]$ over-approximates $\text{ReLU}(\hat{z})$ on that interval (the standard triangle-relaxation lemma applied with endpoints l_A, u_A). Since $z^c(x) = \text{ReLU}(\hat{z}^c(x))$ under the exact encoding,

$$z^c(x) \leq \frac{u_A}{u_A - l_A} (\hat{z}^c(x) - l_A),$$

which together with $M(1 - b) = 0$ is exactly (19). Constraint (20) has slack $M \cdot b = M$ on its right-hand side; by the choice of M this is at least $u_{m,k}^{N_2, c} \geq z^c(x)$, so the constraint is vacuous.

(ii) $b = 0$. Symmetric: the exact big- M on the sibling enforces $\hat{z}^{\text{pre}}(x) \in [l_{m,k}^{N_2, \text{pre}}, 0]$, hence $\hat{z}^c(x) \in [l_{m,k}^{N_2, \text{pre}} + l_{\text{int}}, u_{\text{int}}] \cap [l_{m,k}^{N_2, c}, u_{m,k}^{N_2, c}] = [l_I, u_I]$ by (22). The triangle envelope on $[l_I, u_I]$ over-approximates ReLU, giving (20) with $M \cdot b = 0$; (19) is vacuous by the same big- M argument with $M(1 - b) = M$.

So x satisfies all four constraints, in either case of b .

Soundness — “both thin”. The added constraint (23) uses $[l_{\text{int}}, u_{\text{int}}]$, which is derived by sound interval-arithmetic propagation of the input perturbation through N_2 in `compute_diff_and_comp_bounds()`. Therefore for every $x \in \mathcal{F}_2$, $\hat{z}^{\text{pert}}(x) - \hat{z}^{\text{org}}(x) \in [l_{\text{int}}, u_{\text{int}}]$ holds by construction.

Why δ_{exact} remains in the search space. Combining the two soundness arguments above with the unchanged tier “neither” and the unchanged simple-triangle / big- M encodings everywhere else gives

$$\mathcal{F}_2 \subseteq \mathcal{F}_2^{(4)}.$$

Let $x^* \in \arg \max_{x \in \mathcal{F}_2} \text{obj}_{N_2}(x)$, so by definition $\text{obj}_{N_2}(x^*) = \delta_{\text{exact}}$. Since $x^* \in \mathcal{F}_2 \subseteq \mathcal{F}_2^{(4)}$, the point x^* is also feasible under Technique 4. The objective function is unchanged, so

$$\delta_{\text{SibGate}} \triangleq \max_{x \in \mathcal{F}_2^{(4)}} \text{obj}_{N_2}(x) \geq \text{obj}_{N_2}(x^*) = \delta_{\text{exact}}.$$

Taking the max over a superset can only raise the value, so δ_{exact} is in the Technique-4 search space and the reported δ_{SibGate} at OPTIMAL is a sound upper bound on δ_{exact} .

When the relaxation is genuinely tighter than Technique 3. The two conditional intervals $[l_A, u_A], [l_I, u_I]$ are subintervals of $[l_{m,k}^{N_2,c}, u_{m,k}^{N_2,c}]$, so each conditional triangle is no fatter than the single triangle Technique 3 emits. The intersection step (21)–(22) shrinks them strictly whenever Source A’s diff-derived bound and Source B’s absolute bound disagree at a given endpoint (Theorem 3). The “both thin” coupling line shrinks the joint LP feasible region of $(\hat{z}^{\text{org}}, \hat{z}^{\text{pert}})$ from a rectangle to a slab. In each case the relaxation is at least as tight as Technique 3 and strictly tighter on neurons where the additional information binds.

Summary of SibGate’s added constraints. On top of Technique 3’s tiered emission, SibGate writes at most three extra linear constraints per relaxed neuron — and *only* on neurons that Technique 3 already classified as “thin enough” for some level of relaxation:

- *Both-thin tier* (both binaries already dropped by Technique 3): SibGate adds the single pre-activation coupling line (23), $l_{\text{int}} \leq \hat{z}_{m,k}^{\text{pert}} - \hat{z}_{m,k}^{\text{org}} \leq u_{\text{int}}$. No binary, no big- M — just one slab tying the two preacts.
- *One-thin tier* (binary dropped on copy c , kept on sibling pre): SibGate *replaces* the simple Wong–Kolter triangle on copy c with the two big- M conditional upper bounds (19)–(20), each gated on the surviving sibling binary $a_{m,k}^{\text{pre}}$. The two lower bounds (18) are unchanged.
- *Neither-thin tier*: SibGate touches nothing; Technique 3 leaves both copies with their exact big- M encoding.

Why $\delta_{\text{SibGate}} = \delta_{\text{exact}}$ at $\tau = 0$. At threshold $\tau = 0$ Technique 3’s tiered rule collapses significantly. Recall the gap formula (15): for any *split* neuron ($l < 0 < u$) the triangle area $\text{gap}(l, u) > 0$ strictly; for any *stable* neuron ($l \geq 0$ or $u \leq 0$) it is exactly 0. So the only way for a gap to satisfy $g \leq \tau = 0$ is for the neuron to be stable on that copy. Walking the three tiers under this condition:

- *Both-thin* requires $\max(g^{\text{org}}, g^{\text{pert}}) \leq 0$, i.e. both copies stable. But a neuron stable on both copies has no binary on either side (the big- M encoding short-circuits on $l \geq 0$ or $u \leq 0$ before ever allocating a), so the decision-precompute step (`compute_n2_relax_decision!`, line 85 of `utils/n2_relax_decision.jl`) skips it via an early `continue`. **The both-thin tier is empty.**

- *One-thin* requires $\min(g^{\text{org}}, g^{\text{pert}}) \leq 0$ with the other > 0 . The relaxed copy c is the stable side; the sibling is genuinely split and keeps its binary. But on the stable side `relu()` short-circuits at `core_ops.jl:304/316` (returns 0 for $u \leq 0$ or x for $l \geq 0$) *before* the Technique 3 triangle path at line 562. No triangle is emitted on the stable copy — and correspondingly `n2_relu_state[(..., c)]` stays `nothing`. `apply_sibgate_constraints!` sees `st_c === nothing` at line 64 and continues without writing any constraint. **The one-thin tier emits nothing too.**
- *Neither-thin* is the default for every genuinely split neuron ($\text{gap} > 0 > \tau$), and the standard exact big- M encoding is used.

Every neuron of the N_2 MIP therefore receives the exact big- M encoding it would have received without any of Technique 3 or Technique 4 active. The bound globals SibGate populates (`relu_n2pert_*_bounds`, `n2_preact_*_bounds` via `compute_n2_pert_relaxation_bounds` at `run.jl:971-973`) are consulted only by SibGate’s own post-encoding pass (which we just established is a no-op at $\tau = 0$) and by unrelated niche flags (`--no_n1_binaries_and_relaxations_only_on_n2`, `--constrain_n2_xp_via_n1_zonotope`); `intersect_per_copy_bounds` (Sources A/B/C) never reads them. Consequently

$$\mathcal{F}_2^{(\text{SibGate}, \tau=0)} = \mathcal{F}_2^{(\tau=0)} = \mathcal{F}_2,$$

and at termination OPTIMAL

$$\delta_{\text{SibGate}}|_{\tau=0} = \delta_{\text{BTPR}}|_{\tau=0} = \delta_{\text{exact}}.$$

Equality is *constraint-level*: the same MIP is handed to Gurobi in both modes. Reported wall-clock t_{adv} and the incumbent returned at `TIME.LIMIT` may still differ between the two because Gurobi’s parallel branch-and-bound is nondeterministic at the default settings used here (`Threads=32`, `MIPFocus=3`, no fixed `Seed`); two re-runs of either configuration alone exhibit the same wallclock variance.

5 Algorithms

All 25 safe combinations share the same two-phase structure (Algorithm 1 and Algorithm 2). They differ only in which optional sub-steps are enabled, as indicated by the Boolean flag `ZONO`, the five-valued mode flag `VARHINT` $\in \{\text{off}, \text{prev}, \text{direct}, \text{prev_pgd}, \text{direct_pgd}\}$, and the threshold τ .

Algorithm 1 Phase 1: Solve N_1 and extract state (shared by all safe combos)

Require: Network N_1 , perturbation (P, ε) , class pairs (c_s, c_t) , timeout T

Ensure: State directory $\mathcal{S}$ on disk

- 1: **for** each class pair (c_s, c_t) **do**
 - 2: $x_{\text{pgd}} \leftarrow \text{HyperAttack}(N_1, c_s, c_t, P, \varepsilon)$ {PGD warm-start}
 - 3: Build MIP $\mathcal{M}_1$ for N_1 with standard encoding
 - 4: Apply PGD hints to $\mathcal{M}_1$
 - 5: Add VHAGaR dependency constraints to $\mathcal{M}_1$
 - 6: Add perturbed interval constraints to $\mathcal{M}_1$
 - 7: `optimize!`($\mathcal{M}_1, T$)
 - 8: Extract primal solution $(\text{var}_i, \text{val}_i)_i$
 - 9: Extract neuron bounds $(l_{m,k}, u_{m,k})_{\forall m,k}$ from `layers.info`
 - 10: Serialise to $\mathcal{S}/(c_s, c_t)$
 - 11: **end for**
-

Algorithm 2 Phase 2: Solve N_2 with advstd techniques (parameterised)

Require: Networks N_1, N_2 , perturbation (P, ε) , class pairs, timeout T , N1 state $\mathcal{S}$

Require: Flags: $\text{ZONO} \in \{\text{false}, \text{true}\}$, $\text{VARHINT} \in \{\text{off}, \text{prev}, \text{direct}, \text{prev_pgd}, \text{direct_pgd}\}$, $\tau \in \{\text{off}, 0.01, 0.1, 0.5, 1.0\}$

Ensure: Global robustness bound $\delta_{N_2}^*$ for each (c_s, c_t)

```

1: // Technique 1: Precompute N2 bounds from N1 (once, before the loop)
2: Load  $N_1$  diff bounds from  $\mathcal{S}$ 
3: if ZONO then
4:   Compute zonotope diff bounds (Source A) between  $N_1$  and  $N_2$ 
5:   Compute N1-tightened absolute N2 zonotope (Source B), intersect with Source A
6: else
7:   Compute interval-arithmetic diff bounds between  $N_1$  and  $N_2$ 
8: end if
9: for each class pair  $(c_s, c_t)$  do
10:  // Skip if result already exists for this pair
11:  if result file contains  $(c_s, c_t)$  then
12:    continue
13:  end if
14:  Load  $N_1$  state from  $\mathcal{S}/(c_s, c_t)$ : primal  $(\text{var}_i, \text{val}_i)$ , bounds  $(l_{m,k}, u_{m,k})$ 
15:  // Technique 1: Set N1 neuron bounds for N2 encoding
16:  Inject  $(l_{m,k}, u_{m,k})$  as initial N2 bounds {fixes stable ReLUs, tightens big- $M$ }
17:  if  $\tau \neq \text{off}$  then
18:    // Technique 3 (BoundTightPertRelax): tiered per-copy N2 relaxation
19:    Call compute_n2_relax_decision!( $\tau$ ) {fills dict  $(m, k) \rightarrow (\text{relax\_org}, \text{relax\_pert})$ }
20:    for each ReLU neuron  $(m, k)$  in N2, separately for copy  $\in \{\text{org}, \text{pert}\}$  do
21:      Compute  $g_{m,k}^{\text{copy}} = \text{gap}(l_{m,k}^{N_2, \text{copy}}, u_{m,k}^{N_2, \text{copy}})$  from per-copy final bounds
22:      Set  $(\text{relax\_org}, \text{relax\_pert}) \leftarrow \begin{cases} (\text{true}, \text{true}) & \max(g^{\text{org}}, g^{\text{pert}}) \leq \tau \\ (\text{true}, \text{false}) & g^{\text{org}} \leq g^{\text{pert}}, g^{\text{org}} \leq \tau \\ (\text{false}, \text{true}) & g^{\text{pert}} < g^{\text{org}}, g^{\text{pert}} \leq \tau \\ (\text{false}, \text{false}) & \text{else} \end{cases}$ 
23:    end for
24:    During MIP build: for each  $(m, k, \text{copy})$ , if flag true emit triangle LP relaxation on  $[l_{m,k}^{N_2, \text{copy}}, u_{m,k}^{N_2, \text{copy}}]$ , else emit exact big- $M$ 
25:  end if
26:  if --adv_std_n2_sibling_gate and  $\tau \neq \text{off}$  then
27:    // Technique 4 (SibGate): override emission per tier
28:    for each ReLU neuron  $(m, k)$  in N2 do
29:      if tier is both thin ( $\text{relax\_org} \wedge \text{relax\_pert}$ ) then
30:        Emit simple triangle on each copy as in Technique 3
31:        Emit pre-act coupling:  $l_{\text{int}} \leq \hat{z}_{m,k}^{\text{pert}} - \hat{z}_{m,k}^{\text{org}} \leq u_{\text{int}}$  {Eq. (23); no binary, no big- $M$ }
32:      else if tier is one thin (exactly one of  $\text{relax\_org}, \text{relax\_pert}$  is true) then
33:        Let  $c$  = relaxed copy, pre = sibling (kept exact);  $a_{m,k}^{\text{pre}}$  = sibling binary
34:        Compute  $(l_A, u_A, l_I, u_I)$  via Eqs. (21)–(22) {intersect Source-A conditional bounds with Tech-1 unconditional  $(l_{m,k}^{N_2, c}, u_{m,k}^{N_2, c})$ }
35:        Emit Eqs. (18)–(20) on  $c$ ; sibling keeps exact big- $M$ 
36:      else
37:        Tier neither: unchanged (exact big- $M$  on both copies)
38:      end if
39:    end for
40:  end if
41:  Build MIP  $\mathcal{M}_2$  for  $N_2$  {with tightened bounds and relaxed neurons}
42:   $x_{\text{pgd}} \leftarrow \text{HyperAttack}(N_2, c_s, c_t, P, \varepsilon)$ 

```

5.1 Instantiation for Each Safe Combination

[STALE] The 25 combinations below were ranked under the retired `_relaxT`-mode Technique 3. Under the new `BoundTightPertRelax` rule (Section 4.3) the ranking is being regenerated; until then, treat these rows as a historical reference and do not use them to pick safe combos for new experiments.

Table 1 maps each safe combination to its flag settings in Algorithm 2. To reproduce any combination, set the three flags accordingly; all other aspects of the algorithm are identical.

#	VARHINT	ZONO	τ
1	true	true	0.5
2	true	true	1.0
3	true	false	0.5
4	true	false	1.0
5	false	false	0.5
6	false	false	0.5
7	false	true	1.0
8	false	false	1.0
9	false	true	0.1
10	true	true	off
11	false	true	0.5
12	true	true	1.0
13	true	false	1.0
14	false	true	0.5
15	false	true	1.0
16	true	true	off
17	true	false	off
18	true	false	0.1
19	true	false	0.01
20	true	false	0.5
21	false	false	1.0
22	false	true	0.1
23	false	false	0.01
24	true	false	off
25	true	false	0.01

Table 1: Flag instantiation for each safe combination from the full ranking (rows 1–19 are **dominant**, rows 20–25 are **avg-win**). `boundTight` is always true and is not shown. `n1Probe` is always off and is not shown. $\tau = \text{off}$ disables Technique 3 entirely.

6 Summary of Effects

Table 2 summarises which Gurobi attribute each safe technique uses and whether it modifies the mathematical problem.

7 Command-Line Interface

A minimal invocation using the top-ranked safe combination (#1) is:

```
julia run.jl --mode advanced_standard_n2 \
  --dataset mnist --model_name cnn1 \
```


Technique	Flag (short)	Gurobi API	Modifies MIP?
1. Bound Tightening	<code>boundTight</code>	(linear constr.)	yes [†]
1+. Zonotope Bounds	<code>zono</code>	(linear constr.)	yes [†]
2. Variable Hints	<code>varHint=prev/direct</code>	<code>VarHintVal+VarHintPri</code>	no
2+. VarHints via PGD-consensus Start	<code>varHint=prev_pgd/direct_pgd</code>	<code>Start (set_start_value)</code>	no
3. N2 Relaxation	<code>BTPRelax=τ</code>	(replaces binaries)	yes [‡]
4. Sibling-Gated Refinement	<code>SibGate</code>	(replaces upper bounds, adds coupling line)	yes [‡]

Table 2: Summary of the safe advstd techniques. [†]Adds sound constraints derived from over-approximations; the optimum is preserved. [‡]Replaces binaries with LP relaxations; the reported δ is a sound upper bound ($\geq \delta_{\text{exact}}$).

```

--model_path /path/to/N1/model.p \
--model_path2 /path/to/N2/model.p \
--perturbation_patch --perturbation_size "1,14,14,3" \
--ctag 1 --ct "4,5,6" --timeout 1800 \
--n1_state_dir /path/to/n1_state \
--adv_std_mip_start false \
--adv_std_lp_basis false \
--adv_std_bound_tightening true \
--adv_std_zono_bounds true \
--adv_std_n2_relax_threshold 0.5 \ # tiered per-copy; output tag
    _BoundTightPertRelax0.5
--adv_std_n2_sibling_gate true \ # Technique 4: sibling-gated
    triangle + pre-act coupling; tag _SibGate
--adv_std_var_hint prev # also: off / direct (see
    Technique 2)

```

The Python sweep script can run all 25 safe combinations automatically:

```

python3 run_relaxation_sweep.py --timeout 1800 \
--arch_models cnn1=/path/to/n1 3x10=/path/to/n1 \
--advanced_standard \
--sweep_adv_std_mip_start false \
--sweep_adv_std_lp_basis false \
--sweep_adv_std_bound_tightening true \
--sweep_adv_std_zono_bounds false true \
--sweep_adv_std_n1_probe off \
--sweep_adv_std_n2_relax_threshold off 0.01 0.1 0.5 1.0 \
--sweep_adv_std_n2_sibling_gate false true \
--sweep_adv_std_var_hint off prev direct \
--advstd_safe_combos_only \
    paper_experiments/mnist/advstd_combo_ranking_vs_withPerturbed.csv \
--sweep_gurobi_seed 0 1 2 3 4 --ct 4,5,6

```

The `--advstd_safe_combos_only` flag filters the cross-product to only validated combinations (`perf.tier` $\in$ {dominant, avg-win}), blocking known loser/risky combos while allowing untested new combinations through. Note: with the recent `BoundTightPertRelax` (Technique 3), combos from the legacy ranking are treated as “untested” (the CSV either lacks a `relax_mode` column, or its entries pre-date the current flag schema) and admitted permissively — regenerate the ranking once a new sweep completes.

8 Safe-to-Use Combinations

Tables below list *every* recorded advstd combination at all seeds, all τ values, evaluated per-architecture and broken out by perturbation type and size. Each row corresponds to a single $(c_{\text{src}}, c_{\text{tgt}})$ cell of one combo (rows for the same combo are grouped, with the flag columns shown only on the first cell of the combo). We report wall-clock t_{std} and t_{adv} per cell, the per-cell speedup $\text{sp} = t_{\text{std}}/t_{\text{adv}}$ (higher is better; > 1 means advstd finished faster than the with-perturbed-intervals baseline), $\overline{\text{sp}}_{c_{\text{src}}}$ (mean sp over all tested c_{tgt} for the same c_{src} within a combo, shown once per c_{src} group), and n_{relax} (per-cell number of N2 binary variables removed by Technique 3). The `bound.tight` column labels which Technique 1 variant is used (`interval` / `zono` / `none`; `n1Probe` is always off), and the τ column is the Technique 3 threshold. Within each (perturbation, size) block combos are sorted by descending mean sp (fastest advstd first); cells within a combo are sorted by $(c_{\text{src}}, c_{\text{tgt}})$. `mipStart` and `lpBasis` are always off and omitted for brevity. Cells where both modes hit Gurobi’s `TIME_LIMIT` are shown with their wall-clock times; for those cells, a separate per-arch gap-comparison table reports how often advstd reached a tighter $u - l$ bound on the objective than standard, alongside the mean gap of each method.

#	bound.tight	varHint	τ	$\overline{\text{sp}}$	n_wins	n_cells	win_rate	(arch, pert, p_size) where standard beats advstd on every c_{src}	(arch, c_{src}) where, on <i>every</i> tested (pert, p_size), advstd was either faster on average ($\overline{\text{sp}} > 1$), or both methods averaged at the wall-clock cap ($t_{\text{std}} \approx t_{\text{adv}} \approx T$, so the $\overline{\text{sp}} \leq 1$ verdict is a timeout artefact) AND advstd’s $u - l$ gap on the timeout cells was no worse than standard’s (entries needing this timeout-tied clause for at least one (pert, p_size) are tagged (<i>tied at TIME_LIMIT</i>))
1	zono	prev_pgd	0.5+sg	1.440	228	333	68.47%	<i>none</i>	<3x50,1>*; <cnn1,0>; <3x50,0> (<i>tied at TIME_LIMIT</i>)
2	zono	prev_pgd	0.0+sg	1.338	224	329	68.09%	<i>none</i>	<3x50,1>*; <3x10,1>; <cnn1,0>; <3x50,0> (<i>tied at TIME_LIMIT</i>)
3	interval	prev_pgd	0.5+sg	1.421	222	334	66.47%	<i>none</i>	<3x50,1>*; <3x10,1>; <cnn1,0>; <cnn1,1> (<i>tied at TIME_LIMIT</i>); <3x50,0> (<i>tied at TIME_LIMIT</i>)
4	interval	no	0.5	1.224	89	137	64.96%	<i>none</i>	<i>none</i>
5	zono	direct_pgd	0.5	1.341	83	130	63.85%	<3x10,translation,3,3>; <3x10,occ,5,5,5>	<i>none</i>
6	interval	prev_pgd	0.0+sg	1.387	213	336	63.39%	<i>none</i>	<3x50,1>*; <3x10,1>; <cnn1,1> (<i>tied at TIME_LIMIT</i>); <3x50,0> (<i>tied at TIME_LIMIT</i>)
7	interval	prev_pgd	0.0	1.311	85	137	62.04%	<3x10,translation,3,3>; <3x10,occ,5,5,5>	<cnn1,2>*
8	interval	prev_pgd	0.5	1.350	83	136	61.03%	<3x10,translation,3,3>; <3x10,occ,5,5,5>	<i>none</i>

Table 3: Overall advstd combination ranking at all seeds, all τ values (part 1/2, combos ranked 1–8 of 16 by `win_rate`) — one row per flag combination, aggregated over every (arch, pert, size, c_{src} , c_{tgt}) cell in the per-arch tables below. $\overline{\text{sp}}$: mean per-cell $\text{sp} = t_{\text{std}}/t_{\text{adv}}$; n_{wins} counts cells with $\text{sp} > 1$; $\text{win_rate} = n_{\text{wins}}/n_{\text{cells}}$ (bold when $> 50\%$). losers: (arch, pert, p_size) triples where every tested c_{src} averaged $\text{sp} < 1$. winners: (arch, c_{src}) pairs where every tested (pert, p_size) averaged $\text{sp} > 1$, or $\overline{t_{\text{std}}}$ and $\overline{t_{\text{adv}}}$ both sat within 60s of the observed `TIME_LIMIT` cap *and* the gap rule ($\overline{\text{gap}}_{\text{adv}} \leq \overline{\text{gap}}_{\text{std}}$ or $|\overline{\text{gap}}_{\text{std}} - \overline{\text{gap}}_{\text{adv}}| < 0.05$) holds on the timeout cells (such entries are tagged (*tied at TIME_LIMIT*)); * flags partial coverage (some (pert, p_size) for this arch were not exercised by this combo). Sorted by descending `win_rate`, then $\overline{\text{sp}}$; $\overline{\text{sp}} > 1$ in **bold**. Rows are shaded blue.

#	bound_tight	varHint	τ	$\overline{sp}$	n_wins	n_cells	win_rate	$\langle \text{arch}, \text{pert}, \text{p_size} \rangle$ where standard beats advstd on every c_{src}	$\langle \text{arch}, c_{src} \rangle$ where, on <i>every</i> tested $(\text{pert}, \text{p_size})$, advstd was either faster on average ($\overline{sp} > 1$), or both methods averaged at the wall-clock cap ($\overline{t_{std}} \approx \overline{t_{adv}} \approx T$, so the $\overline{sp} \leq 1$ verdict is a timeout artefact) AND advstd's $u-l$ gap on the timeout cells was no worse than standard's (entries needing this timeout-tied clause for at least one $(\text{pert}, \text{p_size})$ are tagged (<i>tied at TIME_LIMIT</i>))
9	zono	no	0.5	1.171	79	130	60.77%	$\langle \text{cnn1}, \text{translation}, 3, 1 \rangle$	$\langle \text{cnn1}, 2 \rangle^*$
10	interval	direct_pgd	0.5	1.321	82	137	59.85%	$\langle 3 \times 10, \text{translation}, 3, 3 \rangle;$ $\langle 3 \times 10, \text{occ}, 5, 5, 5 \rangle$	$\langle \text{cnn1}, 2 \rangle^*; \langle \text{cnn1}, 0 \rangle$
11	zono	prev_pgd	0.5	1.297	192	325	59.08%	$\langle 3 \times 10, \text{translation}, 3, 3 \rangle$	$\langle 3 \times 50, \text{cnn1}, 2 \rangle^*; \langle \text{cnn1}, 0 \rangle; \langle 3 \times 50, 0 \rangle$ (<i>tied at TIME_LIMIT</i>)
12	zono	direct_pgd	0.0	1.216	79	136	58.09%	$\langle 3 \times 10, \text{translation}, 3, 3 \rangle;$ $\langle 3 \times 10, \text{occ}, 5, 5, 5 \rangle$	$\langle \text{cnn1}, 2 \rangle^*$
13	zono	prev_pgd	0.0	1.241	191	329	58.05%	$\langle 3 \times 10, \text{translation}, 3, 3 \rangle;$ $\langle 3 \times 10, \text{occ}, 5, 5, 5 \rangle$	$\langle 3 \times 50, 2 \rangle^*; \langle \text{cnn1}, 0 \rangle; \langle 3 \times 50, 0 \rangle$ (<i>tied at TIME_LIMIT</i>)
14	zono	no	0.0	1.130	74	130	56.92%	<i>none</i>	$\langle \text{cnn1}, 2 \rangle^*$
15	interval	direct_pgd	0.0	1.194	77	137	56.20%	$\langle 3 \times 10, \text{translation}, 3, 3 \rangle$	$\langle \text{cnn1}, 2 \rangle^*$
16	interval	no	0.0	1.160	76	136	55.88%	$\langle 3 \times 10, \text{translation}, 3, 3 \rangle$	$\langle \text{cnn1}, 2 \rangle^*$

Table 4: Overall advstd combination ranking (part 2/2, combos ranked 9–16 of 16 by win_rate); continuation of Table 3. Column meanings: see Table 3.

Per-perturbation histograms: $\overline{sp}$ paired with win-rate

The figures below answer “which advstd combination should I pick on this perturbation $\times$ architecture?” from *two* angles at once. Each row of paired charts shows a single $\langle \text{arch}, \text{pert} \rangle$ slice twice: the **left** chart reports the combo’s mean speed-up (how fast it is on average), the **right** chart reports its win rate (how often it actually beats standard). Combos occupy a **stable x position** across both charts of a row *and* across every row of an arch (alphabetical on the combo abbreviation), so a reader can scan one column down the page and watch a combo’s behaviour change as the perturbation type changes in either metric.

Left chart — bar height = the arithmetic mean of $\overline{sp}$ across every $c_{\text{src}} \times c_{\text{tgt}} \times \text{seed}$ cell across all p_sizes of the perturbation type. **Each bar is vertically split** into a *solid bottom* and a *faded + diagonally hatched top* in the same combo colour: the solid fraction equals $\text{wins}/\text{p_sizes}$, where a p_size’s win is defined by a *hybrid* rule that mirrors the orange gap-comparison table’s treatment of the two cell populations — a p_size in which every cell finished (no `TIME LIMIT`) wins iff $\overline{sp} > 1$; a p_size in which every cell hit `TIME LIMIT` wins iff $\overline{\text{gap}}_{\text{adv}} < \overline{\text{gap}}_{\text{std}}$ across its timeout cells; a p_size with cells of both kinds is judged *per cell* and wins iff a majority of cells win on their own type (each non-timeout cell votes *yes* when $sp > 1$, each timeout cell votes *yes* when its $u - l$ gap on advstd is tighter than on standard). Bars above 3.0 are clipped at that height and the true value is printed in **bold** above the bar; the dashed line marks the $\overline{sp} = 1$ breakeven.

Right chart — each column encodes three quantities by nested rectangles on an absolute-count y-axis (scale $0 \rightarrow n_{\text{max}}$, where n_{max} is the largest n_cells of any combo on the chart, marked by a dashed reference line): a faint *envelope* of height n_{max} (maximum coverage anywhere on the chart); a mid-grey *tests* rectangle of height n_cells (this combo’s actual coverage on the slice); and a coloured *wins* bar inside, split into a *solid bottom* = finished wins (cells where the run finished and advstd was faster, $sp > 1$) and a *faded + hatched top* = timeout wins (cells where both modes hit `TIME LIMIT` and advstd’s $\overline{\text{gap}}_{\text{adv}} < \overline{\text{gap}}_{\text{std}}$). The combo’s **win rate** is therefore the fraction of the mid-grey rectangle that is coloured; an under-tested combo’s column is visibly shorter than its envelope. A combo with no timeout cells shows a fully solid coloured bar, and a combo whose wins all came from tighter-gap timeout cells shows only a hatched top.

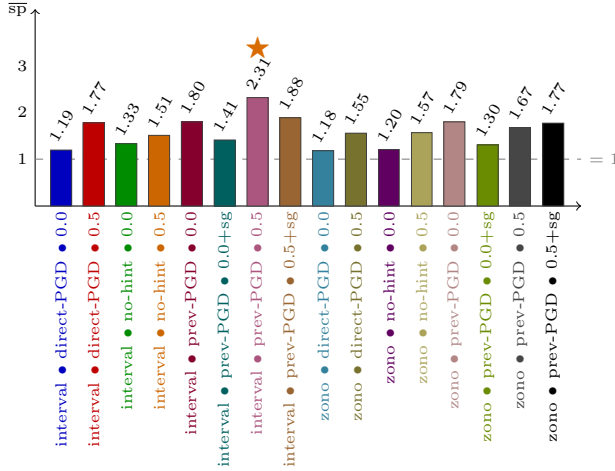
The ★ above each chart marks the best combo on that chart (highest $\overline{sp}$ on the left, highest win rate on the right — the two stars may pick different combos when the orderings disagree). A best-to-worst ranking summary line sits above each chart with the per-combo metric value and the relevant breakdown ($\text{wins}/\text{p_sizes}$ for sp , $(\text{finished} + \text{timeout})/n$ for win rate). Cells with mismatched `--timeout` caps between the two modes are excluded (same rule as the per-arch perturbation tables and the blue overall ranking).

Architecture cnn1 — $\overline{sp}$ paired with win-rate

This page shows 4 paired rows, one per **perturbation type** measured on **cnn1**. Each row repeats the same $\langle \text{arch}, \text{pert} \rangle$ slice twice: the *left* chart is the mean-speedup view (section intro above); the *right* chart is the win-rate view (same intro, different bar definition). Combo columns occupy the same x position in both charts of a row, so a column can be scanned down the page to follow one combo across perturbation types in both metrics.

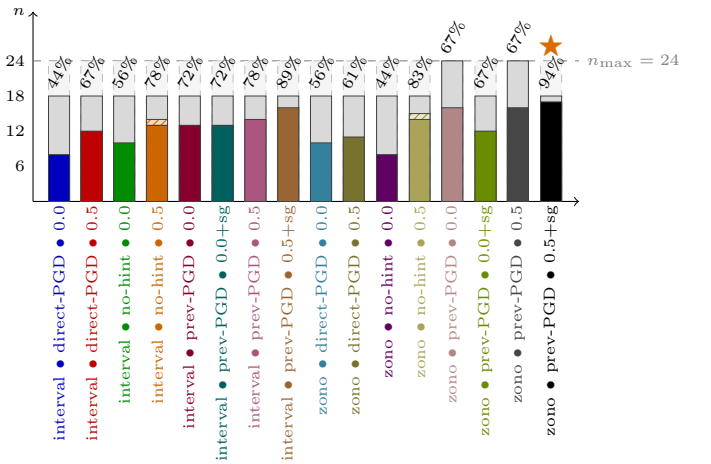
sp **cnn1** | pert **patch** (p-sizes merged: 1,14,14,3) | 16
combos, total $n = 300$ cells.

Ranking (best $\rightarrow$ worst): ★ 2.31 (1/1) 1.88 (1/1)
 1.80 (1/1) 1.79 (1/1) 1.77 (1/1) 1.77 (1/1)
 1.67 (1/1) 1.57 (1/1) 1.55 (1/1) 1.51 (1/1)
 1.41 (1/1) 1.33 (1/1) 1.30 (1/1) 1.20 (1/1)
 1.19 (1/1) 1.18 (1/1)



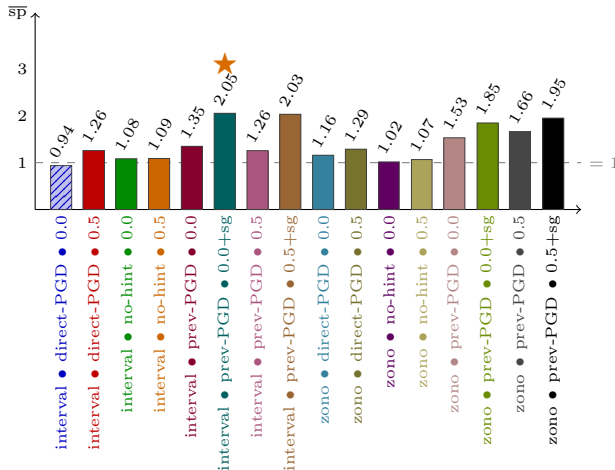
win **cnn1** | pert **patch** (p-sizes merged: 1,14,14,3) | 16
combos, total $n = 300$ cells, $n_{\max} = 24$.

Ranking (best $\rightarrow$ worst): ★ 94% (17f+0t/18)
 89% (16f+0t/18) 83% (14f+1t/18) 78% (13f+1t/18)
 78% (14f+0t/18) 72% (13f+0t/18) 72% (13f+0t/18)
 67% (12f+0t/18) 67% (16f+0t/24) 67% (12f+0t/18)
 67% (16f+0t/24) 61% (11f+0t/18) 56% (10f+0t/18)
 56% (10f+0t/18) 44% (8f+0t/18) 44% (8f+0t/18)



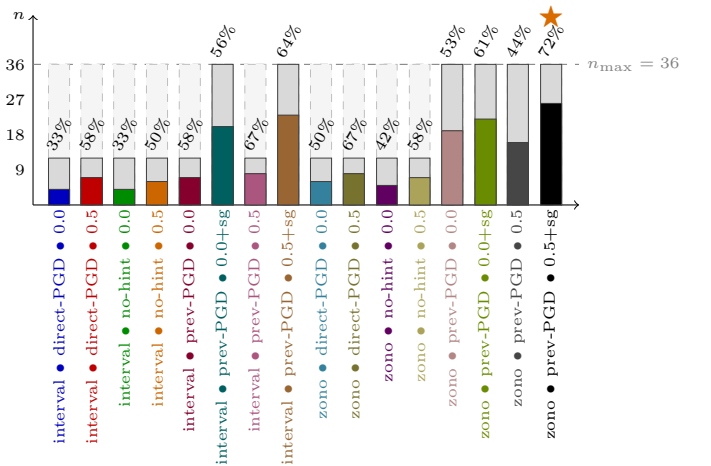
sp **cnn1** | pert **occ** (p-sizes merged: 3,3,5,5,5,5) | 16
combos, total $n = 336$ cells.

Ranking (best $\rightarrow$ worst): ★ 2.05 (2/2) 2.03 (2/2)
 1.95 (2/2) 1.85 (2/2) 1.66 (2/2) 1.53 (2/2)
 1.35 (1/1) 1.29 (1/1) 1.26 (1/1) 1.26 (1/1)
 1.16 (1/1) 1.09 (1/1) 1.08 (1/1) 1.07 (1/1)
 1.02 (1/1) 0.94 (0/1)



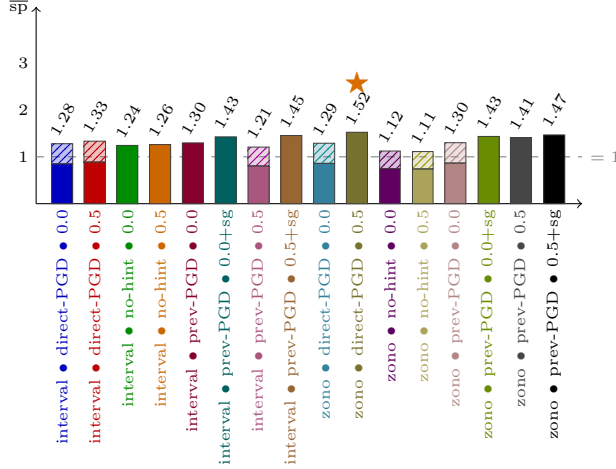
win **cnn1** | pert **occ** (p-sizes merged: 3,3,5,5,5,5) | 16
combos, total $n = 336$ cells, $n_{\max} = 36$.

Ranking (best $\rightarrow$ worst): ★ 72% (26f+0t/36)
 67% (8f+0t/12) 67% (8f+0t/12) 64% (23f+0t/36)
 61% (22f+0t/36) 58% (7f+0t/12) 58% (7f+0t/12)
 58% (7f+0t/12) 56% (20f+0t/36) 53% (19f+0t/36)
 50% (6f+0t/12) 50% (6f+0t/12) 44% (16f+0t/36)
 42% (5f+0t/12) 33% (4f+0t/12) 33% (4f+0t/12)



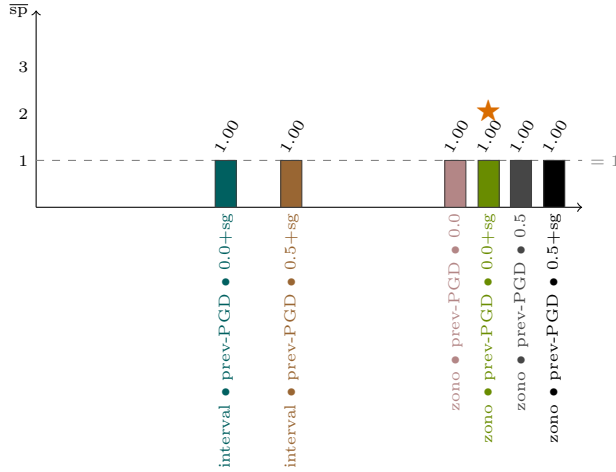
sp cnn1 | pert translation (p-sizes merged: 1,1, 3,1, 3,3) | 16 combos, total $n = 766$ cells.

Ranking (best → worst): ★ 1.52 (3/3) ■ 1.47 (3/3)
 ■ 1.45 (3/3) ■ 1.43 (3/3) ■ 1.43 (3/3) ■ 1.41 (3/3)
 ■ 1.33 (2/3) ■ 1.30 (2/3) ■ 1.30 (3/3) ■ 1.29 (2/3)
 ■ 1.28 (2/3) ■ 1.26 (3/3) ■ 1.24 (3/3) ■ 1.21 (2/3)
 ■ 1.12 (2/3) ■ 1.11 (2/3)



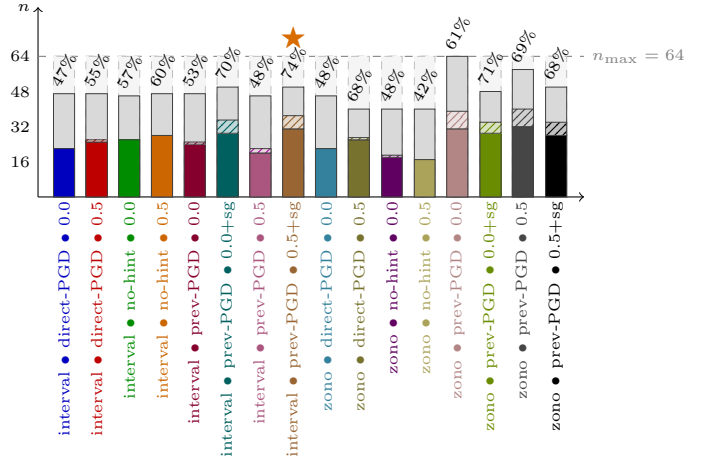
sp cnn1 | pert linf (p-sizes merged: 0.1) | 6 combos, total $n = 86$ cells.

Ranking (best → worst): ★ 1.00 (1/1) ■ 1.00 (1/1)
 ■ 1.00 (1/1) ■ 1.00 (1/1) ■ 1.00 (1/1) ■ 1.00 (1/1)



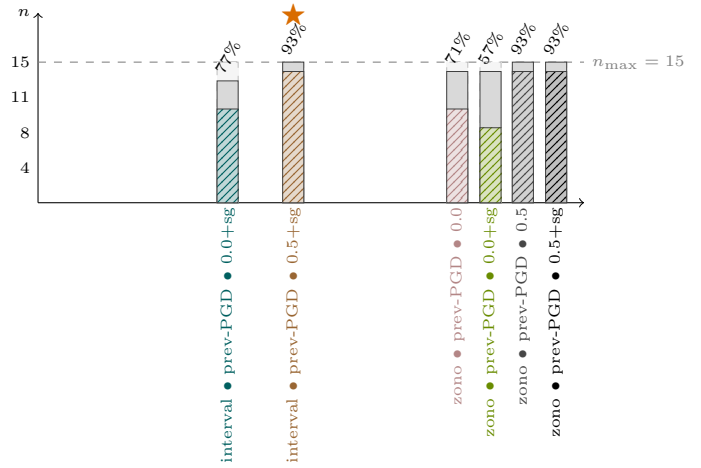
win cnn1 | pert translation (p-sizes merged: 1,1, 3,1, 3,3) | 16 combos, total $n = 766$ cells, $n_{\max} = 64$.

Ranking (best → worst): ★ 74% (31f+6t/50)
 ■ 71% (29f+5t/48) ■ 70% (29f+6t/50) ■ 69% (32f+8t/58)
 ■ 68% (28f+6t/50) ■ 68% (26f+1t/40) ■ 61% (31f+8t/64)
 ■ 60% (28f+0t/47) ■ 57% (26f+0t/46) ■ 55% (25f+1t/47)
 ■ 53% (24f+1t/47) ■ 48% (20f+2t/46) ■ 48% (22f+0t/46)
 ■ 48% (18f+1t/40) ■ 47% (22f+0t/47) ■ 42% (17f+0t/40)



win cnn1 | pert linf (p-sizes merged: 0.1) | 6 combos, total $n = 86$ cells, $n_{\max} = 15$.

Ranking (best → worst): ★ 93% (0f+14t/15)
 ■ 93% (0f+14t/15) ■ 93% (0f+14t/15) ■ 77% (0f+10t/13)
 ■ 71% (0f+10t/14) ■ 57% (0f+8t/14)

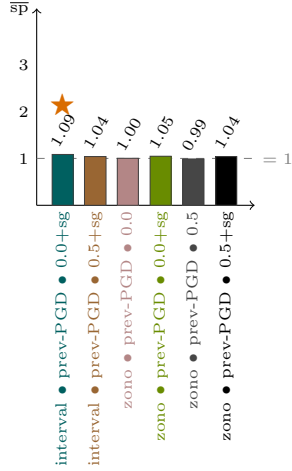


Architecture 3x50 — $\overline{sp}$ paired with win-rate

This page shows 4 paired rows, one per **perturbation type** measured on **3x50**. Each row repeats the same $\langle \text{arch}, \text{pert} \rangle$ slice twice: the *left* chart is the mean-speedup view (section intro above); the *right* chart is the win-rate view (same intro, different bar definition). Combo columns occupy the same x position in both charts of a row, so a column can be scanned down the page to follow one combo across perturbation types in both metrics.

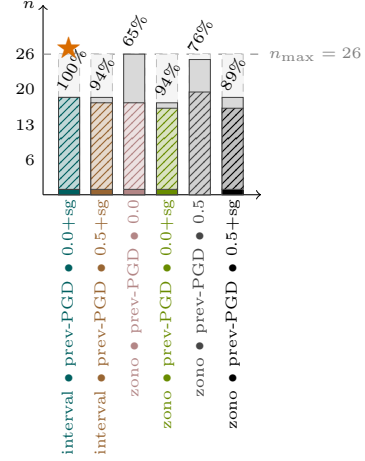
sp 3x50 | pert **patch** (p.sizes merged: 1,14,14,3) | 6 combos, total $n = 122$ cells.

Ranking (best $\rightarrow$ worst): ★ 1.09 (1/1) 1.05 (1/1)
1.04 (1/1) 1.04 (1/1) 1.00 (1/1) 0.99 (1/1)



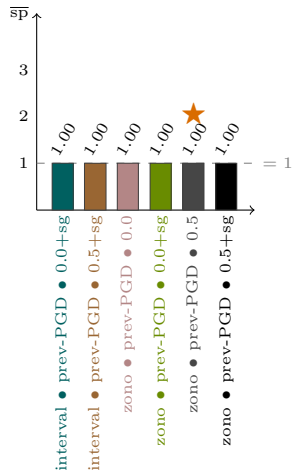
win 3x50 | pert **patch** (p.sizes merged: 1,14,14,3) | 6 combos, total $n = 122$ cells, $n_{\max} = 26$.

Ranking (best $\rightarrow$ worst): ★ 100% (1f+17t/18)
94% (1f+16t/18) 94% (1f+15t/17) 89% (1f+15t/18)
76% (0f+19t/25) 65% (1f+16t/26)



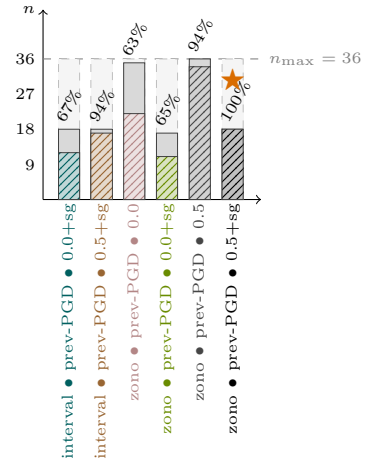
sp 3x50 | pert **occ** (p.sizes merged: 3,3,5, 5,5,5) | 6 combos, total $n = 142$ cells.

Ranking (best $\rightarrow$ worst): ★ 1.00 (2/2) 1.00 (2/2)
1.00 (2/2) 1.00 (2/2) 1.00 (2/2) 1.00 (2/2)



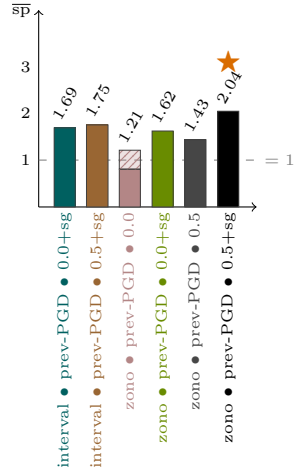
win 3x50 | pert **occ** (p.sizes merged: 3,3,5, 5,5,5) | 6 combos, total $n = 142$ cells, $n_{\max} = 36$.

Ranking (best $\rightarrow$ worst): ★ 100% (0f+18t/18)
94% (0f+17t/18) 94% (0f+34t/36) 67% (0f+12t/18)
65% (0f+11t/17) 63% (0f+22t/35)



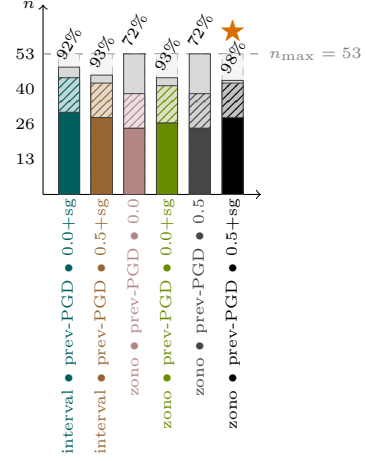
sp 3x50 | pert **translation** (p-sizes merged: 1,1, 3,1, 3,3) | 6 combos, total $n = 286$ cells.

Ranking (best → worst): ★ 2.04 (3/3) 1.75 (3/3) 1.69 (3/3) 1.62 (3/3) 1.43 (3/3) 1.21 (2/3)



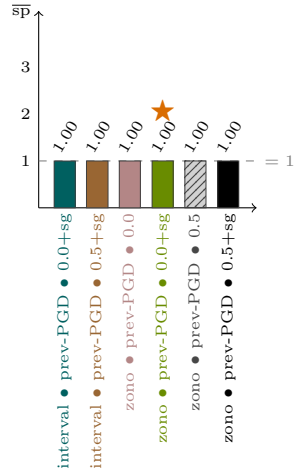
win 3x50 | pert **translation** (p-sizes merged: 1,1, 3,1, 3,3) | 6 combos, total $n = 286$ cells, $n_{\max} = 53$.

Ranking (best → worst): ★ 98% (29f+13t/43) 93% (29f+13t/45) 93% (27f+14t/44) 92% (31f+13t/48) 72% (25f+13t/53) 72% (25f+13t/53)



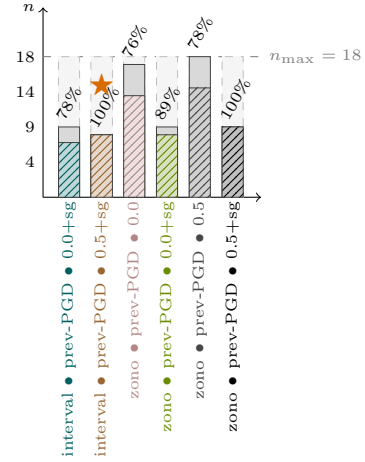
sp 3x50 | pert **linf** (p-sizes merged: 0.1) | 6 combos, total $n = 70$ cells.

Ranking (best → worst): ★ 1.00 (1/1) 1.00 (0/1) 1.00 (1/1) 1.00 (1/1) 1.00 (1/1) 1.00 (1/1)



win 3x50 | pert **linf** (p-sizes merged: 0.1) | 6 combos, total $n = 70$ cells, $n_{\max} = 18$.

Ranking (best → worst): ★ 100% (0f+8t/8) 100% (0f+9t/9) 89% (0f+8t/9) 78% (0f+7t/9) 78% (0f+14t/18) 76% (0f+13t/17)

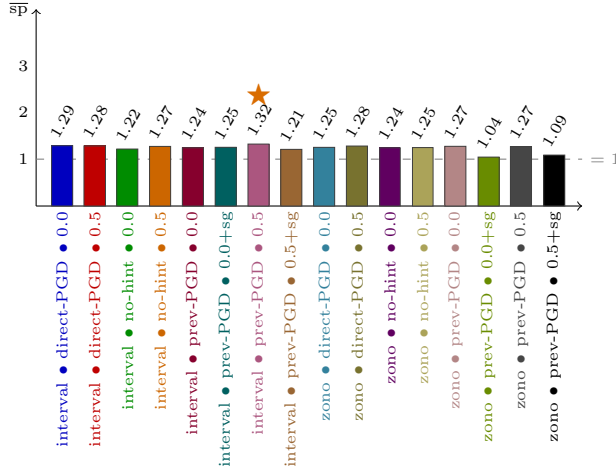


Architecture 3x10 — $\overline{sp}$ paired with win-rate

This page shows 4 paired rows, one per **perturbation type** measured on **3x10**. Each row repeats the same $\langle \text{arch}, \text{pert} \rangle$ slice twice: the *left* chart is the mean-speedup view (section intro above); the *right* chart is the win-rate view (same intro, different bar definition). Combo columns occupy the same x position in both charts of a row, so a column can be scanned down the page to follow one combo across perturbation types in both metrics.

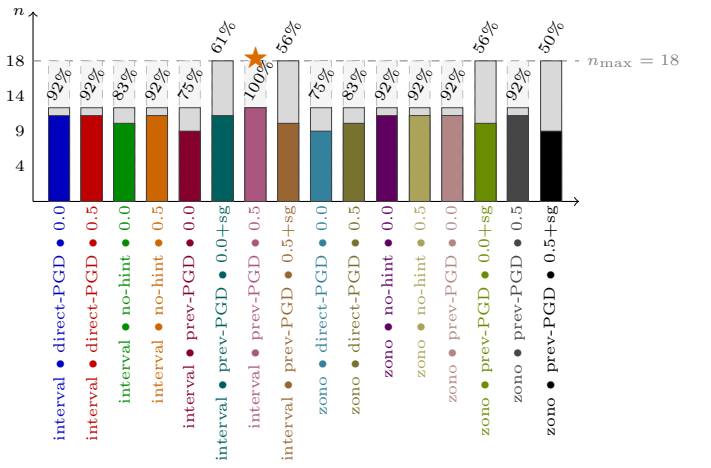
sp 3x10 | pert **patch** (p-sizes merged: 1,14,14,3) | 16
combos, total $n = 216$ cells.

Ranking (best $\rightarrow$ worst): ★ 1.32 (1/1) ■ 1.29 (1/1)
 ■ 1.28 (1/1) ■ 1.28 (1/1) ■ 1.27 (1/1) ■ 1.27 (1/1)
 ■ 1.27 (1/1) ■ 1.25 (1/1) ■ 1.25 (1/1) ■ 1.25 (1/1)
 ■ 1.24 (1/1) ■ 1.24 (1/1) ■ 1.22 (1/1) ■ 1.21 (1/1)
 ■ 1.09 (1/1) ■ 1.04 (1/1)



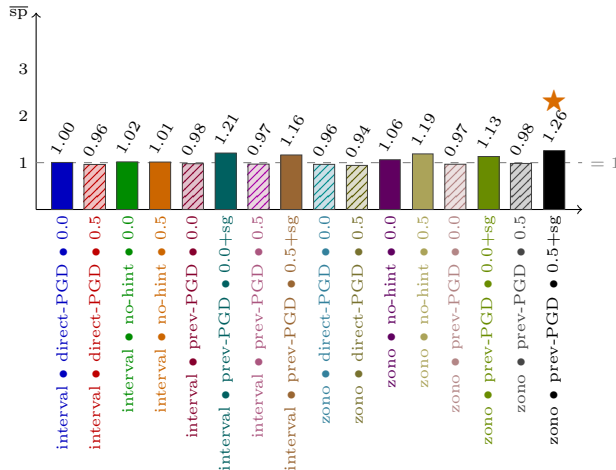
win 3x10 | pert **patch** (p-sizes merged: 1,14,14,3) | 16
combos, total $n = 216$ cells, $n_{\max} = 18$.

Ranking (best $\rightarrow$ worst): ★ 100% (12f+0t/12)
 ■ 92% (11f+0t/12) ■ 92% (11f+0t/12) ■ 92% (11f+0t/12)
 ■ 92% (11f+0t/12) ■ 92% (11f+0t/12) ■ 92% (11f+0t/12)
 ■ 92% (11f+0t/12) ■ 83% (10f+0t/12) ■ 83% (10f+0t/12)
 ■ 75% (9f+0t/12) ■ 75% (9f+0t/12) ■ 61% (11f+0t/18)
 ■ 56% (10f+0t/18) ■ 56% (10f+0t/18) ■ 50% (9f+0t/18)



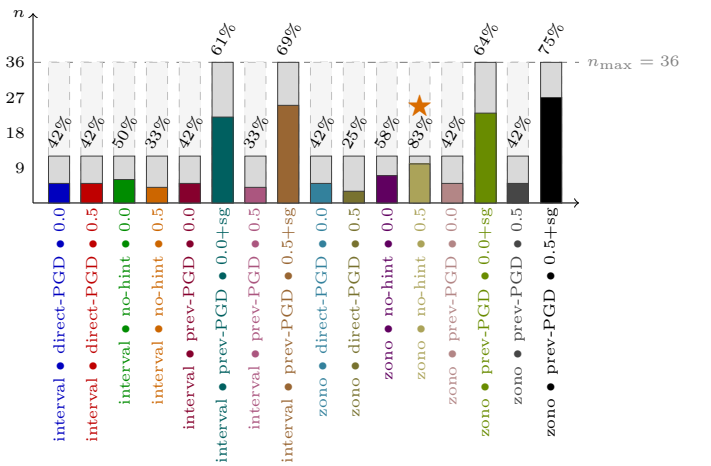
sp 3x10 | pert **occ** (p-sizes merged: 3,3,5,5,5,5) | 16
combos, total $n = 288$ cells.

Ranking (best $\rightarrow$ worst): ★ 1.26 (2/2) ■ 1.21 (2/2)
 ■ 1.19 (1/1) ■ 1.16 (2/2) ■ 1.13 (2/2) ■ 1.06 (1/1)
 ■ 1.02 (1/1) ■ 1.01 (1/1) ■ 1.00 (1/1) ■ 0.98 (0/1)
 ■ 0.98 (0/1) ■ 0.97 (0/1) ■ 0.97 (0/1) ■ 0.96 (0/1)
 ■ 0.96 (0/1) ■ 0.94 (0/1)



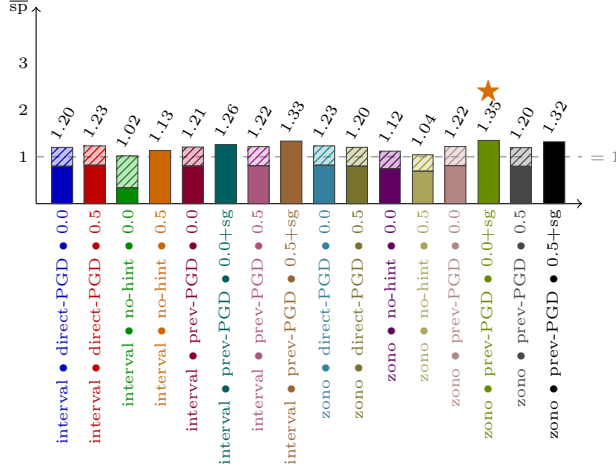
win 3x10 | pert **occ** (p-sizes merged: 3,3,5,5,5,5) | 16
combos, total $n = 288$ cells, $n_{\max} = 36$.

Ranking (best $\rightarrow$ worst): ★ 83% (10f+0t/12)
 ■ 75% (27f+0t/36) ■ 69% (25f+0t/36) ■ 64% (23f+0t/36)
 ■ 61% (22f+0t/36) ■ 58% (7f+0t/12) ■ 50% (6f+0t/12)
 ■ 42% (5f+0t/12) ■ 42% (5f+0t/12) ■ 42% (5f+0t/12)
 ■ 42% (5f+0t/12) ■ 42% (5f+0t/12) ■ 42% (5f+0t/12)
 ■ 33% (4f+0t/12) ■ 33% (4f+0t/12) ■ 25% (3f+0t/12)



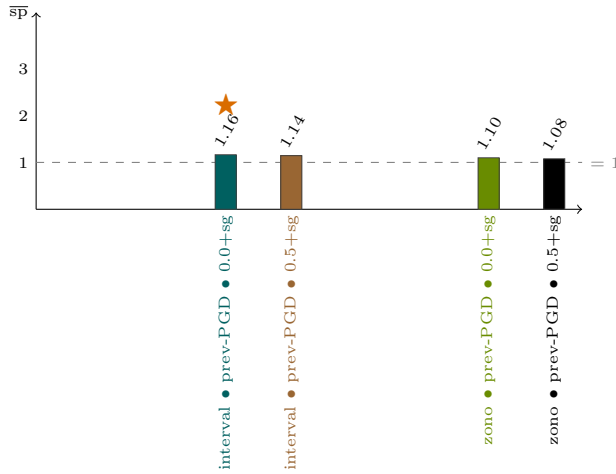
sp 3x10 | pert translation (p-sizes merged: 1,1, 3,1, 3,3) | 16 combos, total $n = 648$ cells.

Ranking (best → worst): ★ 1.35 (3/3) 1.33 (3/3)
 1.32 (3/3) 1.26 (3/3) 1.23 (2/3) 1.23 (2/3)
 1.22 (2/3) 1.22 (2/3) 1.21 (2/3) 1.20 (2/3)
 1.20 (2/3) 1.20 (2/3) 1.13 (3/3) 1.12 (2/3)
 1.04 (2/3) 1.02 (1/3)



sp 3x10 | pert linf (p-sizes merged: 0.1) | 4 combos, total $n = 72$ cells.

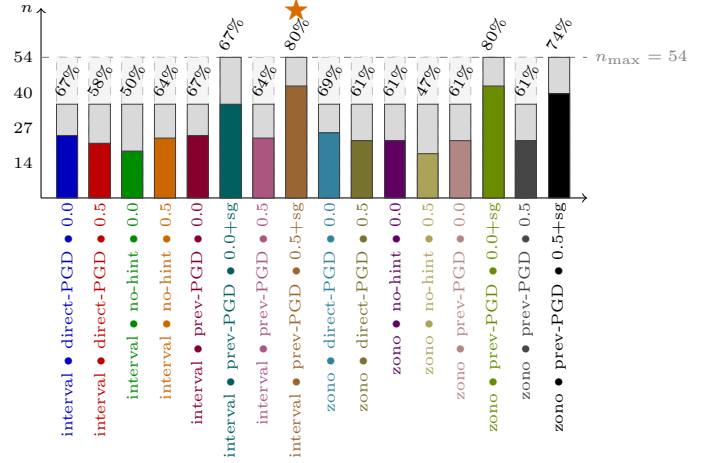
Ranking (best → worst): ★ 1.16 (1/1) 1.14 (1/1)
 1.10 (1/1) 1.08 (1/1)



win 3x10 | pert translation (p-sizes merged: 1,1, 3,1, 3,3) | 16 combos, total $n = 648$ cells, $n_{\max} = 54$.

Ranking (best → worst): ★ 80% (43f+0t/54)

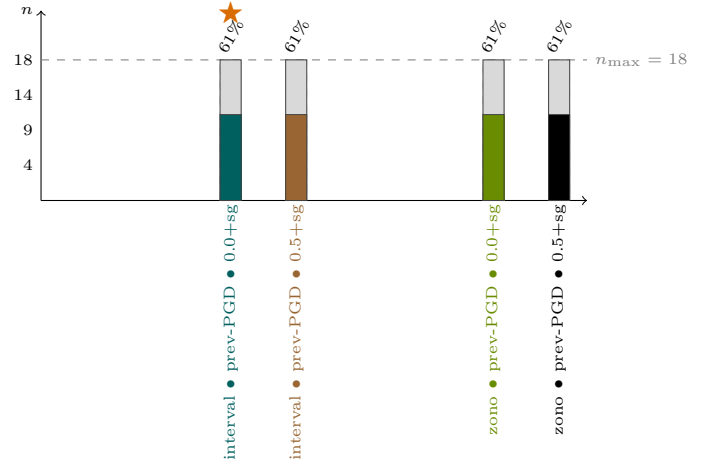
80% (43f+0t/54) 74% (40f+0t/54) 69% (25f+0t/36)
 67% (24f+0t/36) 67% (24f+0t/36) 67% (36f+0t/54)
 64% (23f+0t/36) 64% (23f+0t/36) 61% (22f+0t/36)
 61% (22f+0t/36) 61% (22f+0t/36) 61% (22f+0t/36)
 58% (21f+0t/36) 50% (18f+0t/36) 47% (17f+0t/36)



win 3x10 | pert linf (p-sizes merged: 0.1) | 4 combos, total $n = 72$ cells, $n_{\max} = 18$.

Ranking (best → worst): ★ 61% (11f+0t/18)

61% (11f+0t/18) 61% (11f+0t/18) 61% (11f+0t/18)

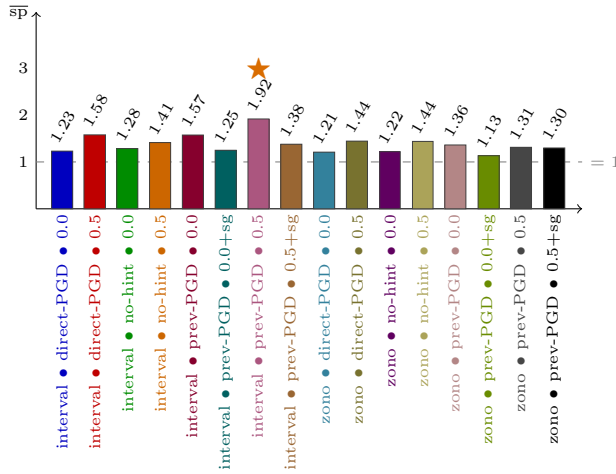


Per-perturbation histograms across all architectures: $\overline{sp}$ paired with win-rate

This section pools every $(c_{src}, c_{tgt}, seed)$ cell across *all* measured architectures, so each paired row reports a combo's behaviour on one perturbation type aggregated over every arch we ran. Bar definitions, the best-combo marker, and the rules used to split each bar into solid + hatched segments are exactly the same as in the per-arch section above. Architecture is shown as **ALL** in each chart's caption.

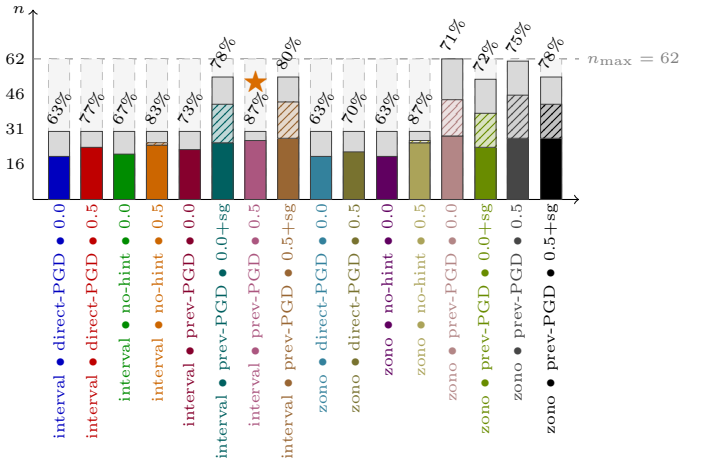
sp ALL | pert **patch** (p-sizes merged: 1,14,14,3) | 16 combos, total $n = 638$ cells.

Ranking (best → worst): ★ 1.92 (1/1) 1.58 (1/1)
 1.57 (1/1) 1.44 (1/1) 1.41 (1/1)
 1.38 (1/1) 1.36 (1/1) 1.31 (1/1) 1.30 (1/1)
 1.28 (1/1) 1.25 (1/1) 1.23 (1/1) 1.22 (1/1)
 1.21 (1/1) 1.13 (1/1)



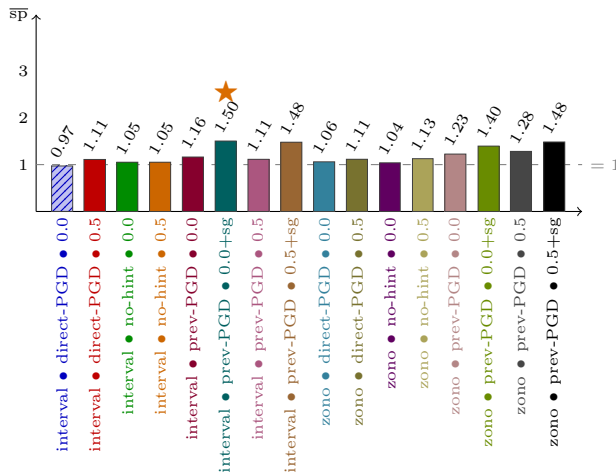
win ALL | pert **patch** (p-sizes merged: 1,14,14,3) | 16 combos, total $n = 638$ cells, $n_{max} = 62$.

Ranking (best → worst): ★ 87% (26f+0t/30)
 87% (25f+1t/30) 83% (24f+1t/30) 80% (27f+16t/54)
 78% (25f+17t/54) 78% (27f+15t/54)
 77% (23f+0t/30) 75% (27f+19t/61) 73% (22f+0t/30)
 72% (23f+15t/53) 71% (28f+16t/62)
 70% (21f+0t/30) 67% (20f+0t/30) 63% (19f+0t/30)
 63% (19f+0t/30) 63% (19f+0t/30)



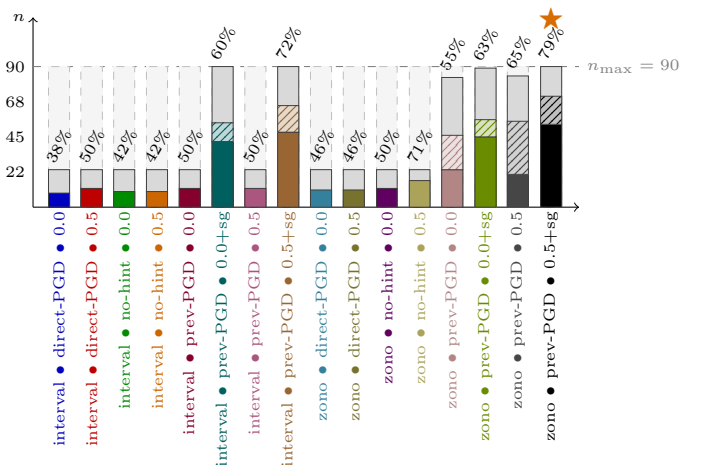
sp ALL | pert **occ** (p-sizes merged: 3,3,5,5,5,5) | 16 combos, total $n = 766$ cells.

Ranking (best → worst): ★ 1.50 (2/2) 1.48 (2/2)
 1.48 (2/2) 1.40 (2/2) 1.28 (2/2) 1.23 (2/2)
 1.16 (1/1) 1.13 (1/1) 1.11 (1/1) 1.11 (1/1)
 1.11 (1/1) 1.06 (1/1) 1.05 (1/1) 1.05 (1/1)
 1.04 (1/1) 0.97 (0/1)



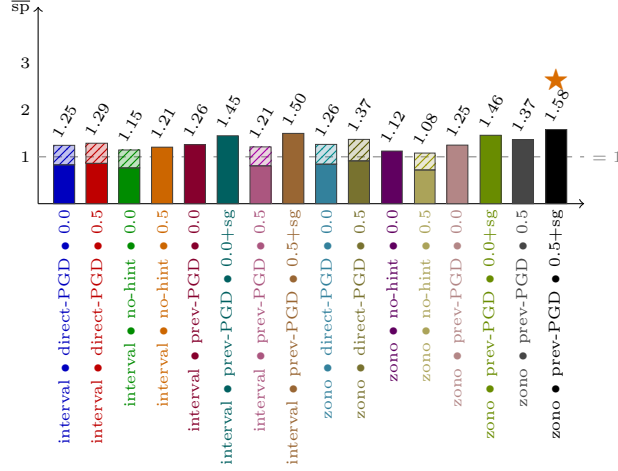
win ALL | pert **occ** (p-sizes merged: 3,3,5,5,5,5) | 16 combos, total $n = 766$ cells, $n_{max} = 90$.

Ranking (best → worst): ★ 79% (53f+18t/90)
 72% (48f+17t/90) 71% (17f+0t/24)
 65% (21f+34t/84) 63% (45f+11t/89)
 60% (42f+12t/90) 55% (24f+22t/83)
 50% (12f+0t/24) 50% (12f+0t/24) 50% (12f+0t/24)
 50% (12f+0t/24) 46% (11f+0t/24) 46% (11f+0t/24)
 42% (10f+0t/24) 42% (10f+0t/24) 38% (9f+0t/24)



sp ALL | pert translation (p.sizes merged: 1,1, 3,1, 3,3) | 16 combos, total $n = 1700$ cells.

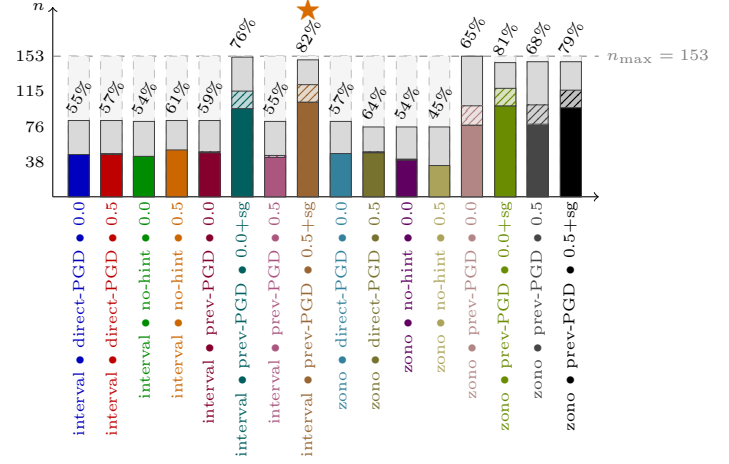
Ranking (best → worst): ★ 1.58 (3/3) 1.50 (3/3)
 1.46 (3/3) 1.45 (3/3) 1.37 (2/3) 1.37 (3/3)
 1.29 (2/3) 1.26 (2/3) 1.26 (3/3) 1.25 (3/3)
 1.25 (2/3) 1.21 (2/3) 1.21 (3/3) 1.15 (2/3)
 1.12 (3/3) 1.08 (2/3)



win ALL | pert translation (p.sizes merged: 1,1, 3,1, 3,3) | 16 combos, total $n = 1700$ cells, $n_{\max} = 153$.

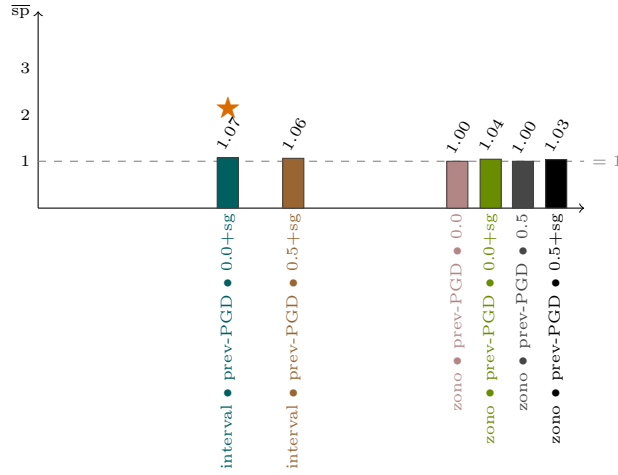
Ranking (best → worst): ★ 82% (103f+19t/149)

81% (99f+19t/146) 79% (97f+19t/147)
 76% (96f+19t/152) 68% (79f+21t/147)
 65% (78f+21t/153) 64% (48f+1t/76)
 61% (51f+0t/83) 59% (48f+1t/83) 57% (47f+0t/82)
 57% (46f+1t/83) 55% (46f+0t/83) 55% (43f+2t/82)
 54% (40f+1t/76) 54% (44f+0t/82) 45% (34f+0t/76)



sp ALL | pert linf (p.sizes merged: 0.1) | 6 combos, total $n = 228$ cells.

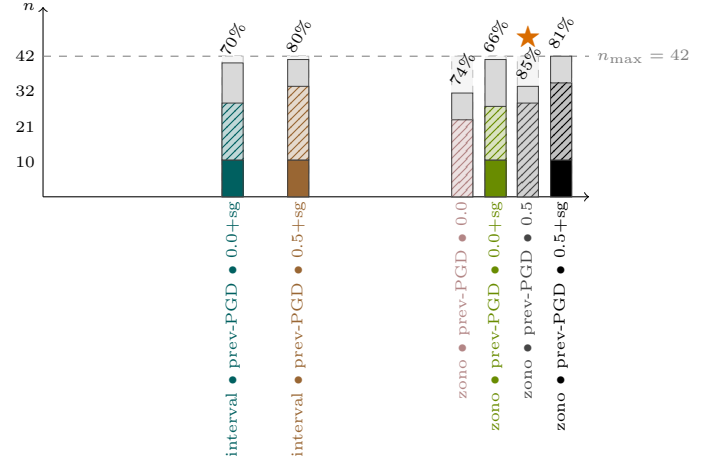
Ranking (best → worst): ★ 1.07 (1/1) 1.06 (1/1)
 1.04 (1/1) 1.03 (1/1) 1.00 (1/1) 1.00 (1/1)



win ALL | pert linf (p.sizes merged: 0.1) | 6 combos, total $n = 228$ cells, $n_{\max} = 42$.

Ranking (best → worst): ★ 85% (0f+28t/33)

81% (11f+23t/42) 80% (11f+22t/41)
 74% (0f+23t/31) 70% (11f+17t/40)
 66% (11f+16t/41)



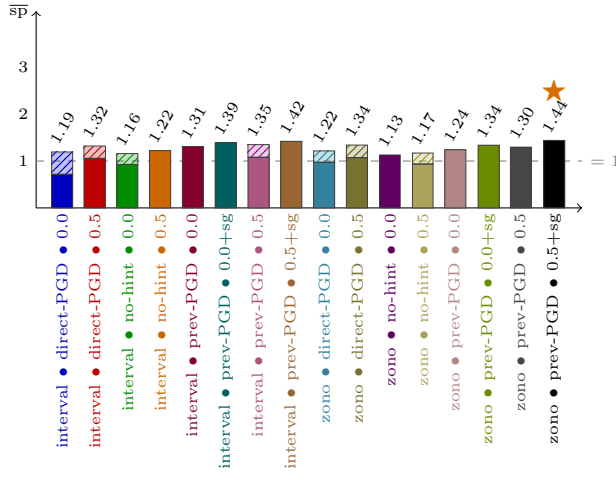
Overall histograms across all architectures and perturbations: $\overline{sp}$ paired with win-rate

This section pools every $(c_{src}, c_{tgt}, seed)$ cell across *all* architectures *and* *all* perturbation types into a single paired row — one bar per combo summarising every cell on record. Bar definitions are the same as in the sections above; both the architecture and the perturbation are shown as **ALL** in each chart’s caption.

sp ALL | pert ALL (p.sizes merged: 0.1, 1, 1, 1, 14, 14, 3, 3, 1, 3, 3, 3, 3, 5, 5, 5, 5) | 16 combos, total $n = 3332$ cells.

Ranking (best $\rightarrow$ worst): ★ 1.44 (7/7) 1.42 (7/7)

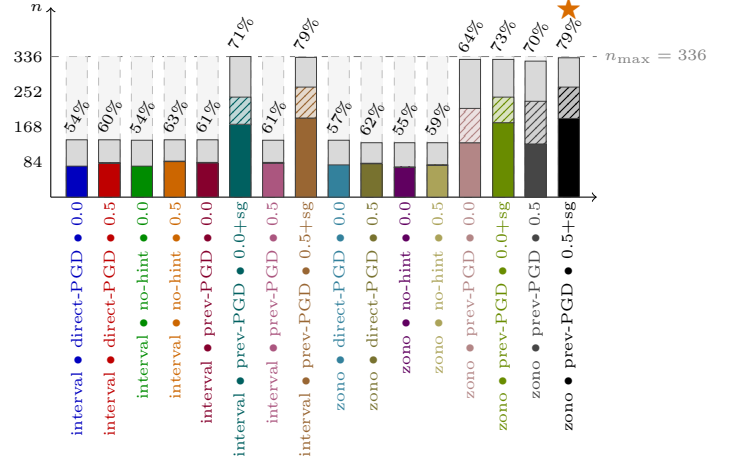
1.39 (7/7) 1.35 (4/5) 1.34 (4/5) 1.34 (7/7)
 1.32 (4/5) 1.31 (5/5) 1.30 (7/7) 1.24 (7/7)
 1.22 (5/5) 1.22 (4/5) 1.19 (3/5) 1.17 (4/5)
 1.16 (4/5) 1.13 (5/5)



win ALL | pert ALL (p.sizes merged: 0.1, 1, 1, 1, 14, 14, 3, 3, 1, 3, 3, 3, 3, 5, 5, 5, 5) | 16 combos, total $n = 3332$ cells, $n_{max} = 336$.

Ranking (best $\rightarrow$ worst): ★ 79% (188f+75t/333)

79% (189f+74t/334) 73% (178f+61t/329)
 71% (174f+65t/336) 70% (127f+102t/325)
 64% (130f+82t/329) 63% (85f+1t/137)
 62% (80f+1t/130) 61% (81f+2t/136)
 61% (82f+1t/137) 60% (81f+1t/137)
 59% (76f+1t/130) 57% (77f+0t/136)
 55% (71f+1t/130) 54% (74f+0t/137)



p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
patch	1,14,14,3	zono	no	0.5	0	3	164	96.17	51.01	1.885	1.524	0.04
					0	4	164	328.21	101.25	3.242		
					0	5	164	62.33	49.39	1.262		
					0	6	164	34.38	42.42	0.810		
					0	7	164	71.72	97.50	0.736		
					0	8	164	55.30	45.77	1.208		
interval	prev_pgd	0.5+sg	0	1	164	81.52	60.37	1.350	1.334	0.02		
			0	2	164	45.49	33.13	1.373				
			0	3	164	96.17	55.77	1.724				
			0	4	164	328.21	175.30	1.872				
			0	5	164	62.33	40.12	1.554				
			0	6	164	34.38	27.43	1.253				
			0	7	164	71.72	121.01	0.593				
			0	8	164	55.30	51.07	1.083				
zono	prev_pgd	0.0	0	9	164	65.33	54.39	1.201				
			0	1	0	81.52	49.94	1.632	1.293	0.02		
			0	2	0	45.49	27.91	1.630				
			0	3	0	96.17	41.72	2.305				
			0	4	0	328.21	276.30	1.188				
			0	5	0	62.33	47.36	1.316				
			0	6	0	34.38	45.06	0.763				
			0	7	0	71.72	77.54	0.925				
zono	prev_pgd	0.5+sg	0	8	0	55.30	58.76	0.941				
			0	9	0	65.33	70.02	0.933				
			0	1	164	81.52	65.97	1.236	1.292	0.02		
			0	2	164	45.49	36.07	1.261				
			0	3	164	96.17	54.79	1.755				
			0	4	164	328.21	182.41	1.799				
			0	5	164	62.33	39.50	1.578				
			0	6	164	34.38	28.45	1.208				
interval	prev_pgd	0.0+sg	0	7	164	71.72	125.19	0.573				
			0	8	164	55.30	50.03	1.105				
			0	9	164	65.33	58.68	1.113				
			0	1	0	81.52	44.35	1.838	1.264	0.03		
			0	2	0	45.49	29.94	1.519				
			0	3	0	96.17	44.34	2.169				
			0	4	0	328.21	323.13	1.016				
			0	5	0	62.33	56.11	1.111				
zono	prev_pgd	0.0+sg	0	6	0	34.38	49.68	0.692				
			0	7	0	71.72	72.22	0.993				
			0	8	0	55.30	49.69	1.113				
			0	9	0	65.33	70.60	0.925				
			0	1	0	81.52	43.94	1.855	1.252	0.03		
			0	2	0	45.49	38.96	1.168				
			0	3	0	96.17	41.34	2.326				
			0	4	0	328.21	258.18	1.271				
zono	prev_pgd	0.0+sg	0	5	0	62.33	57.72	1.080				
			0	6	0	34.38	54.04	0.636				
			0	7	0	71.72	77.27	0.928				
			0	8	0	55.30	51.10	1.082				
			0	9	0	65.33	71.03	0.920				

Table 5: Architecture **cnn1** — every recorded advstd combination at all seeds, all τ values, perturbation **patch**(1,14,14,3), $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 1/3, combos ranked 1–6 of 16 by mean **sp**) (6 combos, 51 cell rows; see Tables labelled **tab:safe-cnn1.*** for the other perturbations / c_{src} / τ -buckets of this arch). Rows are tinted by c_{src} (one distinct color per source class); **sp** > 1 (advstd faster) in **bold**. Combos sorted by descending mean **sp**; cells within a combo by (c_{src}, c_{tgt}) . Total 1519 cell rows across this architecture. Timeout-cell gaps: Table 95.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
patch	1,14,14,3	interval	no	0.5	0	3	164	96.17	51.79	1.857	1.208	0.04
					0	4	164	328.21	161.03	2.038		
					0	5	164	62.33	62.86	0.992		
					0	6	164	34.38	52.81	0.651		
					0	7	164	71.72	124.15	0.578		
					0	8	164	55.30	48.84	1.132		
		interval	prev_pgd	0.5	0	3	164	96.17	76.72	1.254	1.114	0.02
					0	4	164	328.21	205.69	1.596		
					0	5	164	62.33	40.95	1.522		
					0	6	164	34.38	27.32	1.258		
					0	7	164	71.72	116.56	0.615		
					0	8	164	55.30	126.11	0.439		
		zono	prev_pgd	0.5	0	1	164	81.52	71.88	1.134	1.104	0.02
					0	2	164	45.49	42.76	1.064		
					0	3	164	96.17	84.10	1.144		
					0	4	164	328.21	366.58	0.895		
					0	5	164	62.33	45.80	1.361		
					0	6	164	34.38	26.99	1.274		
					0	7	164	71.72	109.11	0.657		
					0	8	164	55.30	42.52	1.301		
					0	9	164	65.33	58.89	1.109		
		interval	direct_pgd	0.5	0	3	164	96.17	66.52	1.446	1.044	0.03
					0	4	164	328.21	301.39	1.089		
					0	5	164	62.33	49.17	1.268		
					0	6	164	34.38	43.55	0.789		
					0	7	164	71.72	87.56	0.819		
					0	8	164	55.30	64.92	0.852		
		interval	no	0.0	0	3	0	96.17	53.51	1.797	0.918	0.02
					0	4	0	328.21	482.18	0.681		
					0	5	0	62.33	603.28	0.103		
					0	6	0	34.38	56.96	0.604		
					0	7	0	71.72	47.44	1.512		
					0	8	0	55.30	67.96	0.814		
		interval	prev_pgd	0.0	0	3	0	96.17	56.12	1.714	0.897	0.03
					0	4	0	328.21	428.98	0.765		
					0	5	0	62.33	57.21	1.089		
					0	6	0	34.38	89.39	0.385		
					0	7	0	71.72	115.10	0.623		
					0	8	0	55.30	68.82	0.804		

Table 6: Architecture **cnn1** — perturbation **patch(1,14,14,3)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 2/3, combos ranked 7–12 of 16 by mean **sp**); continuation of Table 55 (6 combos, 39 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
patch	1,14,14,3	zono	direct_pgd	0.5	0	3	164	96.17	67.40	1.427	<i>0.811</i>	<i>0.03</i>
					0	4	164	328.21	374.86	0.876		
					0	5	164	62.33	59.24	1.052		
					0	6	164	34.38	54.43	0.632		
					0	7	164	71.72	205.70	0.349		
					0	8	164	55.30	103.94	0.532		
		zono	no	0.0	0	3	0	96.17	76.55	1.256	<i>0.692</i>	<i>0.02</i>
					0	4	0	328.21	525.50	0.625		
					0	5	0	62.33	744.70	0.084		
					0	6	0	34.38	68.74	0.500		
					0	7	0	71.72	94.25	0.761		
					0	8	0	55.30	59.63	0.927		
		zono	direct_pgd	0.0	0	3	0	96.17	1816.27	0.053	<i>0.564</i>	<i>0.51</i>
					0	4	0	328.21	579.79	0.566		
					0	5	0	62.33	54.46	1.145		
					0	6	0	34.38	84.39	0.407		
					0	7	0	71.72	98.45	0.728		
					0	8	0	55.30	113.47	0.487		
		interval	direct_pgd	0.0	0	3	0	96.17	1813.17	0.053	<i>0.533</i>	<i>0.51</i>
					0	4	0	328.21	636.96	0.515		
					0	5	0	62.33	74.07	0.842		
					0	6	0	34.38	62.84	0.547		
					0	7	0	71.72	88.39	0.811		
					0	8	0	55.30	129.35	0.428		

Table 7: Architecture **cnn1** — perturbation **patch(1,14,14,3)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 3/3, combos ranked 13–16 of 16 by mean **sp**); continuation of Table 55 (4 combos, 24 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_src}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_src}$
patch	1,14,14,3	interval	prev_pgd	0.5+sg	1	0	164	56.34	63.35	0.889	2.435	0.02
					1	2	164	54.24	54.18	1.001		
					1	3	164	94.60	22.92	4.127		
					1	4	164	113.19	67.25	1.683		
					1	5	164	138.25	40.35	3.426		
					1	6	164	206.61	140.19	1.474		
					1	7	164	83.24	35.05	2.375		
					1	8	164	67.32	20.62	3.265		
		interval	prev_pgd	0.5	1	9	164	102.50	27.92	3.671	2.280	0.02
					1	3	164	94.60	22.74	4.160		
					1	4	164	113.19	73.66	1.537		
					1	5	164	138.25	70.21	1.969		
					1	6	164	206.61	133.40	1.549		
					1	7	164	83.24	37.29	2.232		
					1	8	164	67.32	30.14	2.234		
		zono	prev_pgd	0.5+sg	1	0	164	56.34	46.50	1.212	2.244	0.02
					1	2	164	54.24	36.01	1.506		
					1	3	164	94.60	26.15	3.618		
					1	4	164	113.19	49.23	2.299		
					1	5	164	138.25	67.20	2.057		
					1	6	164	206.61	122.51	1.686		
					1	7	164	83.24	33.22	2.506		
					1	8	164	67.32	44.88	1.500		
		interval	no	0.5	1	9	164	102.50	26.89	3.812	1.911	0.01
					1	3	164	94.60	30.30	3.122		
					1	4	164	113.19	84.52	1.339		
					1	5	164	138.25	76.75	1.801		
					1	6	164	206.61	115.30	1.792		
					1	7	164	83.24	49.21	1.692		
					1	8	164	67.32	39.18	1.718		
		interval	prev_pgd	0.0	1	3	0	94.60	60.23	1.571	1.711	0.02
					1	4	0	113.19	67.97	1.665		
					1	5	0	138.25	61.73	2.240		
					1	6	0	206.61	115.20	1.793		
					1	7	0	83.24	44.04	1.890		
					1	8	0	67.32	60.71	1.109		
		zono	no	0.5	1	3	164	94.60	51.75	1.828	1.646	0.02
					1	4	164	113.19	69.92	1.619		
					1	5	164	138.25	68.04	2.032		
					1	6	164	206.61	149.88	1.379		
					1	7	164	83.24	52.87	1.574		
					1	8	164	67.32	46.60	1.445		

Table 8: Architecture **cnn1** — perturbation **patch**(1,14,14,3), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5\}$ (part 1/3, combos ranked 1–6 of 16 by mean sp); continuation of Table 55 (6 combos, 42 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
patch	1,14,14,3	interval	no	0.0	1	3	0	94.60	71.27	1.327	1.612	0.03
					1	4	0	113.19	56.19	2.014		
					1	5	0	138.25	78.56	1.760		
					1	6	0	206.61	105.37	1.961		
					1	7	0	83.24	66.38	1.254		
					1	8	0	67.32	49.69	1.355		
		interval	prev_pgd	0.0+sg	1	0	0	56.34	39.82	1.415	1.555	0.02
					1	2	0	54.24	127.86	0.424		
					1	3	0	94.60	24.81	3.813		
					1	4	0	113.19	69.83	1.621		
					1	5	0	138.25	82.29	1.680		
					1	6	0	206.61	138.34	1.493		
					1	7	0	83.24	100.81	0.826		
					1	8	0	67.32	43.19	1.559		
					1	9	0	102.50	88.31	1.161		
		interval	direct_pgd	0.5	1	3	164	94.60	33.34	2.837	1.542	0.54
					1	4	164	113.19	1813.29	0.062		
					1	5	164	138.25	78.59	1.759		
					1	6	164	206.61	141.01	1.465		
					1	7	164	83.24	50.22	1.658		
					1	8	164	67.32	45.81	1.470		
		zono	direct_pgd	0.5	1	3	164	94.60	38.58	2.452	1.517	0.52
					1	4	164	113.19	1815.95	0.062		
					1	5	164	138.25	75.23	1.838		
					1	6	164	206.61	137.68	1.501		
					1	7	164	83.24	47.68	1.746		
					1	8	164	67.32	44.70	1.506		
		zono	no	0.0	1	3	0	94.60	60.39	1.566	1.471	0.01
					1	4	0	113.19	169.45	0.668		
					1	5	0	138.25	104.89	1.318		
					1	6	0	206.61	116.67	1.771		
					1	7	0	83.24	59.83	1.391		
					1	8	0	67.32	31.85	2.114		
		zono	prev_pgd	0.0	1	0	0	56.34	60.50	0.931	1.448	0.03
					1	2	0	54.24	96.94	0.560		
					1	3	0	94.60	54.97	1.721		
					1	4	0	113.19	55.31	2.046		
					1	5	0	138.25	39.19	3.528		
					1	6	0	206.61	203.15	1.017		
					1	7	0	83.24	128.46	0.648		
					1	8	0	67.32	52.12	1.292		
					1	9	0	102.50	79.60	1.288		

Table 9: Architecture **cnn1** — perturbation **patch(1,14,14,3)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$ (part 2/3, combos ranked 7–12 of 16 by mean sp); continuation of Table 55 (6 combos, 42 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
patch	1,14,14,3	zono	prev_pgd	0.5	1	0	164	56.34	63.16	0.892	1.392	<i>0.32</i>
					1	2	164	54.24	85.27	0.636		
					1	3	164	94.60	24.94	3.793		
					1	4	164	113.19	66.56	1.701		
					1	5	164	138.25	136.87	1.010		
					1	6	164	206.61	197.30	1.047		
					1	7	164	83.24	46.86	1.776		
					1	8	164	67.32	40.90	1.646		
					1	9	164	102.50	3614.35	0.028		
		interval	direct_pgd	0.0	1	3	0	94.60	50.45	1.875	1.377	<i>0.01</i>
					1	4	0	113.19	73.22	1.546		
					1	5	0	138.25	74.33	1.860		
					1	6	0	206.61	145.70	1.418		
					1	7	0	83.24	125.07	0.666		
					1	8	0	67.32	74.84	0.900		
		zono	prev_pgd	0.0+sg	1	0	0	56.34	41.71	1.351	1.356	<i>0.02</i>
					1	2	0	54.24	66.38	0.817		
					1	3	0	94.60	49.35	1.917		
					1	4	0	113.19	45.67	2.478		
					1	5	0	138.25	48.83	2.831		
					1	6	0	206.61	2654.58	0.078		
					1	7	0	83.24	189.58	0.439		
					1	8	0	67.32	60.46	1.113		
					1	9	0	102.50	87.14	1.176		
		zono	direct_pgd	0.0	1	3	0	94.60	46.31	2.043	1.344	<i>0.01</i>
					1	4	0	113.19	83.31	1.359		
					1	5	0	138.25	77.27	1.789		
					1	6	0	206.61	154.97	1.333		
					1	7	0	83.24	199.19	0.418		
					1	8	0	67.32	60.07	1.121		

Table 10: Architecture **cnn1** — perturbation **patch(1,14,14,3)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$ (part 3/3, combos ranked 13–16 of 16 by mean sp); continuation of Table 55 (4 combos, 30 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_{\text{src}}}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_{\text{src}}}$
patch	1,14,14,3	interval	prev_pgd	0.5	2	3	164	129.59	63.16	2.052	3.546	1.24
					2	4	164	130.79	177.97	0.735		
					2	5	164	161.86	151.47	1.069		
					2	6	164	1819.45	160.26	11.353		
					2	7	164	369.96	71.68	5.161		
					2	8	164	88.80	98.11	0.905		
		zono	prev_pgd	0.0	2	3	0	129.59	36.84	3.518	3.069	1.24
					2	4	0	130.79	154.25	0.848		
					2	5	0	161.86	98.93	1.636		
					2	6	0	1819.45	272.30	6.682		
					2	7	0	369.96	88.85	4.164		
					2	8	0	88.80	56.78	1.564		
		zono	prev_pgd	0.5	2	3	164	129.59	72.80	1.780	2.942	1.24
					2	4	164	130.79	222.25	0.588		
					2	5	164	161.86	175.97	0.920		
					2	6	164	1819.45	202.85	8.969		
					2	7	164	369.96	80.70	4.584		
					2	8	164	88.80	109.96	0.808		
		interval	prev_pgd	0.0	2	3	0	129.59	38.87	3.334	2.778	1.24
					2	4	0	130.79	169.63	0.771		
					2	5	0	161.86	111.27	1.455		
					2	6	0	1819.45	298.28	6.100		
					2	7	0	369.96	96.60	3.830		
					2	8	0	88.80	75.15	1.182		
		interval	direct_pgd	0.5	2	3	164	129.59	41.18	3.147	2.739	1.23
					2	4	164	130.79	158.30	0.826		
					2	5	164	161.86	90.58	1.787		
					2	6	164	1819.45	242.11	7.515		
					2	7	164	369.96	151.03	2.450		
					2	8	164	88.80	125.26	0.709		
		zono	direct_pgd	0.5	2	3	164	129.59	50.01	2.591	2.333	1.23
					2	4	164	130.79	170.01	0.769		
					2	5	164	161.86	99.61	1.625		
					2	6	164	1819.45	299.99	6.065		
					2	7	164	369.96	166.57	2.221		
					2	8	164	88.80	122.44	0.725		

Table 11: Architecture **cnn1** — perturbation **patch(1,14,14,3)**, $c_{\text{src}} = 2$, $\tau \in \{0.0, 0.5\}$ (part 1/2, combos ranked 1–6 of 12 by mean sp); continuation of Table 55 (6 combos, 36 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_{\text{src}}}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_{\text{src}}}$
patch	1,14,14,3	interval	direct_pgd	0.0	2	3	0	129.59	62.57	2.071	1.671	3.24
					2	4	0	130.79	102.33	1.278		
					2	5	0	161.86	1814.67	0.089		
					2	6	0	1819.45	350.27	5.194		
					2	7	0	369.96	1814.71	0.204		
					2	8	0	88.80	74.67	1.189		
					2	3	0	129.59	70.73	1.832	1.631	3.24
					2	4	0	130.79	103.01	1.270		
					2	5	0	161.86	1815.61	0.089		
					2	6	0	1819.45	346.93	5.244		
		zono	direct_pgd	0.0	2	7	0	369.96	1814.23	0.204		
					2	8	0	88.80	77.52	1.146		
					2	3	164	129.59	59.74	2.169	1.526	0.48
					2	4	164	130.79	197.14	0.663		
					2	5	164	161.86	84.70	1.911		
		zono	no	0.5	2	6	164	1819.45	1814.59	1.003		
					2	7	164	369.96	230.05	1.608		
					2	8	164	88.80	49.23	1.804		
					2	3	0	129.59	129.99	0.997	1.461	1.74
					2	4	0	130.79	181.54	0.720		
		interval	no	0.0	2	5	0	161.86	227.31	0.712		
					2	6	0	1819.45	361.37	5.035		
					2	7	0	369.96	1813.99	0.204		
					2	8	0	88.80	80.88	1.098		
		zono	no	0.0	2	3	0	129.59	133.12	0.973	1.449	1.74
					2	4	0	130.79	130.90	0.999		
					2	5	0	161.86	242.77	0.667		
					2	6	0	1819.45	397.00	4.583		
					2	7	0	369.96	1812.87	0.204		
		interval	no	0.5	2	8	0	88.80	70.19	1.265	1.404	0.49
					2	3	164	129.59	57.71	2.246		
					2	4	164	130.79	182.58	0.716		
					2	5	164	161.86	102.32	1.582		
					2	6	164	1819.45	1815.11	1.002		
					2	7	164	369.96	245.11	1.509		
					2	8	164	88.80	64.85	1.369		

Table 12: Architecture **cnn1** — perturbation **patch(1,14,14,3)**, $c_{\text{src}} = 2$, $\tau \in \{0.0, 0.5\}$ (part 2/2, combos ranked 7–12 of 12 by mean **sp**); continuation of Table 55 (6 combos, 36 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
occ	3,3,5	interval	prev_pgd	0.5+sg	0	1	165	261.44	63.88	4.093	1.852	0.67
					0	2	165	79.60	38.97	2.043		
					0	3	165	191.10	74.61	2.561		
					0	4	165	367.16	325.20	1.129		
					0	5	165	134.16	65.05	2.062		
					0	6	165	39.08	39.45	0.991		
					0	7	165	107.68	3613.84	0.030		
					0	8	165	41.26	32.01	1.289		
					0	9	165	144.97	58.64	2.472		
	zono	prev_pgd	0.5+sg	0	1	166	261.44	64.62	4.046	1.668	0.04	
				0	2	166	79.60	36.38	2.188			
				0	3	166	191.10	95.18	2.008			
				0	4	166	367.16	238.30	1.541			
				0	5	166	134.16	107.92	1.243			
				0	6	166	39.08	46.32	0.844			
				0	7	166	107.68	143.30	0.751			
				0	8	166	41.26	52.50	0.786			
				0	9	166	144.97	90.12	1.609			
	interval	prev_pgd	0.0+sg	0	1	0	261.44	105.72	2.473	1.596	0.34	
				0	2	0	79.60	50.13	1.588			
				0	3	0	191.10	130.53	1.464			
				0	4	0	367.16	3616.55	0.102			
				0	5	0	134.16	43.77	3.065			
				0	6	0	39.08	46.23	0.845			
				0	7	0	107.68	51.24	2.101			
				0	8	0	41.26	43.97	0.938			
				0	9	0	144.97	81.13	1.787			
	zono	prev_pgd	0.5	0	1	166	261.44	72.79	3.592	1.551	0.04	
				0	2	166	79.60	52.79	1.508			
				0	3	166	191.10	119.00	1.606			
				0	4	166	367.16	743.95	0.494			
				0	5	166	134.16	235.66	0.569			
				0	6	166	39.08	75.39	0.518			
				0	7	166	107.68	75.40	1.428			
				0	8	166	41.26	41.56	0.993			
				0	9	166	144.97	44.62	3.249			
	zono	prev_pgd	0.0+sg	0	1	0	261.44	155.79	1.678	1.339	0.34	
				0	2	0	79.60	59.14	1.346			
				0	3	0	191.10	96.70	1.976			
				0	4	0	367.16	3612.90	0.102			
				0	5	0	134.16	62.84	2.135			
				0	6	0	39.08	57.10	0.684			
				0	7	0	107.68	63.85	1.686			
				0	8	0	41.26	48.21	0.856			
				0	9	0	144.97	91.15	1.590			
	zono	prev_pgd	0.0	0	1	0	261.44	151.44	1.726	1.221	0.34	
				0	2	0	79.60	56.27	1.415			
				0	3	0	191.10	155.69	1.227			
				0	4	0	367.16	3620.51	0.101			
				0	5	0	134.16	61.40	2.185			
				0	6	0	39.08	64.13	0.609			
				0	7	0	107.68	70.13	1.535			
				0	8	0	41.26	54.40	0.758			
				0	9	0	144.97	101.29	1.431			

Table 13: Architecture **cnn1** — perturbation **occ(3,3,5)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$; continuation of Table 55 (6 combos, 54 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
occ	3,3,5	interval	prev_pgd	0.0+sg	1	0	0	304.94	3614.25	0.084	4.413	0.37
					1	2	0	63.44	58.69	1.081		
					1	3	0	47.05	31.68	1.485		
					1	4	0	51.14	52.24	0.979		
					1	5	0	26.16	60.78	0.430		
					1	6	0	1811.91	57.75	31.375		
					1	7	0	100.71	42.90	2.348		
					1	8	0	40.61	106.44	0.382		
					1	9	0	326.37	210.75	1.549		
		interval	prev_pgd	0.5+sg	1	0	165	304.94	103.18	2.955	4.138	0.33
					1	2	165	63.44	28.25	2.246		
					1	3	165	47.05	47.47	0.991		
					1	4	165	51.14	45.50	1.124		
					1	5	165	26.16	40.24	0.650		
					1	6	165	1811.91	74.36	24.367		
					1	7	165	100.71	33.10	3.043		
					1	8	165	40.61	90.81	0.447		
		zono	prev_pgd	0.5+sg	1	0	166	304.94	85.45	3.569	3.597	0.31
					1	2	166	63.44	36.77	1.725		
					1	3	166	47.05	33.16	1.419		
					1	4	166	51.14	37.84	1.351		
					1	5	166	26.16	35.06	0.746		
					1	6	166	1811.91	108.84	16.647		
					1	7	166	100.71	25.48	3.953		
					1	8	166	40.61	46.48	0.874		
		zono	prev_pgd	0.0+sg	1	0	0	304.94	3612.14	0.084	3.459	0.38
					1	2	0	63.44	34.03	1.864		
					1	3	0	47.05	68.37	0.688		
					1	4	0	51.14	49.67	1.030		
					1	5	0	26.16	44.96	0.582		
					1	6	0	1811.91	81.06	22.353		
					1	7	0	100.71	43.67	2.306		
					1	8	0	40.61	59.37	0.684		
		zono	prev_pgd	0.5	1	0	166	304.94	138.13	2.208	3.053	0.31
					1	2	166	63.44	58.11	1.092		
					1	3	166	47.05	67.89	0.693		
					1	4	166	51.14	86.78	0.589		
					1	5	166	26.16	41.73	0.627		
					1	6	166	1811.91	87.94	20.604		
					1	7	166	100.71	128.27	0.785		
					1	8	166	40.61	301.23	0.135		
		zono	prev_pgd	0.0	1	0	0	304.94	3617.31	0.084	2.617	0.38
					1	2	0	63.44	105.93	0.599		
					1	3	0	47.05	109.62	0.429		
					1	4	0	51.14	93.99	0.544		
					1	5	0	26.16	80.93	0.323		
					1	6	0	1811.91	94.02	19.272		
					1	7	0	100.71	66.35	1.518		
					1	8	0	40.61	214.68	0.189		
					1	9	0	326.37	546.56	0.597		

Table 14: Architecture **cnn1** — perturbation **occ(3,3,5)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$; continuation of Table 55 (6 combos, 54 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
occ	5,5,5	zono	prev_pgd	0.0+sg	0	1	0	59.49	25.55	2.328	1.424	0.03
					0	2	0	58.98	41.38	1.425		
					0	3	0	42.18	52.01	0.811		
					0	4	0	536.01	293.18	1.828		
					0	5	0	110.91	57.99	1.913		
					0	6	0	33.37	35.27	0.946		
					0	7	0	56.20	43.84	1.282		
					0	8	0	57.73	44.89	1.286		
					0	9	0	89.23	89.33	0.999		
		interval	direct_pgd	0.5	0	3	173	42.18	34.82	1.211	1.399	0.02
					0	4	173	536.01	335.80	1.596		
					0	5	173	110.91	48.63	2.281		
					0	6	173	33.37	41.37	0.807		
					0	7	173	56.20	48.60	1.156		
					0	8	173	57.73	42.98	1.343		
		zono	direct_pgd	0.5	0	3	173	42.18	34.73	1.215	1.381	0.02
					0	4	173	536.01	375.96	1.426		
					0	5	173	110.91	46.90	2.365		
					0	6	173	33.37	40.99	0.814		
					0	7	173	56.20	49.72	1.130		
					0	8	173	57.73	43.15	1.338		
		interval	prev_pgd	0.0	0	3	0	42.18	75.61	0.558	1.348	0.03
					0	4	0	536.01	323.39	1.657		
					0	5	0	110.91	56.61	1.959		
					0	6	0	33.37	35.94	0.928		
					0	7	0	56.20	36.94	1.521		
					0	8	0	57.73	39.51	1.461		
		zono	direct_pgd	0.0	0	3	0	42.18	41.00	1.029	1.324	0.01
					0	4	0	536.01	332.14	1.614		
					0	5	0	110.91	51.62	2.149		
					0	6	0	33.37	51.67	0.646		
					0	7	0	56.20	58.31	0.964		
					0	8	0	57.73	37.46	1.541		
		zono	prev_pgd	0.0	0	1	0	59.49	31.37	1.896	1.323	0.03
					0	2	0	58.98	49.22	1.198		
					0	3	0	42.18	53.85	0.783		
					0	4	0	536.01	325.55	1.646		
					0	5	0	110.91	57.06	1.944		
					0	6	0	33.37	34.23	0.975		
					0	7	0	56.20	44.09	1.275		
					0	8	0	57.73	38.37	1.505		
					0	9	0	89.23	131.01	0.681		

Table 15: Architecture **cnn1** — perturbation **occ(5,5,5)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 1/3, combos ranked 1–6 of 16 by mean sp); continuation of Table 55 (6 combos, 42 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
occ	5,5,5	interval	prev_pgd	0.5	0	3	173	42.18	28.19	1.496	1.296	0.03
					0	4	173	536.01	460.90	1.163		
					0	5	173	110.91	72.10	1.538		
					0	6	173	33.37	37.94	0.880		
					0	7	173	56.20	30.40	1.849		
					0	8	173	57.73	67.72	0.852		
		interval	no	0.5	0	3	173	42.18	38.63	1.092	1.188	0.03
					0	4	173	536.01	315.44	1.699		
					0	5	173	110.91	53.53	2.072		
					0	6	173	33.37	65.73	0.508		
					0	7	173	56.20	92.77	0.606		
					0	8	173	57.73	50.07	1.153		
		interval	no	0.0	0	3	0	42.18	65.05	0.648	1.178	0.02
					0	4	0	536.01	348.88	1.536		
					0	5	0	110.91	47.94	2.314		
					0	6	0	33.37	44.16	0.756		
					0	7	0	56.20	62.06	0.906		
					0	8	0	57.73	63.37	0.911		
		zono	prev_pgd	0.5+sg	0	1	173	59.49	36.92	1.611	1.174	0.32
					0	2	173	58.98	36.98	1.595		
					0	3	173	42.18	51.03	0.827		
					0	4	173	536.01	3611.39	0.148		
					0	5	173	110.91	72.11	1.538		
					0	6	173	33.37	38.41	0.869		
					0	7	173	56.20	3122.22	0.018		
					0	8	173	57.73	37.23	1.551		
					0	9	173	89.23	37.06	2.408		
		zono	no	0.5	0	3	173	42.18	39.09	1.079	1.145	0.03
					0	4	173	536.01	341.61	1.569		
					0	5	173	110.91	52.15	2.127		
					0	6	173	33.37	67.59	0.494		
					0	7	173	56.20	100.10	0.561		
					0	8	173	57.73	55.67	1.037		
		interval	direct_pgd	0.0	0	3	0	42.18	64.93	0.650	1.142	0.01
					0	4	0	536.01	338.13	1.585		
					0	5	0	110.91	50.78	2.184		
					0	6	0	33.37	63.26	0.528		
					0	7	0	56.20	64.03	0.878		
					0	8	0	57.73	56.18	1.028		

Table 16: Architecture **cnn1** — perturbation **occ**(5,5,5), $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 2/3, combos ranked 7–12 of 16 by mean **sp**); continuation of Table 55 (6 combos, 39 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_{\text{src}}}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_{\text{src}}}$
occ	5,5,5	zono	prev_pgd	0.5	0	1	173	59.49	74.31	0.801	1.079	0.03
					0	2	173	58.98	111.33	0.530		
					0	3	173	42.18	31.56	1.337		
					0	4	173	536.01	461.06	1.163		
					0	5	173	110.91	70.63	1.570		
					0	6	173	33.37	38.78	0.860		
					0	7	173	56.20	33.16	1.695		
					0	8	173	57.73	62.44	0.925		
					0	9	173	89.23	107.49	0.830		
		interval	prev_pgd	0.5+sg	0	1	173	59.49	35.59	1.672	1.050	0.32
					0	2	173	58.98	39.26	1.502		
					0	3	173	42.18	49.97	0.844		
					0	4	173	536.01	3612.38	0.148		
					0	5	173	110.91	84.64	1.310		
					0	6	173	33.37	41.88	0.797		
					0	7	173	56.20	3186.88	0.018		
					0	8	173	57.73	39.65	1.456		
					0	9	173	89.23	52.47	1.701		
		zono	no	0.0	0	3	0	42.18	122.06	0.346	0.926	0.02
					0	4	0	536.01	444.89	1.205		
					0	5	0	110.91	64.69	1.714		
					0	6	0	33.37	54.35	0.614		
					0	7	0	56.20	67.59	0.831		
					0	8	0	57.73	68.23	0.846		
		interval	prev_pgd	0.0+sg	0	1	0	59.49	43.91	1.355	0.898	0.03
					0	2	0	58.98	69.88	0.844		
					0	3	0	42.18	77.33	0.545		
					0	4	0	536.01	541.92	0.989		
					0	5	0	110.91	88.56	1.252		
					0	6	0	33.37	49.63	0.672		
					0	7	0	56.20	101.98	0.551		
					0	8	0	57.73	54.47	1.060		
					0	9	0	89.23	109.47	0.815		

Table 17: Architecture **cnn1** — perturbation **occ(5,5,5)**, $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5\}$ (part 3/3, combos ranked 13–16 of 16 by mean **sp**); continuation of Table 55 (4 combos, 33 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_src}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_src}$
occ	5,5,5	zono	prev_pgd	0.5+sg	1	0	173	67.88	61.40	1.106	1.359	0.04
					1	2	173	55.31	52.45	1.055		
					1	3	173	73.18	28.98	2.525		
					1	4	173	35.85	39.42	0.909		
					1	5	173	35.61	22.92	1.554		
					1	6	173	41.49	29.14	1.424		
					1	7	173	38.38	31.18	1.231		
					1	8	173	40.33	39.20	1.029		
		interval	prev_pgd	0.0	1	9	173	379.24	271.09	1.399		
					1	3	0	73.18	24.81	2.950	1.348	0.03
					1	4	0	35.85	39.50	0.908		
					1	5	0	35.61	50.26	0.709		
					1	6	0	41.49	38.01	1.092		
					1	7	0	38.38	40.74	0.942		
					1	8	0	40.33	27.05	1.491		
		interval	prev_pgd	0.0+sg	1	0	0	67.88	100.45	0.676	1.307	0.03
					1	2	0	55.31	59.25	0.934		
					1	3	0	73.18	31.63	2.314		
					1	4	0	35.85	32.49	1.103		
					1	5	0	35.61	21.60	1.649		
					1	6	0	41.49	23.79	1.744		
					1	7	0	38.38	88.80	0.432		
					1	8	0	40.33	39.80	1.013		
		interval	prev_pgd	0.5	1	9	0	379.24	199.82	1.898		
					1	3	173	73.18	32.71	2.237	1.218	0.04
					1	4	173	35.85	40.76	0.880		
					1	5	173	35.61	27.73	1.284		
					1	6	173	41.49	39.45	1.052		
					1	7	173	38.38	55.80	0.688		
					1	8	173	40.33	34.62	1.165		
		zono	direct_pgd	0.5	1	3	173	73.18	32.79	2.232	1.193	0.03
					1	4	173	35.85	42.36	0.846		
					1	5	173	35.61	31.22	1.141		
					1	6	173	41.49	43.46	0.955		
					1	7	173	38.38	34.96	1.098		
					1	8	173	40.33	45.53	0.886		
		zono	prev_pgd	0.0+sg	1	0	0	67.88	133.40	0.509	1.166	0.02
					1	2	0	55.31	66.19	0.836		
					1	3	0	73.18	50.82	1.440		
					1	4	0	35.85	50.58	0.709		
					1	5	0	35.61	37.22	0.957		
					1	6	0	41.49	24.40	1.700		
					1	7	0	38.38	33.07	1.161		
					1	8	0	40.33	21.97	1.836		
					1	9	0	379.24	281.06	1.349		

Table 18: Architecture **cnn1** — perturbation **occ(5,5,5)**, $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5\}$ (part 1/3, combos ranked 1–6 of 16 by mean **sp**); continuation of Table 55 (6 combos, 45 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
occ	5,5,5	interval	direct_pgd	0.5	1	3	173	73.18	33.69	2.172	1.118	0.02
					1	4	173	35.85	43.22	0.829		
					1	5	173	35.61	40.50	0.879		
					1	6	173	41.49	55.21	0.751		
					1	7	173	38.38	31.14	1.232		
					1	8	173	40.33	47.95	0.841		
		zono	no	0.0	1	3	0	73.18	40.60	1.802	1.106	0.01
					1	4	0	35.85	49.65	0.722		
					1	5	0	35.61	37.87	0.940		
					1	6	0	41.49	40.64	1.021		
					1	7	0	38.38	39.79	0.965		
					1	8	0	40.33	34.03	1.185		
		interval	prev_pgd	0.5+sg	1	0	173	67.88	49.92	1.360	1.086	0.04
					1	2	173	55.31	48.07	1.151		
					1	3	173	73.18	39.52	1.852		
					1	4	173	35.85	54.25	0.661		
					1	5	173	35.61	42.45	0.839		
					1	6	173	41.49	50.80	0.817		
					1	7	173	38.38	43.01	0.892		
					1	8	173	40.33	39.92	1.010		
					1	9	173	379.24	316.99	1.196		
		zono	direct_pgd	0.0	1	3	0	73.18	44.11	1.659	<i>0.992</i>	0.01
					1	4	0	35.85	41.46	0.865		
					1	5	0	35.61	40.23	0.885		
					1	6	0	41.49	54.65	0.759		
					1	7	0	38.38	49.07	0.782		
					1	8	0	40.33	40.24	1.002		
		interval	no	0.5	1	3	173	73.18	49.58	1.476	<i>0.990</i>	0.01
					1	4	173	35.85	37.47	0.957		
					1	5	173	35.61	34.20	1.041		
					1	6	173	41.49	58.00	0.715		
					1	7	173	38.38	41.41	0.927		
					1	8	173	40.33	48.83	0.826		
		interval	no	0.0	1	3	0	73.18	43.24	1.692	<i>0.990</i>	0.01
					1	4	0	35.85	43.11	0.832		
					1	5	0	35.61	48.13	0.740		
					1	6	0	41.49	41.09	1.010		
					1	7	0	38.38	48.75	0.787		
					1	8	0	40.33	45.90	0.879		

Table 19: Architecture **cnn1** — perturbation **occ**(5,5,5), $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$ (part 2/3, combos ranked 7–12 of 16 by mean **sp**); continuation of Table 55 (6 combos, 39 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_{\text{src}}}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_{\text{src}}}$
occ	5,5,5	zono	no	0.5	1	3	173	73.18	49.39	1.482	0.988	0.01
					1	4	173	35.85	60.99	0.588		
					1	5	173	35.61	34.24	1.040		
					1	6	173	41.49	41.05	1.011		
					1	7	173	38.38	41.95	0.915		
					1	8	173	40.33	45.21	0.892		
		zono	prev_pgd	0.5	1	0	173	67.88	102.58	0.662	0.971	0.03
					1	2	173	55.31	84.51	0.654		
					1	3	173	73.18	59.29	1.234		
					1	4	173	35.85	49.17	0.729		
					1	5	173	35.61	26.54	1.342		
					1	6	173	41.49	43.22	0.960		
					1	7	173	38.38	33.92	1.131		
					1	8	173	40.33	40.37	0.999		
					1	9	173	379.24	370.71	1.023		
		zono	prev_pgd	0.0	1	0	0	67.88	159.02	0.427	0.963	0.03
					1	2	0	55.31	88.21	0.627		
					1	3	0	73.18	43.33	1.689		
					1	4	0	35.85	55.05	0.651		
					1	5	0	35.61	35.74	0.996		
					1	6	0	41.49	40.13	1.034		
					1	7	0	38.38	38.28	1.003		
					1	8	0	40.33	36.13	1.116		
					1	9	0	379.24	337.38	1.124		
		interval	direct_pgd	0.0	1	3	0	73.18	51.82	1.412	0.733	0.02
					1	4	0	35.85	60.74	0.590		
					1	5	0	35.61	924.61	0.039		
					1	6	0	41.49	42.70	0.972		
					1	7	0	38.38	48.51	0.791		
					1	8	0	40.33	68.17	0.592		

Table 20: Architecture **cnn1** — perturbation **occ(5,5,5)**, $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5\}$ (part 3/3, combos ranked 13–16 of 16 by mean sp); continuation of Table 55 (4 combos, 30 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_{\text{src}}}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_{\text{src}}}$
translation	1,1	zono	direct_pgd	0.0	0	3	0	344.88	288.50	1.195	2.171	0.03
					0	4	0	442.26	259.88	1.702		
					0	5	0	393.83	96.14	4.096		
					0	6	0	526.31	217.26	2.422		
					0	7	0	568.70	263.45	2.159		
					0	8	0	1392.84	959.93	1.451		
					0	1	172	1043.60	243.13	4.292		
					0	2	172	755.45	136.38	5.539		
					0	3	172	344.88	279.62	1.233		
					0	4	172	442.26	396.50	1.115		
		zono	prev_pgd	0.5+sg	0	5	172	393.83	478.19	0.824		
					0	6	172	526.31	824.30	0.638		
					0	7	172	568.70	440.72	1.290		
					0	8	172	1392.84	572.91	2.431		
					0	9	172	3623.14	3611.83	1.003		
		interval	direct_pgd	0.0	0	3	0	344.88	253.52	1.360	2.021	0.03
					0	4	0	442.26	344.95	1.282		
					0	5	0	393.83	113.37	3.474		
					0	6	0	526.31	198.91	2.646		
					0	7	0	568.70	278.84	2.040		
					0	8	0	1392.84	1049.74	1.327		
		zono	direct_pgd	0.5	0	3	172	344.88	207.34	1.663	1.976	0.07
					0	4	172	442.26	234.89	1.883		
					0	5	172	393.83	209.41	1.881		
					0	6	172	526.31	200.65	2.623		
					0	7	172	568.70	263.08	2.162		
					0	8	172	1392.84	847.80	1.643		
		zono	prev_pgd	0.5	0	1	172	1043.60	505.28	2.065	1.959	0.26
					0	2	172	755.45	306.73	2.463		
					0	3	172	344.88	131.45	2.624		
					0	4	172	442.26	291.64	1.516		
					0	5	172	393.83	398.06	0.989		
					0	6	172	526.31	1812.37	0.290		
					0	7	172	568.70	151.25	3.760		
					0	8	172	1392.84	477.64	2.916		
					0	9	172	3623.14	3613.82	1.003		
		zono	prev_pgd	0.0	0	1	0	1043.60	384.60	2.713	1.787	0.02
					0	2	0	755.45	470.51	1.606		
					0	3	0	344.88	174.09	1.981		
					0	4	0	442.26	300.31	1.473		
					0	5	0	393.83	285.28	1.381		
					0	6	0	526.31	171.34	3.072		
					0	7	0	568.70	355.62	1.599		
					0	8	0	1392.84	1110.92	1.254		
					0	9	0	3623.14	3614.01	1.003		

Table 21: Architecture **cnn1** — perturbation **translation(1,1)**, $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5\}$ (part 1/3, combos ranked 1–6 of 16 by mean **sp**); continuation of Table 55 (6 combos, 45 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	1,1	interval	prev_pgd	0.0	0	3	0	344.88	213.91	1.612	1.719	<i>0.02</i>
					0	4	0	442.26	323.31	1.368		
					0	5	0	393.83	272.72	1.444		
					0	6	0	526.31	171.29	3.073		
					0	7	0	568.70	357.55	1.591		
					0	8	0	1392.84	1137.33	1.225		
					0	1	0	1043.60	330.03	3.162	1.582	<i>0.03</i>
					0	2	0	755.45	451.94	1.672		
		interval	prev_pgd	0.0+sg	0	3	0	344.88	194.20	1.776		
					0	4	0	442.26	363.26	1.217		
					0	5	0	393.83	361.12	1.091		
					0	6	0	526.31	255.94	2.056		
					0	7	0	568.70	469.86	1.210		
					0	8	0	1392.84	1323.92	1.052		
					0	9	0	3623.14	3614.79	1.002		
		zono	prev_pgd	0.0+sg	0	1	0	1043.60	399.78	2.610	1.485	<i>0.03</i>
					0	2	0	755.45	526.89	1.434		
					0	3	0	344.88	249.56	1.382		
					0	4	0	442.26	419.90	1.053		
					0	5	0	393.83	343.13	1.148		
					0	6	0	526.31	241.76	2.177		
					0	7	0	568.70	434.29	1.309		
					0	8	0	1392.84	1119.32	1.244		
					0	9	0	3623.14	3612.29	1.003		
		interval	prev_pgd	0.5+sg	0	1	171	1043.60	413.94	2.521	1.435	<i>0.02</i>
					0	2	171	755.45	358.04	2.110		
					0	3	171	344.88	401.56	0.859		
					0	4	171	442.26	615.67	0.718		
					0	5	171	393.83	332.27	1.185		
					0	6	171	526.31	392.17	1.342		
					0	7	171	568.70	334.51	1.700		
					0	8	171	1392.84	941.05	1.480		
					0	9	171	3623.14	3612.29	1.003		
		zono	no	0.5	0	3	172	344.88	296.98	1.161	1.345	<i>0.03</i>
					0	4	172	442.26	484.69	0.912		
					0	5	172	393.83	311.73	1.263		
					0	6	172	526.31	228.73	2.301		
					0	7	172	568.70	629.88	0.903		
					0	8	172	1392.84	910.59	1.530		
		interval	no	0.5	0	3	171	344.88	260.07	1.326	1.334	<i>0.02</i>
					0	4	171	442.26	336.89	1.313		
					0	5	171	393.83	326.17	1.207		
					0	6	171	526.31	313.88	1.677		
					0	7	171	568.70	336.40	1.691		
					0	8	171	1392.84	1757.53	0.792		

Table 22: Architecture **cnn1** — perturbation **translation(1,1)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 2/3, combos ranked 7–12 of 16 by mean sp); continuation of Table 55 (6 combos, 45 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	1,1	interval	direct_pgd	0.5	0	3	171	344.88	302.42	1.140	1.268	0.72
					0	4	171	442.26	321.04	1.378		
					0	5	171	393.83	1811.96	0.217		
					0	6	171	526.31	282.73	1.862		
					0	7	171	568.70	253.32	2.245		
					0	8	171	1392.84	1822.33	0.764		
		interval	prev_pgd	0.5	0	3	171	344.88	436.69	0.790	1.155	0.02
					0	4	171	442.26	310.01	1.427		
					0	5	171	393.83	463.68	0.849		
					0	6	171	526.31	432.89	1.216		
					0	7	171	568.70	325.99	1.745		
					0	8	171	1392.84	1545.65	0.901		
		interval	no	0.0	0	3	0	344.88	502.15	0.687	0.970	0.29
					0	4	0	442.26	348.09	1.271		
					0	5	0	393.83	413.77	0.952		
					0	6	0	526.31	1812.54	0.290		
					0	7	0	568.70	419.22	1.357		
					0	8	0	1392.84	1104.32	1.261		
		zono	no	0.0	0	3	0	344.88	554.54	0.622	0.895	0.29
					0	4	0	442.26	366.02	1.208		
					0	5	0	393.83	426.76	0.923		
					0	6	0	526.31	1811.14	0.291		
					0	7	0	568.70	391.80	1.452		
					0	8	0	1392.84	1590.32	0.876		

Table 23: Architecture **cnn1** — perturbation **translation(1,1)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 3/3, combos ranked 13–16 of 16 by mean **sp**); continuation of Table 55 (4 combos, 24 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_src}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_src}$
translation	1,1	zono	prev_pgd	0.0+sg	1	0	0	3296.90	1380.31	2.389	1.421	0.02
					1	2	0	3615.34	3611.66	1.001		
					1	3	0	298.04	437.28	0.682		
					1	4	0	932.29	1266.65	0.736		
					1	5	0	226.98	118.01	1.923		
					1	6	0	336.06	278.58	1.206		
					1	7	0	776.98	386.13	2.012		
					1	8	0	1820.97	3611.74	<i>timeouts differ; 2nd attempt required ($T_{\text{std}} = 1821\text{ s}$, $T_{\text{adv}} = 3612\text{ s}$)</i>		
interval	no	no	0.5	1	3	171	298.04	173.55	1.717	1.380	1.17	
				1	4	171	932.29	614.35	1.518			
				1	5	171	226.98	444.97	0.510			
				1	6	171	336.06	108.23	3.105			
				1	7	171	776.98	1835.29	0.423			
				1	8	171	1820.97	1812.79	1.005			
zono	prev_pgd	0.5+sg	1	0	172	3296.90	2026.11	1.627	1.347	0.03		
			1	2	172	3615.34	3624.60	0.997				
			1	3	172	298.04	270.63	1.101				
			1	4	172	932.29	502.41	1.856				
			1	5	172	226.98	154.11	1.473				
			1	6	172	336.06	346.33	0.970				
			1	7	172	776.98	451.89	1.719				
			1	8	172	1820.97	1540.83	1.182				
interval	prev_pgd	0.5+sg	1	9	172	3619.49	3021.35	1.198		0.05		
			1	0	171	3296.90	2575.71	1.280	1.330			
			1	2	171	3615.34	3612.50	1.001				
			1	3	171	298.04	326.98	0.911				
			1	4	171	932.29	527.23	1.768				
			1	5	171	226.98	301.81	0.752				
			1	6	171	336.06	209.51	1.604				
			1	7	171	776.98	284.98	2.726				
interval	prev_pgd	0.5	1	8	171	1820.97	2052.29	0.887		0.01		
			1	9	171	3619.49	3468.98	1.043				
			1	3	171	298.04	417.83	0.713	1.198			
			1	4	171	932.29	832.20	1.120				
			1	5	171	226.98	309.06	0.734				
			1	6	171	336.06	329.16	1.021				
interval	no	0.0	1	7	171	776.98	299.67	2.593		0.02		
			1	8	171	1820.97	1814.79	1.003				
			1	3	0	298.04	381.75	0.781	1.182			
			1	4	0	932.29	1169.82	0.797				
			1	5	0	226.98	165.19	1.374				
			1	6	0	336.06	204.49	1.643				
interval	no	0.0	1	7	0	776.98	519.38	1.496		0.02		
			1	8	0	1820.97	1820.06	1.000				

Table 24: Architecture **cnn1** — perturbation **translation**(1,1), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5\}$ (part 1/3, combos ranked 1–6 of 16 by mean **sp**); continuation of Table 55 (6 combos, 44 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	1,1	interval	prev_pgd	0.0+sg	1	0	0	3296.90	2634.47	1.251	1.166	<i>0.02</i>
					1	2	0	3615.34	3616.94	1.000		
					1	3	0	298.04	738.03	0.404		
					1	4	0	932.29	1258.76	0.741		
					1	5	0	226.98	194.16	1.169		
					1	6	0	336.06	129.82	2.589		
					1	7	0	776.98	468.70	1.658		
					1	8	0	1820.97	2687.43	0.678		
		zono	direct_pgd	0.5	1	9	0	3619.49	3611.98	1.002		
					1	3	172	298.04	364.56	0.818	1.160	<i>0.04</i>
					1	4	172	932.29	557.51	1.672		
					1	5	172	226.98	264.20	0.859		
					1	6	172	336.06	251.07	1.339		
					1	7	172	776.98	611.92	1.270		
		zono	prev_pgd	0.5	1	8	172	1820.97	1815.10	1.003		
					1	0	172	3296.90	3544.58	0.930	1.108	<i>0.02</i>
					1	2	172	3615.34	3615.80	1.000		
					1	3	172	298.04	349.65	0.852		
					1	4	172	932.29	846.05	1.102		
					1	5	172	226.98	159.90	1.420		
					1	6	172	336.06	353.84	0.950		
					1	7	172	776.98	483.64	1.607		
		interval	prev_pgd	0.0	1	8	172	1820.97	1816.26	1.003		
					1	3	0	298.04	409.60	0.728	1.081	<i>0.02</i>
					1	4	0	932.29	1269.03	0.735		
					1	5	0	226.98	154.32	1.471		
					1	6	0	336.06	361.04	0.931		
					1	7	0	776.98	479.96	1.619		
		interval	direct_pgd	0.5	1	8	0	1820.97	1814.30	1.004		
					1	3	171	298.04	488.40	0.610	1.030	<i>0.05</i>
					1	4	171	932.29	831.07	1.122		
					1	5	171	226.98	275.05	0.825		
					1	6	171	336.06	529.76	0.634		
					1	7	171	776.98	391.35	1.985		
		zono	no	0.0	1	8	171	1820.97	1813.87	1.004		
					1	3	0	298.04	459.45	0.649	<i>0.945</i>	<i>0.04</i>
					1	4	0	932.29	870.50	1.071		
					1	5	0	226.98	230.31	0.986		
					1	6	0	336.06	442.83	0.759		
					1	7	0	776.98	644.45	1.206		
					1	8	0	1820.97	1817.80	1.002		

Table 25: Architecture **cnn1** — perturbation **translation(1,1)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$ (part 2/3, combos ranked 7–12 of 16 by mean sp); continuation of Table 55 (6 combos, 41 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_{\text{src}}}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_{\text{src}}}$
translation	1,1	zono	prev_pgd	0.0	1	0	0	3296.90	3554.63	0.927	<i>0.936</i>	<i>0.04</i>
					1	2	0	3615.34	3614.15	1.000		
					1	3	0	298.04	292.76	1.018		
					1	4	0	932.29	1135.71	0.821		
					1	5	0	226.98	270.13	0.840		
					1	6	0	336.06	321.11	1.047		
					1	7	0	776.98	1013.88	0.766		
					1	8	0	1820.97	1819.69	1.001		
					1	9	0	3619.49	3613.89	1.002		
		zono	no	0.5	1	3	172	298.04	303.71	0.981	<i>0.916</i>	<i>0.04</i>
					1	4	172	932.29	1029.63	0.905		
					1	6	172	336.06	392.08	0.857		
					1	7	172	776.98	1416.25	0.549		
					1	8	172	1820.97	1411.65	1.290		
		zono	direct_pgd	0.0	1	3	0	298.04	379.46	0.785	<i>0.802</i>	<i>0.03</i>
					1	4	0	932.29	1816.50	0.513		
					1	5	0	226.98	274.36	0.827		
					1	6	0	336.06	347.39	0.967		
					1	7	0	776.98	1089.86	0.713		
					1	8	0	1820.97	1814.85	1.003		
		interval	direct_pgd	0.0	1	3	0	298.04	670.24	0.445	<i>0.717</i>	<i>0.06</i>
					1	4	0	932.29	1805.24	0.516		
					1	5	0	226.98	293.86	0.772		
					1	6	0	336.06	378.70	0.887		
					1	7	0	776.98	1150.00	0.676		
					1	8	0	1820.97	1814.94	1.003		

Table 26: Architecture **cnn1** — perturbation **translation**(1,1), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5\}$ (part 3/3, combos ranked 13–16 of 16 by mean **sp**); continuation of Table 55 (4 combos, 26 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	1,1	zono	prev_pgd	0.5	2	3	172	953.53	809.38	1.178	1.405	0.90
					2	4	172	1436.87	966.56	1.487		
					2	5	172	596.85	531.78	1.122		
					2	6	172	712.37	988.11	0.721		
					2	7	172	1815.01	632.94	2.868		
					2	8	172	773.34	731.42	1.057		
		interval	no	0.0	2	3	0	953.53	768.94	1.240	1.295	0.95
					2	4	0	1436.87	1267.89	1.133		
					2	5	0	596.85	644.75	0.926		
					2	6	0	712.37	1711.89	0.416		
					2	7	0	1815.01	611.58	2.968		
					2	8	0	773.34	709.59	1.090		
		zono	prev_pgd	0.0	2	3	0	953.53	557.23	1.711	1.246	0.90
					2	4	0	1436.87	1335.09	1.076		
					2	5	0	596.85	1114.12	0.536		
					2	6	0	712.37	787.44	0.905		
					2	7	0	1815.01	764.65	2.374		
					2	8	0	773.34	884.92	0.874		
		interval	prev_pgd	0.5	2	3	171	953.53	968.88	0.984	1.203	0.90
					2	4	171	1436.87	764.13	1.880		
					2	5	171	596.85	683.29	0.873		
					2	6	171	712.37	908.35	0.784		
					2	7	171	1815.01	944.41	1.922		
					2	8	171	773.34	1000.55	0.773		
		zono	direct_pgd	0.0	2	3	0	953.53	742.64	1.284	1.199	0.96
					2	4	0	1436.87	1813.97	0.792		
					2	5	0	596.85	543.04	1.099		
					2	6	0	712.37	1074.73	0.663		
					2	7	0	1815.01	743.01	2.443		
					2	8	0	773.34	844.35	0.916		
		interval	prev_pgd	0.0	2	3	0	953.53	672.68	1.418	1.182	0.91
					2	4	0	1436.87	1677.46	0.857		
					2	5	0	596.85	819.48	0.728		
					2	6	0	712.37	913.72	0.780		
					2	7	0	1815.01	744.91	2.437		
					2	8	0	773.34	883.95	0.875		

Table 27: Architecture **cnn1** — perturbation **translation(1,1)**, $c_{src} = 2$, $\tau \in \{0.0, 0.5\}$ (part 1/2, combos ranked 1–6 of 12 by mean **sp**); continuation of Table 55 (6 combos, 36 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	1,1	interval	direct_pgd	0.5	2	3	171	953.53	798.95	1.193	1.116	0.95
					2	4	171	1436.87	1813.77	0.792		
					2	5	171	596.85	703.01	0.849		
					2	6	171	712.37	1218.08	0.585		
					2	7	171	1815.01	789.43	2.299		
					2	8	171	773.34	791.24	0.977		
		interval	no	0.5	2	3	171	953.53	849.21	1.123	1.112	1.86
					2	4	171	1436.87	1701.85	0.844		
					2	5	171	596.85	790.23	0.755		
					2	6	171	712.37	1816.26	0.392		
					2	7	171	1815.01	767.61	2.364		
					2	8	171	773.34	647.80	1.194		
		interval	direct_pgd	0.0	2	3	0	953.53	1123.02	0.849	1.040	0.94
					2	4	0	1436.87	1814.26	0.792		
					2	5	0	596.85	1052.38	0.567		
					2	6	0	712.37	1577.13	0.452		
					2	7	0	1815.01	713.02	2.546		
					2	8	0	773.34	747.93	1.034		
		zono	no	0.5	2	3	172	953.53	860.04	1.109	1.030	0.94
					2	4	172	1436.87	1813.92	0.792		
					2	5	172	596.85	752.57	0.793		
					2	6	172	712.37	1221.18	0.583		
					2	7	172	1815.01	1099.27	1.651		
					2	8	172	773.34	617.09	1.253		
		zono	no	0.0	2	3	0	953.53	1814.80	0.525	1.021	2.02
					2	4	0	1436.87	1814.54	0.792		
					2	5	0	596.85	608.57	0.981		
					2	6	0	712.37	1815.38	0.392		
					2	7	0	1815.01	794.47	2.285		
					2	8	0	773.34	672.25	1.150		
		zono	direct_pgd	0.5	2	3	172	953.53	1049.46	0.909	0.993	0.91
					2	4	172	1436.87	1821.11	0.789		
					2	5	172	596.85	830.78	0.718		
					2	6	172	712.37	1517.95	0.469		
					2	7	172	1815.01	1057.38	1.717		
					2	8	172	773.34	569.92	1.357		

Table 28: Architecture **cnn1** — perturbation **translation(1,1)**, $c_{src} = 2$, $\tau \in \{0.0, 0.5\}$ (part 2/2, combos ranked 7–12 of 12 by mean **sp**); continuation of Table 55 (6 combos, 36 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	3,1	zono	direct_pgd	0.5	0	3	190	458.41	225.16	2.036	1.738	0.02
					0	4	190	270.41	211.84	1.276		
					0	5	190	285.88	115.96	2.465		
					0	6	190	258.24	239.06	1.080		
					0	7	190	323.42	317.45	1.019		
					0	8	190	296.48	116.28	2.550		
					0	1	190	365.11	176.78	2.065		
					0	2	190	531.41	110.69	4.801		
					0	3	190	458.41	246.37	1.861		
					0	4	190	270.41	337.89	0.800		
		zono	prev_pgd	0.5+sg	0	5	190	285.88	322.52	0.886	1.595	0.03
					0	6	190	258.24	260.24	0.992		
					0	7	190	323.42	344.66	0.938		
					0	8	190	296.48	571.62	0.519		
					0	9	190	3618.45	2424.69	1.492		
		zono	direct_pgd	0.0	0	3	0	458.41	229.76	1.995	1.574	0.05
					0	4	0	270.41	278.69	0.970		
					0	5	0	285.88	118.80	2.406		
					0	6	0	258.24	118.90	2.172		
					0	7	0	323.42	323.40	1.000		
					0	8	0	296.48	328.32	0.903		
		interval	direct_pgd	0.0	0	3	0	458.41	226.43	2.025	1.555	0.06
					0	4	0	270.41	262.12	1.032		
					0	5	0	285.88	121.21	2.359		
					0	6	0	258.24	118.51	2.179		
					0	7	0	323.42	297.66	1.087		
					0	8	0	296.48	454.33	0.653		
		zono	prev_pgd	0.0+sg	0	1	0	365.11	386.33	0.945	1.291	0.02
					0	2	0	531.41	181.01	2.936		
					0	3	0	458.41	228.26	2.008		
					0	4	0	270.41	459.90	0.588		
					0	5	0	285.88	366.10	0.781		
					0	6	0	258.24	300.41	0.860		
					0	7	0	323.42	277.97	1.164		
					0	8	0	296.48	221.55	1.338		
		interval	prev_pgd	0.5+sg	0	9	0	3618.45	3611.86	1.002	1.212	0.04
					0	1	181	365.11	349.91	1.043		
					0	2	181	531.41	410.24	1.295		
					0	3	181	458.41	273.58	1.676		
					0	4	181	270.41	293.06	0.923		
					0	5	181	285.88	334.19	0.855		
					0	6	181	258.24	204.63	1.262		
					0	7	181	323.42	337.60	0.958		
					0	8	181	296.48	161.51	1.836		
					0	9	181	3618.45	3409.12	1.061		

Table 29: Architecture **cnn1** — perturbation **translation(3,1)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 1/3, combos ranked 1–6 of 16 by mean **sp**); continuation of Table 55 (6 combos, 45 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	3,1	interval	prev_pgd	0.0+sg	0	1	0	365.11	334.47	1.092	1.199	0.02
					0	2	0	531.41	180.50	2.944		
					0	3	0	458.41	221.01	2.074		
					0	4	0	270.41	488.09	0.554		
					0	5	0	285.88	392.89	0.728		
					0	6	0	258.24	344.43	0.750		
					0	7	0	323.42	307.83	1.051		
					0	8	0	296.48	496.73	0.597		
		interval	direct_pgd	0.5	0	3	181	458.41	249.76	1.835	1.197	0.02
					0	4	181	270.41	298.42	0.906		
					0	5	181	285.88	266.16	1.074		
					0	6	181	258.24	301.82	0.856		
					0	7	181	323.42	298.44	1.084		
					0	8	181	296.48	208.11	1.425		
					0	1	0	365.11	436.52	0.836	1.123	
					0	2	0	531.41	205.78	2.582		
		zono	prev_pgd	0.0	0	3	0	458.41	249.86	1.835		0.02
					0	4	0	270.41	465.54	0.581		
					0	5	0	285.88	378.39	0.756		
					0	6	0	258.24	328.08	0.787		
					0	7	0	323.42	294.08	1.100		
					0	8	0	296.48	471.74	0.628		
					0	9	0	3618.45	3613.89	1.001		
					0	1	190	365.11	265.55	1.375	1.097	0.03
		zono	prev_pgd	0.5	0	2	190	531.41	538.16	0.987		
					0	3	190	458.41	275.19	1.666		
					0	4	190	270.41	241.81	1.118		
					0	5	190	285.88	348.23	0.821		
					0	6	190	258.24	308.46	0.837		
					0	7	190	323.42	529.48	0.611		
					0	8	190	296.48	203.91	1.454		
					0	9	190	3618.45	3615.49	1.001		
		interval	no	0.0	0	3	0	458.41	279.15	1.642	1.010	0.01
					0	4	0	270.41	488.16	0.554		
					0	5	0	285.88	241.79	1.182		
					0	6	0	258.24	304.00	0.849		
					0	7	0	323.42	340.76	0.949		
					0	8	0	296.48	336.22	0.882		
		interval	prev_pgd	0.0	0	3	0	458.41	225.74	2.031	1.006	0.02
					0	4	0	270.41	475.66	0.568		
					0	5	0	285.88	367.51	0.778		
					0	6	0	258.24	296.39	0.871		
					0	7	0	323.42	275.71	1.173		
					0	8	0	296.48	483.81	0.613		

Table 30: Architecture **cnn1** — perturbation **translation(3,1)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 2/3, combos ranked 7–12 of 16 by mean **sp**); continuation of Table 55 (6 combos, 45 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	3,1	interval	no	0.5	0	3	181	458.41	440.50	1.041	<i>0.868</i>	<i>0.07</i>
					0	4	181	270.41	343.01	0.788		
					0	5	181	285.88	259.82	1.100		
					0	6	181	258.24	461.41	0.560		
					0	7	181	323.42	482.64	0.670		
					0	8	181	296.48	282.36	1.050		
	zono		no	0.0	0	3	0	458.41	404.77	1.133	<i>0.812</i>	<i>0.01</i>
					0	4	0	270.41	376.75	0.718		
					0	5	0	285.88	286.57	0.998		
					0	6	0	258.24	502.61	0.514		
					0	7	0	323.42	479.76	0.674		
					0	8	0	296.48	355.67	0.834		
	interval		prev_pgd	0.5	0	3	181	458.41	1817.39	0.252	<i>0.693</i>	<i>0.31</i>
					0	4	181	270.41	472.20	0.573		
					0	5	181	285.88	227.51	1.257		
					0	6	181	258.24	532.65	0.485		
					0	7	181	323.42	403.74	0.801		
					0	8	181	296.48	373.95	0.793		
	zono		no	0.5	0	3	190	458.41	241.72	1.896	<i>0.655</i>	<i>0.72</i>
					0	4	190	270.41	1813.27	0.149		
					0	5	190	285.88	625.34	0.457		
					0	6	190	258.24	454.30	0.568		
					0	7	190	323.42	463.93	0.697		
					0	8	190	296.48	1813.48	0.163		

Table 31: Architecture **cn1** — perturbation **translation(3,1)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 3/3, combos ranked 13–16 of 16 by mean **sp**); continuation of Table 55 (4 combos, 24 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_src}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_src}$
translation	3,1	interval	prev_pgd	0.0+sg	1	0	0	275.97	195.38	1.412	1.791	0.09
					1	2	0	433.07	291.89	1.484		
					1	3	0	285.99	355.86	0.804		
					1	4	0	111.60	91.98	1.213		
					1	5	0	501.62	76.96	6.518		
					1	6	0	249.93	172.79	1.446		
					1	7	0	245.31	315.16	0.778		
					1	8	0	264.15	156.82	1.684		
		interval	prev_pgd	0.5+sg	1	9	0	625.05	802.57	0.779		
					1	0	181	275.97	138.13	1.998	1.687	0.77
					1	2	181	433.07	257.89	1.679		
					1	3	181	285.99	279.80	1.022		
					1	4	181	111.60	83.53	1.336		
					1	5	181	501.62	89.55	5.602		
					1	6	181	249.93	3611.48	0.069		
					1	7	181	245.31	392.25	0.625		
		interval	no	0.5	1	8	181	264.15	132.98	1.986		
					1	9	181	625.05	721.03	0.867		
					1	3	181	285.99	124.76	2.292	1.631	0.13
					1	4	181	111.60	148.45	0.752		
					1	5	181	501.62	130.20	3.853		
					1	6	181	249.93	241.73	1.034		
					1	7	181	245.31	345.19	0.711		
					1	8	181	264.15	230.79	1.145		
		zono	prev_pgd	0.0+sg	1	0	0	275.97	258.69	1.067	1.494	0.08
					1	2	0	433.07	271.13	1.597		
					1	3	0	285.99	130.18	2.197		
					1	4	0	111.60	166.50	0.670		
					1	5	0	501.62	126.07	3.979		
					1	6	0	249.93	171.00	1.462		
					1	7	0	245.31	276.57	0.887		
					1	8	0	264.15	526.03	0.502		
		interval	prev_pgd	0.5	1	9	0	625.05	574.96	1.087		
					1	3	181	285.99	168.95	1.693	1.484	0.14
					1	4	181	111.60	108.92	1.025		
					1	5	181	501.62	202.35	2.479		
					1	6	181	249.93	140.68	1.777		
					1	7	181	245.31	411.83	0.596		
					1	8	181	264.15	197.64	1.337		
		interval	direct_pgd	0.5	1	3	181	285.99	327.27	0.874	1.445	0.10
					1	4	181	111.60	141.92	0.786		
					1	5	181	501.62	157.57	3.183		
					1	6	181	249.93	127.73	1.957		
					1	7	181	245.31	187.92	1.305		
					1	8	181	264.15	469.13	0.563		

Table 32: Architecture **cnn1** — perturbation **translation(3,1)**, $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5\}$ (part 1/3, combos ranked 1–6 of 16 by mean sp); continuation of Table 55 (6 combos, 45 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	3,1	zono	prev_pgd	0.0	1	0	0	275.97	284.70	0.969	1.433	0.08
					1	2	0	433.07	436.37	0.992		
					1	3	0	285.99	242.06	1.181		
					1	4	0	111.60	147.10	0.759		
					1	5	0	501.62	121.32	4.135		
					1	6	0	249.93	169.21	1.477		
					1	7	0	245.31	240.75	1.019		
					1	8	0	264.15	153.94	1.716		
					1	9	0	625.05	968.70	0.645		
		zono	direct_pgd	0.5	1	3	190	285.99	361.59	0.791	1.431	0.14
					1	4	190	111.60	336.76	0.331		
					1	5	190	501.62	184.61	2.717		
					1	6	190	249.93	88.95	2.810		
					1	7	190	245.31	178.60	1.374		
					1	8	190	264.15	471.44	0.560		
		interval	prev_pgd	0.0	1	3	0	285.99	648.09	0.441	1.389	0.12
					1	4	0	111.60	165.79	0.673		
					1	5	0	501.62	133.79	3.749		
					1	6	0	249.93	360.44	0.693		
					1	7	0	245.31	183.05	1.340		
					1	8	0	264.15	183.51	1.439		
		zono	no	0.0	1	3	0	285.99	225.87	1.266	1.389	0.10
					1	4	0	111.60	266.23	0.419		
					1	5	0	501.62	191.86	2.615		
					1	6	0	249.93	105.87	2.361		
					1	7	0	245.31	497.23	0.493		
					1	8	0	264.15	223.71	1.181		
		interval	no	0.0	1	3	0	285.99	194.40	1.471	1.256	0.09
					1	4	0	111.60	218.18	0.512		
					1	5	0	501.62	334.48	1.500		
					1	6	0	249.93	327.83	0.762		
					1	7	0	245.31	102.06	2.404		
					1	8	0	264.15	297.86	0.887		
		zono	prev_pgd	0.5	1	0	190	275.97	392.20	0.704	1.050	0.09
					1	2	190	433.07	519.61	0.833		
					1	3	190	285.99	404.25	0.707		
					1	4	190	111.60	202.48	0.551		
					1	5	190	501.62	333.21	1.505		
					1	6	190	249.93	193.54	1.291		
					1	7	190	245.31	199.66	1.229		
					1	8	190	264.15	182.86	1.445		
					1	9	190	625.05	526.06	1.188		

Table 33: Architecture **cnn1** — perturbation **translation(3,1)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$ (part 2/3, combos ranked 7–12 of 16 by mean sp); continuation of Table 55 (6 combos, 42 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	3,1	zono	prev_pgd	0.5+sg	1	0	190	275.97	381.41	0.724	1.007	<i>0.09</i>
					1	2	190	433.07	408.60	1.060		
					1	3	190	285.99	572.12	0.500		
					1	4	190	111.60	143.67	0.777		
					1	5	190	501.62	176.92	2.835		
					1	6	190	249.93	461.89	0.541		
					1	7	190	245.31	664.53	0.369		
					1	8	190	264.15	350.46	0.754		
		interval	direct_pgd	0.0	1	9	190	625.05	414.52	1.508		
					1	3	0	285.99	359.95	0.795	<i>0.861</i>	<i>0.14</i>
					1	4	0	111.60	586.27	0.190		
					1	5	0	501.62	197.57	2.539		
					1	6	0	249.93	451.87	0.553		
					1	7	0	245.31	416.52	0.589		
		zono	no	0.5	1	8	0	264.15	527.36	0.501		
					1	3	190	285.99	406.70	0.703	<i>0.833</i>	<i>0.15</i>
					1	4	190	111.60	298.54	0.374		
					1	5	190	501.62	310.09	1.618		
					1	6	190	249.93	258.10	0.968		
					1	7	190	245.31	514.64	0.477		
		zono	direct_pgd	0.0	1	8	190	264.15	308.21	0.857		
					1	3	0	285.99	546.00	0.524	<i>0.631</i>	<i>0.11</i>
					1	4	0	111.60	417.92	0.267		
					1	5	0	501.62	505.20	0.993		
					1	6	0	249.93	443.52	0.564		
					1	7	0	245.31	292.67	0.838		
					1	8	0	264.15	439.55	0.601		

Table 34: Architecture **cnn1** — perturbation **translation(3,1)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$ (part 3/3, combos ranked 13–16 of 16 by mean **sp**); continuation of Table 55 (4 combos, 27 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	3,1	interval	no	0.0	2	3	0	1520.47	572.05	2.658	1.456	0.05
					2	4	0	1813.81	970.70	1.869		
					2	5	0	642.29	570.00	1.127		
					2	6	0	732.39	813.37	0.900		
					2	7	0	586.50	584.42	1.004		
					2	8	0	949.03	806.55	1.177		
		interval	direct_pgd	0.5	2	3	181	1520.47	959.24	1.585	1.291	0.09
					2	4	181	1813.81	1012.40	1.792		
					2	5	181	642.29	611.02	1.051		
					2	6	181	732.39	1056.48	0.693		
					2	7	181	586.50	808.80	0.725		
					2	8	181	949.03	499.07	1.902		
		interval	prev_pgd	0.0	2	3	0	1520.47	682.18	2.229	1.115	0.08
					2	4	0	1813.81	1026.87	1.766		
					2	5	0	642.29	1360.90	0.472		
					2	6	0	732.39	1395.71	0.525		
					2	7	0	586.50	763.40	0.768		
					2	8	0	949.03	1020.94	0.930		
		zono	direct_pgd	0.0	2	3	0	1520.47	899.04	1.691	1.030	0.10
					2	4	0	1813.81	1153.78	1.572		
					2	5	0	642.29	944.59	0.680		
					2	6	0	732.39	1487.54	0.492		
					2	7	0	586.50	1197.61	0.490		
					2	8	0	949.03	757.33	1.253		

Table 35: Architecture **cnn1** — perturbation **translation(3,1)**, $c_{src} = 2$, $\tau \in \{0.0, 0.5\}$ (part 1/2, combos ranked 1–4 of 8 by mean **sp**); continuation of Table 55 (4 combos, 24 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	3,1	interval	direct_pgd	0.0	2	3	0	1520.47	763.21	1.992	1.027	0.09
					2	4	0	1813.81	1168.34	1.552		
					2	5	0	642.29	788.91	0.814		
					2	6	0	732.39	1429.23	0.512		
					2	7	0	586.50	1153.10	0.509		
					2	8	0	949.03	1210.61	0.784		
		interval	no	0.5	2	3	181	1520.47	671.17	2.265	0.998	0.30
					2	4	181	1813.81	1752.28	1.035		
					2	5	181	642.29	1829.06	0.351		
					2	6	181	732.39	1823.50	0.402		
					2	7	181	586.50	1291.30	0.454		
					2	8	181	949.03	640.75	1.481		
		zono	prev_pgd	0.0	2	3	0	1520.47	868.11	1.751	0.914	0.05
					2	4	0	1813.81	1820.06	0.997		
					2	5	0	642.29	822.96	0.780		
					2	6	0	732.39	1294.01	0.566		
					2	7	0	586.50	1323.13	0.443		
					2	8	0	949.03	1001.97	0.947		
		interval	prev_pgd	0.5	2	3	181	1520.47	686.38	2.215	0.904	0.09
					2	4	181	1813.81	1790.70	1.013		
					2	5	181	642.29	1381.69	0.465		
					2	6	181	732.39	1524.68	0.480		
					2	7	181	586.50	1687.02	0.348		

Table 36: Architecture **cnn1** — perturbation **translation(3,1)**, $c_{src} = 2$, $\tau \in \{0.0, 0.5\}$ (part 2/2, combos ranked 5–8 of 8 by mean sp); continuation of Table 55 (4 combos, 23 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_{\text{src}}}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_{\text{src}}}$
translation	3,3	interval	direct_pgd	0.5	0	3	180	800.69	228.83	3.499	2.381	0.01
					0	4	180	1821.48	1817.99	1.002		
					0	5	181	682.97	235.99	2.894		
					0	6	180	765.68	305.42	2.507		
					0	7	180	1231.36	415.64	2.963		
					0	8	180	375.90	264.02	1.424		
		zono	direct_pgd	0.5	0	3	190	800.69	301.29	2.658	2.317	0.03
					0	4	190	1821.48	1406.06	1.295		
					0	5	190	682.97	248.62	2.747		
					0	6	190	765.68	297.86	2.571		
					0	7	190	1231.36	404.42	3.045		
					0	8	190	375.90	236.66	1.588		
		zono	prev_pgd	0.5	0	1	190	431.25	381.28	1.131	2.143	0.23
					0	2	190	427.19	220.50	1.937		
					0	3	190	800.69	350.15	2.287		
					0	4	190	1821.48	1812.28	1.005		
					0	5	190	682.97	141.61	4.823		
					0	6	190	765.68	338.72	2.261		
					0	7	190	1231.36	391.75	3.143		
					0	8	190	375.90	220.73	1.703		
					0	9	190	3612.87	3620.89	0.998		
		interval	prev_pgd	0.5	0	3	181	800.69	887.89	0.902	2.098	0.02
					0	4	180	1821.48	1265.58	1.439		
					0	5	180	682.97	148.72	4.592		
					0	6	180	765.68	388.40	1.971		
					0	7	181	1231.36	446.44	2.758		
					0	8	180	375.90	407.43	0.923		
		interval	direct_pgd	0.0	0	3	0	800.69	289.09	2.770	2.067	0.04
					0	4	0	1821.48	1811.90	1.005		
					0	5	0	682.97	260.32	2.624		
					0	6	0	765.68	514.44	1.488		
					0	7	0	1231.36	866.65	1.421		
					0	8	0	375.90	121.39	3.097		
		zono	no	0.5	0	3	190	800.69	402.39	1.990	2.010	0.03
					0	4	190	1821.48	1341.89	1.357		
					0	5	190	682.97	330.07	2.069		
					0	6	190	765.68	341.53	2.242		
					0	7	190	1231.36	485.35	2.537		
					0	8	190	375.90	201.49	1.866		

Table 37: Architecture **cnn1** — perturbation **translation(3,3)**, $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5\}$ (part 1/3, combos ranked 1–6 of 16 by mean sp); continuation of Table 55 (6 combos, 39 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	3,3	zono	direct_pgd	0.0	0	3	0	800.69	278.16	2.879	1.991	<i>0.04</i>
					0	4	0	1821.48	1816.09	1.003		
					0	5	0	682.97	260.72	2.620		
					0	6	0	765.68	545.35	1.404		
					0	7	0	1231.36	946.91	1.300		
					0	8	0	375.90	137.15	2.741		
		interval	prev_pgd	0.0	0	3	0	800.69	361.59	2.214	1.876	<i>0.01</i>
					0	4	0	1821.48	1811.70	1.005		
					0	5	0	682.97	413.35	1.652		
					0	6	0	765.68	435.35	1.759		
					0	7	0	1231.36	418.63	2.941		
					0	8	0	375.90	223.40	1.683		
		zono	no	0.0	0	3	0	800.69	350.15	2.287	1.793	<i>0.39</i>
					0	4	0	1821.48	1815.03	1.004		
					0	5	0	682.97	264.20	2.585		
					0	6	0	765.68	473.32	1.618		
					0	7	0	1231.36	625.19	1.970		
					0	8	0	375.90	290.05	1.296		
		zono	prev_pgd	0.0	0	1	0	431.25	217.05	1.987	1.776	<i>0.02</i>
					0	2	0	427.19	317.76	1.344		
					0	3	0	800.69	337.58	2.372		
					0	4	0	1821.48	1811.26	1.006		
					0	5	0	682.97	345.88	1.975		
					0	6	0	765.68	363.34	2.107		
					0	7	0	1231.36	510.71	2.411		
					0	8	0	375.90	210.38	1.787		
					0	9	0	3612.87	3613.38	1.000		
		zono	prev_pgd	0.0+sg	0	1	0	431.25	493.26	0.874	1.772	<i>0.04</i>
					0	2	0	427.19	296.37	1.441		
					0	3	0	800.69	354.13	2.261		
					0	4	0	1821.48	1942.63	0.938		
					0	5	0	682.97	368.04	1.856		
					0	6	0	765.68	312.70	2.449		
					0	7	0	1231.36	382.92	3.216		
					0	8	0	375.90	196.12	1.917		
					0	9	0	3612.87	3612.29	1.000		
		interval	no	0.5	0	3	180	800.69	455.08	1.759	1.739	<i>0.01</i>
					0	4	181	1821.48	1816.49	1.003		
					0	5	180	682.97	437.85	1.560		
					0	6	180	765.68	329.28	2.325		
					0	7	180	1231.36	450.77	2.732		
					0	8	181	375.90	355.96	1.056		

Table 38: Architecture **cnn1** — perturbation **translation(3,3)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 2/3, combos ranked 7–12 of 16 by mean **sp**); continuation of Table 55 (6 combos, 42 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	3,3	interval	no	0.0	0	3	0	800.69	340.33	2.353	1.648	<i>0.39</i>
					0	4	0	1821.48	1815.79	1.003		
					0	5	0	682.97	284.48	2.401		
					0	6	0	765.68	543.74	1.408		
					0	7	0	1231.36	621.39	1.982		
					0	8	0	375.90	505.89	0.743		
					0	1	190	431.25	395.71	1.090	1.643	<i>0.18</i>
					0	2	190	427.19	392.98	1.087		
		zono	prev_pgd	0.5+sg	0	3	190	800.69	451.63	1.773		
					0	4	190	1821.48	1472.60	1.237		
					0	5	190	682.97	300.89	2.270		
					0	6	190	765.68	342.35	2.237		
					0	7	190	1231.36	308.48	3.992		
					0	8	190	375.90	3612.02	0.104		
					0	9	190	3612.87	3612.25	1.000		
		interval	prev_pgd	0.5+sg	0	1	180	431.25	291.19	1.481	1.619	<i>0.03</i>
					0	2	180	427.19	457.50	0.934		
					0	3	180	800.69	375.42	2.133		
					0	4	180	1821.48	1162.34	1.567		
					0	5	181	682.97	613.28	1.114		
					0	6	180	765.68	342.23	2.237		
					0	7	180	1231.36	387.05	3.181		
					0	8	180	375.90	405.09	0.928		
		interval	prev_pgd	0.0+sg	0	9	180	3612.87	3615.06	0.999		
					0	1	0	431.25	205.35	2.100	1.594	<i>0.04</i>
					0	2	0	427.19	321.66	1.328		
					0	3	0	800.69	343.85	2.329		
					0	4	0	1821.48	1977.60	0.921		
					0	5	0	682.97	386.43	1.767		
					0	6	0	765.68	359.33	2.131		
					0	7	0	1231.36	680.80	1.809		
					0	8	0	375.90	389.35	0.965		
					0	9	0	3612.87	3624.15	0.997		

Table 39: Architecture **cnn1** — perturbation **translation(3,3)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 3/3, combos ranked 13–16 of 16 by mean **sp**); continuation of Table 55 (4 combos, 33 cell rows). Column meanings: see Table 55.

p-type	p-size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_src}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_src}$
translation	3,3	interval	prev_pgd	0.5+sg	1	0	181	3614.59	3610.08	1.001	1.426	0.31
					1	2	180	3617.57	3611.08	1.002		
					1	3	180	1812.65	3612.04	<i>timeouts differ; 2nd attempt required ($T_{\text{std}} = 1813\text{ s}$, $T_{\text{adv}} = 3612\text{ s}$)</i>		
					1	4	180	1086.37	531.81	2.043		
					1	6	180	1815.56	3611.16	<i>timeouts differ; 2nd attempt required ($T_{\text{std}} = 1816\text{ s}$, $T_{\text{adv}} = 3611\text{ s}$)</i>		
					1	7	180	176.49	84.77	2.082		
					1	8	180	1820.58	3611.71	<i>timeouts differ; 2nd attempt required ($T_{\text{std}} = 1821\text{ s}$, $T_{\text{adv}} = 3612\text{ s}$)</i>		
					1	9	180	3616.82	3611.72	1.001		
		interval	prev_pgd	0.0+sg	1	0	0	3614.59	3612.91	1.000	1.072	0.16
					1	2	0	3617.57	3613.24	1.001		
					1	3	0	1812.65	3618.95	<i>timeouts differ; 2nd attempt required ($T_{\text{std}} = 1813\text{ s}$, $T_{\text{adv}} = 3619\text{ s}$)</i>		
					1	4	0	1086.37	3622.12	0.300		
					1	6	0	1815.56	3619.08	<i>timeouts differ; 2nd attempt required ($T_{\text{std}} = 1816\text{ s}$, $T_{\text{adv}} = 3619\text{ s}$)</i>		
					1	7	0	176.49	85.77	2.058		
					1	8	0	1820.58	3614.02	<i>timeouts differ; 2nd attempt required ($T_{\text{std}} = 1821\text{ s}$, $T_{\text{adv}} = 3614\text{ s}$)</i>		
					1	9	0	3616.82	3612.46	1.001		
		interval	no	0.0	1	3	0	1812.65	1815.72	0.998	1.068	0.10
					1	4	0	1086.37	1820.20	0.597		
					1	7	0	176.49	105.38	1.675		
					1	8	0	1820.58	1814.55	1.003		
zono		zono	prev_pgd	0.5	1	0	190	3614.59	3621.34	0.998	1.040	0.18
					1	2	190	3617.57	3615.27	1.001		
					1	3	190	1812.65	1815.95	0.998		
					1	4	190	1086.37	1326.67	0.819		
					1	6	190	1815.56	1823.05	0.996		
					1	7	190	176.49	116.99	1.509		
					1	8	190	1820.58	1816.18	1.002		
					1	9	190	3616.82	3621.12	0.999		
		interval	no	0.5	1	3	180	1812.65	1815.41	0.998	0.976	0.11
					1	4	180	1086.37	1816.44	0.598		
					1	6	181	1815.56	1814.60	1.001		
					1	7	181	176.49	137.95	1.279		
					1	8	181	1820.58	1817.76	1.002		
		zono	prev_pgd	0.0	1	0	0	3614.59	3620.73	0.998	0.970	0.11
					1	2	0	3617.57	3620.48	0.999		
					1	4	0	1086.37	1813.65	0.599		
					1	6	0	1815.56	1816.59	0.999		
					1	7	0	176.49	147.80	1.194		
					1	8	0	1820.58	1823.02	0.999		
					1	9	0	3616.82	3622.13	0.999		

Table 40: Architecture **cnn1** — perturbation **translation(3,3)**, $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5\}$ (part 1/3, combos ranked 1–6 of 16 by mean **sp**); continuation of Table 55 (6 combos, 40 cell rows). Column meanings: see Table 55.

p.type	p.size	bound.tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_src}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_src}$	
translation	3,3	interval	prev_pgd	0.0	1	3	0	1812.65	1815.38	0.998	0.960	0.07	
					1	4	0	1086.37	1814.67	0.599			
					1	6	0	1815.56	1814.10	1.001			
					1	7	0	176.49	147.30	1.198			
					1	8	0	1820.58	1813.81	1.004			
	zono	no	0.0	1	4	0	1086.37	1813.42	0.599	0.959	0.09		
				1	6	0	1815.56	1813.14	1.001				
				1	7	0	176.49	142.70	1.237				
				1	8	0	1820.58	1819.30	1.001				
	zono	prev_pgd	0.5+sg	1	0	190	3614.59	3624.92	0.997	0.944	0.14		
				1	2	190	3617.57	3624.58	0.998				
				1	3	189	1812.65	3612.86	timeouts differ; 2nd attempt required ($T_{\text{std}} = 1813\text{ s}$, $T_{\text{adv}} = 3613\text{ s}$)				
				1	4	190	1086.37	3612.77	0.301				
				1	6	190	1815.56	3618.22	timeouts differ; 2nd attempt required ($T_{\text{std}} = 1816\text{ s}$, $T_{\text{adv}} = 3618\text{ s}$)				
				1	7	190	176.49	124.23	1.421				
				1	8	190	1820.58	3616.81	timeouts differ; 2nd attempt required ($T_{\text{std}} = 1821\text{ s}$, $T_{\text{adv}} = 3617\text{ s}$)				
				1	9	190	3616.82	3613.35	1.001				
	zono	no	0.5	1	3	190	1812.65	1817.10	0.998	0.934	0.08		
				1	4	190	1086.37	1817.93	0.598				
				1	6	190	1815.56	1815.52	1.000				
				1	7	190	176.49	164.25	1.075				
				1	8	190	1820.58	1816.41	1.002				
	zono	prev_pgd	0.0+sg	1	0	0	3614.59	3612.05	1.001	0.910	1.37		
				1	2	0	3617.57	3612.43	1.001				
				1	3	0	1812.65	3612.58	timeouts differ; 2nd attempt required ($T_{\text{std}} = 1813\text{ s}$, $T_{\text{adv}} = 3613\text{ s}$)				
				1	4	0	1086.37	3613.26	0.301				
				1	6	0	1815.56	3613.74	timeouts differ; 2nd attempt required ($T_{\text{std}} = 1816\text{ s}$, $T_{\text{adv}} = 3614\text{ s}$)				
				1	7	0	176.49	141.74	1.245				
				1	8	0	1820.58	3614.33	timeouts differ; 2nd attempt required ($T_{\text{std}} = 1821\text{ s}$, $T_{\text{adv}} = 3614\text{ s}$)				
				1	9	0	3616.82	3611.94	1.001				
	interval	direct_pgd	0.0	1	3	0	1812.65	1813.61	0.999	0.871	0.10		
				1	4	0	1086.37	1813.57	0.599				
				1	6	0	1815.56	1813.43	1.001				
				1	7	0	176.49	233.95	0.754				
				1	8	0	1820.58	1814.40	1.003				

Table 41: Architecture **cnn1** — perturbation **translation(3,3)**, $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5\}$ (part 2/3, combos ranked 7–12 of 16 by mean sp); continuation of Table 55 (6 combos, 35 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	3,3	interval	direct_pgd	0.5	1	3	181	1812.65	1822.16	0.995	0.862	0.08
					1	4	181	1086.37	1823.43	0.596		
					1	6	181	1815.56	1819.69	0.998		
					1	7	181	176.49	243.61	0.724		
					1	8	181	1820.58	1822.56	0.999		
		zono	direct_pgd	0.5	1	4	190	1086.37	1816.03	0.598	0.823	1.77
					1	6	190	1815.56	1815.10	1.000		
					1	7	190	176.49	254.89	0.692		
					1	8	190	1820.58	1820.42	1.000		
		interval	prev_pgd	0.5	1	3	181	1812.65	1823.00	0.994	0.816	0.06
					1	4	181	1086.37	1823.35	0.596		
					1	6	181	1815.56	1825.91	0.994		
					1	7	181	176.49	357.46	0.494		
					1	8	181	1820.58	1816.23	1.002		
		zono	direct_pgd	0.0	1	3	0	1812.65	1816.32	0.998	0.737	0.06
					1	4	0	1086.37	1817.13	0.598		
					1	7	0	176.49	503.95	0.350		
					1	8	0	1820.58	1819.58	1.001		

Table 42: Architecture **cnn1** — perturbation **translation(3,3)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$ (part 3/3, combos ranked 13–16 of 16 by mean **sp**); continuation of Table 55 (4 combos, 18 cell rows). Column meanings: see Table 55.

p.type	p.size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
linf	0.1	zono	prev_pgd	0.0+sg	0	1	0	3620.83	3611.20	1.003	1.002	2.20
					0	2	0	3623.78	3611.38	1.003		
					0	3	0	1814.89	3611.37	timeouts differ; 2nd attempt required ($T_{std} = 1815\text{ s}$, $T_{adv} = 3611\text{ s}$)		
					0	4	0	1817.37	3611.12	timeouts differ; 2nd attempt required ($T_{std} = 1817\text{ s}$, $T_{adv} = 3611\text{ s}$)		
					0	5	0	1812.49	3611.29	timeouts differ; 2nd attempt required ($T_{std} = 1812\text{ s}$, $T_{adv} = 3611\text{ s}$)		
					0	7	0	3618.34	3611.32	1.002		
					0	8	0	3615.99	3611.69	1.001		
					0	9	0	3616.65	3611.43	1.001		
		zono	prev_pgd	0.5+sg	0	1	164	3620.83	3612.45	1.002	1.002	1.62
					0	2	164	3623.78	3611.63	1.003		
					0	3	164	1814.89	3611.85	timeouts differ; 2nd attempt required ($T_{std} = 1815\text{ s}$, $T_{adv} = 3612\text{ s}$)		
					0	4	164	1817.37	3612.76	timeouts differ; 2nd attempt required ($T_{std} = 1817\text{ s}$, $T_{adv} = 3613\text{ s}$)		
					0	5	164	1812.49	3611.88	timeouts differ; 2nd attempt required ($T_{std} = 1812\text{ s}$, $T_{adv} = 3612\text{ s}$)		
					0	6	164	3616.40	3612.79	1.001		
					0	7	164	3618.34	3612.56	1.002		
					0	8	164	3615.99	3612.98	1.001		
					0	9	164	3616.65	3612.09	1.001		
		interval	prev_pgd	0.5+sg	0	1	164	3620.83	3612.46	1.002	1.001	1.71
					0	2	164	3623.78	3617.39	1.002		
					0	3	164	1814.89	3612.35	timeouts differ; 2nd attempt required ($T_{std} = 1815\text{ s}$, $T_{adv} = 3612\text{ s}$)		
					0	4	164	1817.37	3617.42	timeouts differ; 2nd attempt required ($T_{std} = 1817\text{ s}$, $T_{adv} = 3617\text{ s}$)		
					0	5	164	1812.49	3613.70	timeouts differ; 2nd attempt required ($T_{std} = 1812\text{ s}$, $T_{adv} = 3614\text{ s}$)		
					0	6	164	3616.40	3611.97	1.001		
					0	7	164	3618.34	3621.89	0.999		
					0	8	164	3615.99	3611.69	1.001		
					0	9	164	3616.65	3611.88	1.001		
		interval	prev_pgd	0.0+sg	0	2	0	3623.78	3613.83	1.003	1.001	2.91
					0	3	0	1814.89	3620.03	timeouts differ; 2nd attempt required ($T_{std} = 1815\text{ s}$, $T_{adv} = 3620\text{ s}$)		
					0	4	0	1817.37	3620.91	timeouts differ; 2nd attempt required ($T_{std} = 1817\text{ s}$, $T_{adv} = 3621\text{ s}$)		
					0	5	0	1812.49	3615.15	timeouts differ; 2nd attempt required ($T_{std} = 1812\text{ s}$, $T_{adv} = 3615\text{ s}$)		
					0	7	0	3618.34	3621.47	0.999		
					0	8	0	3615.99	3616.40	1.000		
					0	9	0	3616.65	3611.71	1.001		
		zono	prev_pgd	0.0	0	1	0	3620.83	3615.91	1.001	1.001	1.85
					0	2	0	3623.78	3616.89	1.002		
					0	3	0	1814.89	3617.97	timeouts differ; 2nd attempt required ($T_{std} = 1815\text{ s}$, $T_{adv} = 3618\text{ s}$)		
					0	4	0	1817.37	3616.07	timeouts differ; 2nd attempt required ($T_{std} = 1817\text{ s}$, $T_{adv} = 3616\text{ s}$)		
					0	5	0	1812.49	3617.23	timeouts differ; 2nd attempt required ($T_{std} = 1812\text{ s}$, $T_{adv} = 3617\text{ s}$)		
					0	6	0	3616.40	3615.92	1.000		
					0	7	0	3618.34	3619.95	1.000		
					0	8	0	3615.99	3615.72	1.000		
					0	9	0	3616.65	3613.91	1.001		
		zono	prev_pgd	0.5	0	1	164	3620.83	3618.19	1.001	1.001	1.70
					0	2	164	3623.78	3616.22	1.002		
					0	3	164	1814.89	3617.13	timeouts differ; 2nd attempt required ($T_{std} = 1815\text{ s}$, $T_{adv} = 3617\text{ s}$)		
					0	4	164	1817.37	3616.11	timeouts differ; 2nd attempt required ($T_{std} = 1817\text{ s}$, $T_{adv} = 3616\text{ s}$)		
					0	5	164	1812.49	3616.07	timeouts differ; 2nd attempt required ($T_{std} = 1812\text{ s}$, $T_{adv} = 3616\text{ s}$)		
					0	6	164	3616.40	3617.11	1.000		
					0	7	164	3618.34	3615.84	1.001		
					0	8	164	3615.99	3617.30	1.000		
					0	9	164	3616.65	3615.70	1.000		

Table 43: Architecture **cnn1** — perturbation **linf**(0.1), $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$; continuation of Table 55 (6 combos, 51 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
linf	0.1	zono	prev_pgd	0.5+sg	1	0	164	3621.18	3612.15	1.002	1.001	6.94
					1	2	164	3618.57	3612.33	1.002		
					1	3	164	3615.92	3615.24	1.000		
					1	4	164	3615.75	3611.40	1.001		
					1	5	164	3614.47	3611.86	1.001		
					1	6	164	3613.55	3611.25	1.001		
					1	7	164	3614.64	3612.87	1.000		
					1	8	164	3613.69	3612.00	1.000		
					1	9	164	3613.41	3611.82	1.000		
		zono	prev_pgd	0.0+sg	1	0	0	3621.18	3612.06	1.003	1.001	6.26
					1	2	0	3618.57	3612.88	1.002		
					1	3	0	3615.92	3611.77	1.001		
					1	4	0	3615.75	3613.25	1.001		
					1	5	0	3614.47	3612.90	1.000		
					1	6	0	3613.55	3612.48	1.000		
					1	7	0	3614.64	3613.34	1.000		
					1	8	0	3613.69	3611.43	1.001		
					1	9	0	3613.41	3612.64	1.000		
		interval	prev_pgd	0.0+sg	1	0	0	3621.18	3613.38	1.002	1.000	6.25
					1	2	0	3618.57	3613.50	1.001		
					1	3	0	3615.92	3615.79	1.000		
					1	4	0	3615.75	3614.82	1.000		
					1	5	0	3614.47	3622.41	0.998		
					1	6	0	3613.55	3614.39	1.000		
					1	7	0	3614.64	3616.35	1.000		
					1	8	0	3613.69	3615.62	0.999		
					1	9	0	3613.41	3618.26	0.999		
		zono	prev_pgd	0.0	1	0	0	3621.18	3614.80	1.002	1.000	5.51
					1	2	0	3618.57	3613.69	1.001		
					1	3	0	3615.92	3616.03	1.000		
					1	4	0	3615.75	3618.51	0.999		
					1	5	0	3614.47	3618.19	0.999		
					1	7	0	3614.64	3617.05	0.999		
					1	8	0	3613.69	3616.93	0.999		
					1	9	0	3613.41	3618.28	0.999		
		zono	prev_pgd	0.5	1	0	164	3621.18	3619.51	1.000	0.999	5.76
					1	2	164	3618.57	3615.38	1.001		
					1	3	164	3615.92	3624.90	0.998		
					1	4	164	3615.75	3618.21	0.999		
					1	5	164	3614.47	3613.96	1.000		
					1	6	164	3613.55	3615.56	0.999		
					1	7	164	3614.64	3614.94	1.000		
					1	8	164	3613.69	3617.45	0.999		
					1	9	164	3613.41	3617.57	0.999		
		interval	prev_pgd	0.5+sg	1	0	164	3621.18	3611.89	1.003	0.999	7.04
					1	2	164	3618.57	3624.93	0.998		
					1	3	164	3615.92	3623.97	0.998		
					1	4	164	3615.75	3613.79	1.001		
					1	5	164	3614.47	3618.53	0.999		
					1	6	164	3613.55	3620.03	0.998		
					1	7	164	3614.64	3625.03	0.997		
					1	8	164	3613.69	3614.55	1.000		
					1	9	164	3613.41	3623.33	0.997		

Table 44: Architecture **cnn1** — perturbation **linf**(0.1), $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$; continuation of Table 55 (6 combos, 53 cell rows). Column meanings: see Table 55.

p_type	p_size	bound_tight	varHint	τ	c_src	n_timeout	n_advstd_tighter	avg_gap_std	avg_gap_adv	$ \overline{\text{gap}}_{\text{std}} - \overline{\text{gap}}_{\text{adv}} $
patch	1,14,14,3	zono	no	0.5	2	1	1	14.6465	8.9350	5.7114
translation	1,1	interval	no	0.5	2	1	1	14.6465	8.9350	5.7114
		zono	prev_pgd	0.5+sg	0	1	1	1.0368	0.8000	0.2368
		interval	prev_pgd	0.0+sg	0	1	1	1.0368	0.8790	0.1578
		zono	prev_pgd	0.0+sg	0	1	1	1.0368	0.8975	0.1393
		zono	prev_pgd	0.0	0	1	1	1.0368	1.0063	0.0305
		interval	prev_pgd	0.5+sg	0	1	1	1.0368	0.9231	0.1137
		zono	prev_pgd	0.5	0	1	1	1.0368	0.9033	0.1334
		zono	prev_pgd	0.5+sg	1	1	1	1.4000	1.2838	0.1162
		interval	prev_pgd	0.0+sg	1	2	1	1.2020	1.1591	0.0429
		zono	prev_pgd	0.0+sg	1	1	1	1.4000	1.2953	0.1047
		zono	prev_pgd	0.0	1	3	1	1.0864	1.1148	0.0285
		interval	prev_pgd	0.5+sg	1	1	1	1.4000	0.9835	0.4165
		zono	prev_pgd	0.5	1	2	1	1.1275	1.0881	0.0395
		zono	direct_pgd	0.5	1	1	0	0.8551	1.0302	0.1751
		interval	prev_pgd	0.0	1	1	0	0.8551	0.9293	0.0742
		interval	direct_pgd	0.0	1	1	0	0.8551	0.9989	0.1438
		interval	no	0.5	1	1	0	0.8551	0.9452	0.0901
		zono	direct_pgd	0.0	1	1	0	0.8551	0.9808	0.1257
		interval	direct_pgd	0.5	1	1	0	0.8551	0.9520	0.0970
		interval	prev_pgd	0.5	1	1	0	0.8551	0.8965	0.0414
		interval	no	0.0	1	1	0	0.8551	0.8820	0.0269
		zono	no	0.0	1	1	0	0.8551	0.9702	0.1151
translation	3,1	zono	prev_pgd	0.5	0	1	1	0.7186	0.5716	0.1470
		interval	prev_pgd	0.0+sg	0	1	1	0.7186	0.5746	0.1439
		zono	prev_pgd	0.0+sg	0	1	1	0.7186	0.6075	0.1111
		zono	prev_pgd	0.0	0	1	1	0.7186	0.5987	0.1198
translation	3,3	zono	prev_pgd	0.0	2	1	1	1.0905	1.0000	0.0906
		zono	prev_pgd	0.5	1	6	5	2.4785	2.1352	0.3432
		zono	prev_pgd	0.0	1	5	3	2.5163	2.4536	0.0627
		interval	prev_pgd	0.5+sg	1	3	3	2.6573	1.6197	1.0376
		zono	prev_pgd	0.5+sg	1	3	3	2.6573	2.3186	0.3386
		interval	prev_pgd	0.0+sg	1	3	2	2.6573	2.6416	0.0156
		interval	prev_pgd	0.5	1	3	2	2.2997	2.2728	0.0269
		interval	prev_pgd	0.0+sg	0	1	1	1.5665	1.1223	0.4441
		zono	prev_pgd	0.0	0	1	1	1.5665	1.3996	0.1669
		interval	prev_pgd	0.5+sg	0	1	1	1.5665	1.3468	0.2196
		zono	prev_pgd	0.5+sg	0	1	1	1.5665	1.2466	0.3198
		zono	prev_pgd	0.0+sg	0	1	1	1.5665	1.0617	0.5048
		zono	prev_pgd	0.0+sg	1	3	1	2.6573	6.7066	4.0493
		interval	direct_pgd	0.5	1	3	1	2.2997	2.2363	0.0634
		zono	direct_pgd	0.5	1	2	1	2.3049	9.0515	6.7467
		zono	no	0.0	1	2	1	2.3049	2.3950	0.0901
		interval	prev_pgd	0.0	1	3	1	2.2997	2.3569	0.0572
		zono	prev_pgd	0.5	0	1	0	1.5665	1.6011	0.0346
		zono	direct_pgd	0.0	1	2	0	2.0928	2.2357	0.1430
		interval	direct_pgd	0.0	1	3	0	2.2997	2.4615	0.1618
		interval	no	0.0	1	2	0	2.0928	2.2541	0.1613
		zono	no	0.5	1	3	0	2.2997	2.4861	0.1864
		interval	no	0.5	1	3	0	2.2997	2.5268	0.2271
linf	0.1	zono	prev_pgd	0.5	1	9	8	42.7590	32.1674	10.5916
		interval	prev_pgd	0.5+sg	1	9	8	42.7590	30.8436	11.9154
		zono	prev_pgd	0.5+sg	1	9	8	42.7590	31.1136	11.6454
		zono	prev_pgd	0.0	1	8	8	42.1538	31.1258	11.0281
		interval	prev_pgd	0.0+sg	1	9	8	42.7590	32.8763	9.8827
		zono	prev_pgd	0.0+sg	1	9	7	42.7590	34.1703	8.5887
		zono	prev_pgd	0.5	0	6	6	20.1494	16.7462	3.4033
		interval	prev_pgd	0.5+sg	0	6	6	20.1494	16.7293	3.4201
		zono	prev_pgd	0.5+sg	0	6	6	20.1494	16.9137	3.2358
		zono	prev_pgd	0.0	0	6	2	20.1494	21.6567	1.5072
		interval	prev_pgd	0.0+sg	0	4	2	19.6645	20.6832	1.0187
		zono	prev_pgd	0.0+sg	0	5	1	19.0649	20.9186	1.8537

Table 45: Architecture **cnn1** — gap comparison on cells where both modes hit TIME_LIMIT. One row per (combo, c_{src}), aggregated over all tested c_{tgt} . $n_{\text{advstd_tighter}}$: count where advstd’s $u - l$ gap < standard’s. avg_gap_std : mean $u - l$ per method. $\text{avg_gap_adv} < \text{avg_gap_std}$ in **bold**. Includes every combo with at least one such cell (may include combos absent from Table 55).

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
patch	1,14,14,3	zono	prev_pgd	0.0	0	1	0	3613.40	3615.61	0.999	1.001	0.21
					0	2	0	3618.08	3614.12	1.001		
					0	3	0	3618.72	3614.87	1.001		
					0	4	0	3618.68	3614.33	1.001		
					0	6	0	3618.16	3613.97	1.001		
					0	7	0	3617.37	3616.26	1.000		
					0	8	0	3616.74	3613.47	1.001		
					0	9	0	3618.01	3614.23	1.001		
					interval	prev_pgd	0.5+sg	0	1	21		
0	2	21	3618.08	3615.36				1.001				
0	3	21	3618.72	3614.62				1.001				
0	4	21	3618.68	3614.34				1.001				
0	5	21	3618.15	3614.66				1.001				
0	6	21	3618.16	3615.00				1.001				
0	7	21	3617.37	3616.24				1.000				
0	8	21	3616.74	3614.28				1.001				
0	9	21	3618.01	3614.23				1.001				
zono	prev_pgd	0.5	0	1	23	3613.40	3614.83	1.000	1.001	0.16		
			0	2	23	3618.08	3614.48	1.001				
			0	3	23	3618.72	3614.73	1.001				
			0	4	23	3618.68	3615.05	1.001				
			0	5	23	3618.15	3614.26	1.001				
			0	6	23	3618.16	3614.71	1.001				
			0	7	23	3617.37	3615.19	1.001				
			0	8	23	3616.74	3614.93	1.001				
			0	9	23	3618.01	3614.94	1.001				
interval	prev_pgd	0.0+sg	0	1	0	3613.40	3612.11	1.000	1.001	0.44		
			0	2	0	3618.08	3617.54	1.000				
			0	3	0	3618.72	3615.82	1.001				
			0	4	0	3618.68	3623.14	0.999				
			0	5	0	3618.15	3611.74	1.002				
			0	6	0	3618.16	3613.72	1.001				
			0	7	0	3617.37	3618.46	1.000				
			0	8	0	3616.74	3612.02	1.001				
			0	9	0	3618.01	3611.76	1.002				
zono	prev_pgd	0.0+sg	0	1	0	3613.40	3616.30	0.999	0.999	0.35		
			0	2	0	3618.08	3618.08	1.000				
			0	3	0	3618.72	3619.47	1.000				
			0	4	0	3618.68	3620.48	1.000				
			0	5	0	3618.15	3618.51	1.000				
			0	7	0	3617.37	3621.18	0.999				
			0	8	0	3616.74	3621.69	0.999				
			0	9	0	3618.01	3620.71	0.999				
			zono	prev_pgd	0.5+sg	0	1	23			3613.40	3622.61
0	2	23				3618.08	3622.09	0.999				
0	3	23				3618.72	3620.24	1.000				
0	4	23				3618.68	3621.22	0.999				
0	5	23				3618.15	3620.84	0.999				
0	6	23				3618.16	3620.09	0.999				
0	7	24				3617.37	3625.62	0.998				
0	8	24				3616.74	3622.79	0.998				
0	9	23				3618.01	3623.26	0.999				

Table 46: Architecture **3x50** — every recorded advstd combination at all seeds, all τ values, perturbation **patch**(1,14,14,3), $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (6 combos, 52 cell rows; see Tables labelled **tab:safe_3x50_*** for the other perturbations / c_{src} / τ -buckets of this arch). Rows are tinted by c_{src} (one distinct color per source class); **sp**>1 (advstd faster) in **bold**. Combos sorted by descending mean **sp**; cells within a combo by (c_{src}, c_{tgt}) . Total 620 cell rows across this architecture. Timeout-cell gaps: Table 54.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
patch	1,14,14,3	interval	prev_pgd	0.0+sg	1	0	0	2559.91	1018.63	2.513	1.169	0.38
					1	2	0	3620.66	3612.19	1.002		
					1	3	0	3623.24	3613.49	1.003		
					1	4	0	3619.01	3613.36	1.002		
					1	5	0	3616.74	3612.83	1.001		
					1	6	0	3618.00	3611.94	1.002		
					1	7	0	3616.20	3614.14	1.001		
					1	8	0	3616.24	3613.44	1.001		
		zono	prev_pgd	0.0+sg	1	0	0	2559.91	1431.08	1.789	1.088	0.25
					1	2	0	3620.66	3613.17	1.002		
					1	3	0	3623.24	3615.39	1.002		
					1	4	0	3619.01	3616.05	1.001		
					1	5	0	3616.74	3615.20	1.000		
					1	6	0	3618.00	3615.03	1.001		
					1	7	0	3616.20	3615.62	1.000		
					1	8	0	3616.24	3615.62	1.000		
		zono	prev_pgd	0.5+sg	1	0	23	2559.91	1502.27	1.704	1.079	0.30
					1	2	23	3620.66	3621.53	1.000		
					1	3	23	3623.24	3615.57	1.002		
					1	4	23	3619.01	3612.10	1.002		
					1	5	23	3616.74	3613.19	1.001		
					1	6	23	3618.00	3613.04	1.001		
					1	7	23	3616.20	3612.64	1.001		
					1	8	23	3616.24	3613.44	1.001		
		interval	prev_pgd	0.5+sg	1	0	21	2559.91	1489.14	1.719	1.079	0.24
					1	2	21	3620.66	3617.51	1.001		
					1	3	21	3623.24	3620.42	1.001		
					1	4	21	3619.01	3616.94	1.001		
					1	5	21	3616.74	3620.93	0.999		
					1	6	21	3618.00	3619.65	1.000		
					1	7	21	3616.20	3625.02	0.998		
					1	8	21	3616.24	3625.44	0.997		
		zono	prev_pgd	0.0	1	0	0	2559.91	2399.66	1.067	1.008	0.17
					1	2	0	3620.66	3619.31	1.000		
					1	3	0	3623.24	3617.57	1.002		
					1	4	0	3619.01	3615.59	1.001		
					1	5	0	3616.74	3614.76	1.001		
					1	6	0	3618.00	3614.58	1.001		
					1	7	0	3616.20	3615.88	1.000		
					1	8	0	3616.24	3614.73	1.000		
		zono	prev_pgd	0.5	1	0	23	2559.91	3615.77	0.708	0.964	0.18
					1	2	23	3620.66	3615.00	1.002		
					1	3	23	3623.24	3616.90	1.002		
					1	4	23	3619.01	3613.53	1.002		
					1	5	23	3616.74	3614.71	1.001		
					1	7	23	3616.20	3613.58	1.001		
					1	8	23	3616.24	3614.21	1.001		
					1	9	23	3616.01	3622.82	0.998		

Table 47: Architecture **3x50** — perturbation **patch(1,14,14,3)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$; continuation of Table 39 (6 combos, 53 cell rows). Column meanings: see Table 39.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_src}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_src}$
patch	1,14,14,3	zono	prev_pgd	0.0	2	0	0	3615.42	3614.35	1.000	1.001	0.31
					2	1	0	3615.28	3613.24	1.001		
					2	3	0	3617.35	3613.82	1.001		
					2	4	0	3616.63	3613.35	1.001		
					2	5	0	3615.98	3613.94	1.001		
					2	6	0	3616.01	3613.18	1.001		
					2	7	0	3617.01	3613.41	1.001		
					2	8	0	3617.34	3613.10	1.001		
					2	9	0	3617.45	3613.63	1.001		
		zono	prev_pgd	0.5	2	0	23	3615.42	3614.28	1.000	1.001	0.39
					2	1	23	3615.28	3613.33	1.001		
					2	3	23	3617.35	3613.79	1.001		
					2	4	23	3616.63	3613.05	1.001		
					2	5	23	3615.98	3613.28	1.001		
					2	6	23	3616.01	3612.93	1.001		
					2	7	23	3617.01	3613.67	1.001		
					2	8	23	3617.34	3614.27	1.001		

Table 48: Architecture **3x50** — perturbation **patch(1,14,14,3)**, $c_{\text{src}} = 2$, $\tau \in \{0.0, 0.5\}$; continuation of Table 39 (2 combos, 17 cell rows). Column meanings: see Table 39.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
occ	3,3,5	interval	prev_pgd	0.0+sg	0	1	0	3617.16	3613.13	1.001	1.001	0.02
					0	2	0	3616.52	3612.97	1.001		
					0	3	0	3617.63	3613.18	1.001		
					0	4	0	3616.30	3613.25	1.001		
					0	5	0	3614.28	3613.51	1.000		
					0	6	0	3613.76	3613.53	1.000		
					0	7	0	3614.45	3613.48	1.000		
					0	8	0	3614.15	3613.17	1.000		
					0	9	0	3613.30	3612.22	1.000		
		zono	prev_pgd	0.5+sg	0	1	23	3617.16	3613.82	1.001	1.000	0.04
					0	2	23	3616.52	3614.61	1.001		
					0	3	23	3617.63	3614.92	1.001		
					0	4	23	3616.30	3614.25	1.001		
					0	5	23	3614.28	3614.85	1.000		
					0	6	23	3613.76	3613.91	1.000		
					0	7	23	3614.45	3611.79	1.001		
					0	8	23	3614.15	3613.21	1.000		
					0	9	23	3613.30	3612.40	1.000		
		zono	prev_pgd	0.0	0	1	0	3617.16	3613.82	1.001	1.000	0.01
					0	2	0	3616.52	3614.53	1.001		
					0	3	0	3617.63	3615.49	1.001		
					0	4	0	3616.30	3615.68	1.000		
					0	5	0	3614.28	3615.24	1.000		
					0	6	0	3613.76	3614.99	1.000		
					0	7	0	3614.45	3614.76	1.000		
					0	8	0	3614.15	3615.11	1.000		
					0	9	0	3613.30	3614.53	1.000		
		zono	prev_pgd	0.5	0	1	23	3617.16	3616.80	1.000	1.000	0.03
					0	2	23	3616.52	3619.08	0.999		
					0	3	23	3617.63	3615.89	1.000		
					0	4	23	3616.30	3616.45	1.000		
					0	5	23	3614.28	3615.31	1.000		
					0	6	23	3613.76	3615.10	1.000		
					0	7	23	3614.45	3616.31	0.999		
					0	8	23	3614.15	3616.72	0.999		
					0	9	23	3613.30	3615.43	0.999		
		zono	prev_pgd	0.0+sg	0	1	0	3617.16	3619.92	0.999	0.999	0.01
					0	2	0	3616.52	3621.39	0.999		
					0	3	0	3617.63	3620.93	0.999		
					0	4	0	3616.30	3621.24	0.999		
					0	5	0	3614.28	3617.85	0.999		
					0	6	0	3613.76	3618.37	0.999		
					0	7	0	3614.45	3618.59	0.999		
					0	8	0	3614.15	3618.95	0.999		
					0	9	0	3613.30	3623.08	0.997		
		interval	prev_pgd	0.5+sg	0	1	22	3617.16	3619.28	0.999	0.999	0.04
					0	2	22	3616.52	3620.67	0.999		
					0	3	22	3617.63	3621.29	0.999		
					0	4	22	3616.30	3620.83	0.999		
					0	5	22	3614.28	3621.78	0.998		
					0	6	22	3613.76	3620.62	0.998		
					0	7	22	3614.45	3621.36	0.998		
					0	8	22	3614.15	3619.16	0.999		
					0	9	22	3613.30	3620.28	0.998		

Table 49: Architecture **3x50** — perturbation **occ(3,3,5)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$; continuation of Table 39 (6 combos, 54 cell rows). Column meanings: see Table 39.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
occ	3,3,5	zono	prev_pgd	0.5	1	0	23	3614.65	3616.80	0.999	1.000	0.04
					1	2	23	3616.72	3615.36	1.000		
					1	3	23	3614.32	3614.38	1.000		
					1	4	23	3617.83	3614.02	1.001		
					1	5	23	3617.30	3613.97	1.001		
					1	6	23	3617.82	3615.09	1.001		
					1	7	23	3616.42	3615.07	1.000		
					1	8	23	3616.80	3614.19	1.001		
					1	9	23	3615.23	3613.60	1.000		
		zono	prev_pgd	0.0	1	0	0	3614.65	3616.34	1.000	1.000	0.01
					1	2	0	3616.72	3614.66	1.001		
					1	3	0	3614.32	3615.10	1.000		
					1	4	0	3617.83	3617.32	1.000		
					1	5	0	3617.30	3615.12	1.001		
					1	6	0	3617.82	3616.82	1.000		
					1	7	0	3616.42	3616.61	1.000		
					1	8	0	3616.80	3615.20	1.000		
					1	9	0	3615.23	3615.63	1.000		

Table 50: Architecture **3x50** — perturbation **occ(3,3,5)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$; continuation of Table 39 (2 combos, 18 cell rows). Column meanings: see Table 39.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
occ	5,5,5	zono	prev_pgd	0.5	0	1	23	3615.00	3613.54	1.000	1.000	0.13
					0	2	23	3617.07	3614.46	1.001		
					0	3	23	3615.59	3616.01	1.000		
					0	4	23	3617.41	3616.98	1.000		
					0	5	23	3613.89	3614.94	1.000		
					0	6	23	3613.50	3614.76	1.000		
					0	7	23	3613.21	3614.99	1.000		
					0	8	23	3613.49	3614.44	1.000		
		zono	prev_pgd	0.0+sg	0	1	0	3615.00	3614.65	1.000	1.000	0.08
					0	2	0	3617.07	3614.54	1.001		
					0	3	0	3615.59	3615.05	1.000		
					0	4	0	3617.41	3614.15	1.001		
					0	5	0	3613.89	3614.49	1.000		
					0	6	0	3613.50	3619.43	0.998		
					0	7	0	3613.21	3619.96	0.998		
					0	9	0	3613.05	3616.30	0.999		
		interval	prev_pgd	0.5+sg	0	1	22	3615.00	3612.53	1.001	0.999	0.15
					0	2	22	3617.07	3617.21	1.000		
					0	3	22	3615.59	3618.60	0.999		
					0	4	22	3617.41	3618.82	1.000		
					0	5	22	3613.89	3617.52	0.999		
					0	6	22	3613.50	3618.30	0.999		
					0	7	22	3613.21	3621.70	0.998		
					0	8	22	3613.49	3621.67	0.998		
		zono	prev_pgd	0.0	0	1	0	3615.00	3615.17	1.000	0.999	0.06
					0	2	0	3617.07	3624.41	0.998		
					0	3	0	3615.59	3619.66	0.999		
					0	4	0	3617.41	3618.98	1.000		
					0	5	0	3613.89	3619.65	0.998		
					0	6	0	3613.50	3618.56	0.999		
					0	7	0	3613.21	3619.13	0.998		
					0	8	0	3613.49	3618.71	0.999		
		interval	prev_pgd	0.0+sg	0	1	0	3615.00	3613.44	1.000	0.999	0.10
					0	2	0	3617.07	3619.60	0.999		
					0	3	0	3615.59	3620.91	0.999		
					0	4	0	3617.41	3620.32	0.999		
					0	5	0	3613.89	3618.64	0.999		
					0	6	0	3613.50	3620.62	0.998		
					0	7	0	3613.21	3618.94	0.998		
					0	8	0	3613.49	3620.56	0.998		
		zono	prev_pgd	0.5+sg	0	1	23	3615.00	3618.91	0.999	0.998	0.16
					0	2	23	3617.07	3621.72	0.999		
					0	3	23	3615.59	3621.98	0.998		
					0	4	23	3617.41	3621.35	0.999		
					0	5	23	3613.89	3622.05	0.998		
					0	6	23	3613.50	3621.73	0.998		
					0	7	23	3613.21	3621.58	0.998		
					0	8	23	3613.49	3621.10	0.998		
					0	9	23	3613.05	3621.22	0.998		

Table 51: Architecture **3x50** — perturbation **occ(5,5,5)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$; continuation of Table 39 (6 combos, 53 cell rows). Column meanings: see Table 39.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
occ	5,5,5	zono	prev_pgd	0.5	1	0	23	3623.06	3613.80	1.003	1.001	0.26
					1	2	23	3619.60	3613.44	1.002		
					1	3	23	3620.09	3614.27	1.002		
					1	4	23	3614.67	3613.18	1.000		
					1	5	23	3617.45	3614.97	1.001		
					1	6	23	3617.67	3617.70	1.000		
					1	7	23	3617.14	3616.84	1.000		
					1	8	23	3618.06	3616.82	1.000		
					1	9	23	3616.77	3617.39	1.000		
		zono	prev_pgd	0.0	1	0	0	3623.06	3614.58	1.002	1.000	0.09
					1	3	0	3620.09	3623.61	0.999		
					1	4	0	3614.67	3621.17	0.998		
					1	5	0	3617.45	3621.29	0.999		
					1	6	0	3617.67	3616.43	1.000		
					1	7	0	3617.14	3617.82	1.000		
					1	8	0	3618.06	3618.47	1.000		
					1	9	0	3616.77	3616.67	1.000		

Table 52: Architecture **3x50** — perturbation **occ(5,5,5)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$; continuation of Table 39 (2 combos, 17 cell rows). Column meanings: see Table 39.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	1,1	interval	prev_pgd	0.5+sg	0	1	22	3613.52	3614.30	1.000	1.001	<i>0.37</i>
					0	2	22	3617.74	3614.34	1.001		
					0	3	22	3619.72	3614.58	1.001		
					0	4	22	3617.42	3615.57	1.001		
					0	5	22	3618.73	3613.82	1.001		
					0	6	22	3615.31	3614.80	1.000		
					0	7	22	3619.08	3613.56	1.002		
					0	8	22	3615.77	3613.59	1.001		
					0	9	22	3616.33	3612.96	1.001		
		zono	prev_pgd	0.5	0	1	24	3613.52	3616.59	0.999	1.000	<i>0.24</i>
					0	2	24	3617.74	3617.49	1.000		
					0	3	24	3619.72	3615.15	1.001		
					0	4	24	3617.42	3615.32	1.001		
					0	5	24	3618.73	3615.97	1.001		
					0	6	24	3615.31	3614.34	1.000		
					0	7	24	3619.08	3615.64	1.001		
					0	8	24	3615.77	3617.02	1.000		
					0	9	24	3616.33	3616.08	1.000		
		zono	prev_pgd	0.5+sg	0	1	23	3613.52	3621.61	0.998	1.000	<i>0.34</i>
					0	2	24	3617.74	3623.27	0.998		
					0	3	24	3619.72	3622.96	0.999		
					0	4	24	3617.42	3621.01	0.999		
					0	5	24	3618.73	3611.97	1.002		
					0	6	24	3615.31	3612.42	1.001		
					0	7	24	3619.08	3612.03	1.002		
					0	8	24	3615.77	3617.73	0.999		
					0	9	24	3616.33	3612.20	1.001		
		zono	prev_pgd	0.0	0	1	0	3613.52	3619.92	0.998	1.000	<i>0.18</i>
					0	2	0	3617.74	3616.99	1.000		
					0	3	0	3619.72	3613.59	1.002		
					0	4	0	3617.42	3618.62	1.000		
					0	5	0	3618.73	3621.10	0.999		
					0	7	0	3619.08	3622.81	0.999		
					0	8	0	3615.77	3618.25	0.999		
					0	9	0	3616.33	3614.45	1.001		
		zono	prev_pgd	0.0+sg	0	1	0	3613.52	3613.05	1.000	<i>0.999</i>	<i>0.21</i>
					0	2	0	3617.74	3615.79	1.001		
					0	3	0	3619.72	3615.67	1.001		
					0	4	0	3617.42	3622.29	0.999		
					0	5	0	3618.73	3624.01	0.999		
					0	6	0	3615.31	3625.07	0.997		
					0	7	0	3619.08	3628.42	0.997		
					0	8	0	3615.77	3618.05	0.999		
					0	9	0	3616.33	3624.29	0.998		
		interval	prev_pgd	0.0+sg	0	1	0	3613.52	3612.22	1.000	<i>0.998</i>	<i>0.24</i>
					0	2	0	3617.74	3626.00	0.998		
					0	3	0	3619.72	3624.55	0.999		
					0	4	0	3617.42	3624.61	0.998		
					0	5	0	3618.73	3623.19	0.999		
					0	6	0	3615.31	3623.28	0.998		
					0	7	0	3619.08	3624.56	0.998		
					0	8	0	3615.77	3624.90	0.997		
					0	9	0	3616.33	3626.18	0.997		

Table 53: Architecture **3x50** — perturbation **translation(1,1)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$; continuation of Table 39 (6 combos, 53 cell rows). Column meanings: see Table 39.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	1,1	interval	prev_pgd	0.0+sg	1	0	0	2745.23	2999.62	0.915	1.517	<i>0.02</i>
					1	2	0	3616.54	3619.17	0.999		
					1	3	0	3617.19	3621.75	0.999		
					1	4	0	1263.99	602.11	2.099		
					1	5	0	3617.18	3621.21	0.999		
					1	6	0	3618.79	3553.72	1.018		
					1	7	0	2150.14	1511.97	1.422		
					1	8	0	3617.59	2110.98	1.714		
		zono	prev_pgd	0.5	1	0	0	2745.23	2999.62	0.915	1.517	<i>0.02</i>
					1	2	0	3616.54	3619.17	0.999		
					1	3	0	3617.19	3621.75	0.999		
					1	4	0	1263.99	602.11	2.099		
					1	5	0	3617.18	3621.21	0.999		
					1	6	0	3618.79	3553.72	1.018		
					1	7	0	2150.14	1511.97	1.422		
					1	8	0	3617.59	2110.98	1.714		
		zono	prev_pgd	0.5+sg	1	0	24	2745.23	2786.24	0.985	1.477	<i>0.01</i>
					1	2	24	3616.54	3616.15	1.000		
					1	3	24	3617.19	3618.80	1.000		
					1	4	24	1263.99	702.95	1.798		
					1	6	24	3618.79	3616.69	1.001		
					1	7	24	2150.14	1321.11	1.628		
					1	8	24	3617.59	3180.85	1.137		
					1	9	24	3383.79	1036.82	3.264		
		zono	prev_pgd	0.5+sg	1	0	24	2745.23	2657.94	1.033	1.360	<i>0.02</i>
					1	2	24	3616.54	3615.14	1.000		
					1	3	24	3617.19	3615.10	1.001		
					1	4	24	1263.99	652.21	1.938		
					1	5	24	3617.18	3623.69	0.998		
					1	6	24	3618.79	2541.07	1.424		
					1	7	24	2150.14	1012.84	2.123		
					1	8	24	3617.59	2152.02	1.681		
		interval	prev_pgd	0.5+sg	1	0	22	2745.23	2861.86	0.959	1.328	<i>0.01</i>
					1	2	22	3616.54	3624.31	0.998		
					1	3	22	3617.19	3624.07	0.998		
					1	4	22	1263.99	600.96	2.103		
					1	5	22	3617.18	3620.58	0.999		
					1	6	22	3618.79	3622.94	0.999		
					1	7	22	2150.14	1140.08	1.886		
					1	8	22	3617.59	2152.02	1.681		
		zono	prev_pgd	0.0+sg	1	0	0	2745.23	3055.52	0.898	1.310	<i>0.01</i>
					1	2	0	3616.54	3612.50	1.001		
					1	3	0	3617.19	3612.52	1.001		
					1	4	0	1263.99	613.07	2.062		
					1	5	0	3617.18	3611.40	1.002		
					1	6	0	3618.79	3612.05	1.002		
					1	7	0	2150.14	1156.04	1.860		
					1	8	0	3617.59	2182.23	1.658		
		zono	prev_pgd	0.0	1	0	0	2745.23	2729.61	1.006	1.182	<i>0.01</i>
					1	2	0	3616.54	3613.54	1.001		
					1	3	0	3617.19	3613.38	1.001		
					1	4	0	1263.99	1025.37	1.233		
					1	5	0	3617.18	3614.45	1.001		
					1	6	0	3618.79	3615.28	1.001		
					1	7	0	2150.14	1690.67	1.272		
					1	8	0	3617.59	3287.53	1.100		
		zono	prev_pgd	0.0	1	9	0	3383.79	1670.09	2.026		
					1	0	0	2745.23	2729.61	1.006	1.182	<i>0.01</i>
					1	2	0	3616.54	3613.54	1.001		
					1	3	0	3617.19	3613.38	1.001		
					1	4	0	1263.99	1025.37	1.233		
					1	5	0	3617.18	3614.45	1.001		
					1	6	0	3618.79	3615.28	1.001		
					1	7	0	2150.14	1690.67	1.272		
					1	8	0	3617.59	3287.53	1.100		
					1	9	0	3383.79	1670.09	2.026		

Table 54: Architecture **3x50** — perturbation **translation(1,1)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$; continuation of Table 39 (6 combos, 49 cell rows). Column meanings: see Table 39.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	3,1	interval	prev_pgd	0.5+sg	0	1	22	3616.48	3620.18	0.999	2.461	<i>0.06</i>
					0	2	22	2841.10	695.96	4.082		
					0	3	22	3620.30	1182.79	3.061		
					0	4	22	951.29	572.96	1.660		
					0	5	22	594.58	212.27	2.801		
					0	6	22	368.47	223.20	1.651		
					0	7	22	3618.86	1871.63	1.934		
					0	8	22	654.87	371.60	1.762		
					0	9	22	2494.37	593.71	4.201		
		zono	prev_pgd	0.5+sg	0	1	23	3616.48	3616.00	1.000	2.232	<i>0.06</i>
					0	2	23	2841.10	829.87	3.424		
					0	3	23	3620.30	1602.10	2.260		
					0	4	23	951.29	479.95	1.982		
					0	5	23	594.58	294.61	2.018		
					0	6	23	368.47	362.64	1.016		
					0	7	23	3618.86	931.94	3.883		
					0	8	23	654.87	487.15	1.344		
					0	9	23	2494.37	788.90	3.162		
		interval	prev_pgd	0.0+sg	0	1	0	3616.48	3622.15	0.998	2.191	<i>0.06</i>
					0	2	0	2841.10	1722.85	1.649		
					0	3	0	3620.30	1393.13	2.599		
					0	4	0	951.29	740.31	1.285		
					0	5	0	594.58	278.36	2.136		
					0	6	0	368.47	97.10	3.795		
					0	7	0	3618.86	1220.65	2.965		
					0	8	0	654.87	373.21	1.755		
					0	9	0	2494.37	981.41	2.542		
		zono	prev_pgd	0.5	0	1	23	3616.48	3620.00	0.999	2.007	<i>0.08</i>
					0	2	23	2841.10	937.25	3.031		
					0	3	23	3620.30	1520.63	2.381		
					0	4	23	951.29	692.06	1.375		
					0	5	23	594.58	296.96	2.002		
					0	6	23	368.47	151.81	2.427		
					0	7	23	3618.86	2003.83	1.806		
					0	8	23	654.87	463.82	1.412		
					0	9	23	2494.37	948.86	2.629		
		zono	prev_pgd	0.0+sg	0	1	0	3616.48	3628.04	0.997	1.897	<i>0.06</i>
					0	2	0	2841.10	788.17	3.605		
					0	3	0	3620.30	2549.50	1.420		
					0	4	0	951.29	516.41	1.842		
					0	5	0	594.58	375.18	1.585		
					0	6	0	368.47	305.31	1.207		
					0	7	0	3618.86	1862.28	1.943		
					0	8	0	654.87	423.76	1.545		
					0	9	0	2494.37	852.66	2.925		
		zono	prev_pgd	0.0	0	1	0	3616.48	3616.27	1.000	1.521	<i>0.07</i>
					0	2	0	2841.10	1074.80	2.643		
					0	3	0	3620.30	1503.95	2.407		
					0	4	0	951.29	1371.42	0.694		
					0	5	0	594.58	550.60	1.080		
					0	6	0	368.47	223.48	1.649		
					0	7	0	3618.86	2972.03	1.218		
					0	8	0	654.87	439.35	1.491		
					0	9	0	2494.37	1651.38	1.510		

Table 55: Architecture **3x50** — perturbation **translation(3,1)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$; continuation of Table 39 (6 combos, 54 cell rows). Column meanings: see Table 39.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	3,1	zono	prev_pgd	0.5+sg	1	0	23	2446.84	359.36	6.809	2.671	0.01
					1	2	23	602.86	256.60	2.349		
					1	3	23	778.37	359.48	2.165		
					1	4	23	688.38	291.95	2.358		
					1	5	23	1028.23	466.64	2.203		
					1	6	23	1934.73	546.76	3.539		
					1	7	23	763.76	415.19	1.840		
					1	8	23	611.08	434.45	1.407		
					1	9	23	591.67	433.20	1.366		
		interval	prev_pgd	0.5+sg	1	0	22	2446.84	656.13	3.729	1.848	0.01
					1	2	22	602.86	417.06	1.445		
					1	3	22	778.37	538.07	1.447		
					1	4	22	688.38	377.41	1.824		
					1	5	22	1028.23	690.53	1.489		
					1	6	22	1934.73	720.09	2.687		
					1	7	22	763.76	468.68	1.630		
					1	8	22	611.08	590.63	1.035		
					1	9	22	591.67	440.62	1.343		
		zono	prev_pgd	0.0+sg	1	0	0	2446.84	755.88	3.237	1.688	0.02
					1	2	0	602.86	550.94	1.094		
					1	3	0	778.37	489.12	1.591		
					1	4	0	688.38	671.10	1.026		
					1	5	0	1028.23	865.48	1.188		
					1	6	0	1934.73	592.30	3.266		
					1	7	0	763.76	629.56	1.213		
					1	8	0	611.08	623.35	0.980		
					1	9	0	591.67	370.64	1.596		
		interval	prev_pgd	0.0+sg	1	0	0	2446.84	864.50	2.830	1.623	0.01
					1	2	0	602.86	645.55	0.934		
					1	3	0	778.37	398.52	1.953		
					1	4	0	688.38	489.50	1.406		
					1	5	0	1028.23	553.93	1.856		
					1	6	0	1934.73	1115.22	1.735		
					1	7	0	763.76	380.13	2.009		
					1	8	0	611.08	584.17	1.046		
					1	9	0	591.67	705.20	0.839		
		zono	prev_pgd	0.0	1	0	0	2446.84	945.54	2.588	1.591	0.02
					1	2	0	602.86	518.88	1.162		
					1	3	0	778.37	487.20	1.598		
					1	4	0	688.38	641.22	1.074		
					1	5	0	1028.23	778.98	1.320		
					1	6	0	1934.73	702.71	2.753		
					1	7	0	763.76	388.94	1.964		
					1	8	0	611.08	460.37	1.327		
					1	9	0	591.67	1113.52	0.531		
		zono	prev_pgd	0.5	1	0	23	2446.84	1163.45	2.103	1.341	0.01
					1	2	23	602.86	635.63	0.948		
					1	3	23	778.37	639.51	1.217		
					1	4	23	688.38	825.99	0.833		
					1	5	23	1028.23	825.60	1.245		
					1	6	23	1934.73	777.57	2.488		
					1	7	23	763.76	456.80	1.672		
					1	8	23	611.08	754.48	0.810		
					1	9	23	591.67	786.74	0.752		

Table 56: Architecture **3x50** — perturbation **translation(3,1)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$; continuation of Table 39 (6 combos, 54 cell rows). Column meanings: see Table 39.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	3,3	zono	prev_pgd	0.5+sg	0	1	23	3615.53	3226.85	1.120	2.799	<i>0.02</i>
					0	2	23	474.96	256.57	1.851		
					0	3	23	3616.24	1493.26	2.422		
					0	4	23	130.82	100.84	1.297		
					0	5	23	759.69	244.20	3.111		
					0	6	23	234.28	284.44	0.824		
					0	7	23	746.06	71.58	10.423		
					0	8	23	211.17	76.28	2.768		
					0	9	23	329.82	240.58	1.371		
		zono	prev_pgd	0.0+sg	0	1	0	3615.53	3054.02	1.184	2.174	<i>0.02</i>
					0	2	0	474.96	169.50	2.802		
					0	3	0	3616.24	1385.23	2.611		
					0	4	0	130.82	152.35	0.859		
					0	5	0	759.69	161.82	4.695		
					0	6	0	234.28	197.12	1.189		
					0	7	0	746.06	246.40	3.028		
					0	8	0	211.17	145.75	1.449		
					0	9	0	329.82	188.10	1.753		
		interval	prev_pgd	0.5+sg	0	1	22	3615.53	1603.92	2.254	2.120	<i>0.01</i>
					0	2	22	474.96	190.95	2.487		
					0	3	22	3616.24	1655.76	2.184		
					0	4	22	130.82	117.68	1.112		
					0	5	22	759.69	285.56	2.660		
					0	6	22	234.28	280.72	0.835		
					0	7	22	746.06	167.83	4.445		
					0	8	22	211.17	139.62	1.512		
					0	9	22	329.82	207.65	1.588		
		interval	prev_pgd	0.0+sg	0	1	0	3615.53	2742.04	1.319	2.098	<i>0.01</i>
					0	2	0	474.96	157.31	3.019		
					0	3	0	3616.24	2895.14	1.249		
					0	4	0	130.82	127.89	1.023		
					0	5	0	759.69	129.45	5.869		
					0	6	0	234.28	256.40	0.914		
					0	7	0	746.06	269.52	2.768		
					0	8	0	211.17	153.69	1.374		
					0	9	0	329.82	245.40	1.344		
		zono	prev_pgd	0.5	0	1	23	3615.53	2118.92	1.706	1.670	<i>0.02</i>
					0	2	23	474.96	302.99	1.568		
					0	3	23	3616.24	1521.23	2.377		
					0	4	23	130.82	150.78	0.868		
					0	5	23	759.69	289.17	2.627		
					0	6	23	234.28	355.05	0.660		
					0	7	23	746.06	210.52	3.544		
					0	8	23	211.17	385.03	0.548		
					0	9	23	329.82	290.51	1.135		
		zono	prev_pgd	0.0	0	1	0	3615.53	3613.53	1.001	1.059	<i>0.01</i>
					0	2	0	474.96	350.54	1.355		
					0	3	0	3616.24	3615.12	1.000		
					0	4	0	130.82	317.68	0.412		
					0	5	0	759.69	355.15	2.139		
					0	6	0	234.28	356.72	0.657		
					0	7	0	746.06	447.74	1.666		
					0	8	0	211.17	314.67	0.671		
					0	9	0	329.82	524.00	0.629		

Table 57: Architecture **3x50** — perturbation **translation(3,3)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$; continuation of Table 39 (6 combos, 54 cell rows). Column meanings: see Table 39.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	3,3	interval	prev_pgd	0.0+sg	1	0	0	636.41	392.22	1.623	1.720	<i>0.02</i>
					1	2	0	593.58	278.92	2.128		
					1	3	0	571.64	406.03	1.408		
		interval	prev_pgd	0.5+sg	1	0	22	636.41	486.62	1.308	1.308	<i>0.05</i>
					1	0	23	636.41	687.10	0.926	1.117	<i>0.08</i>
					1	2	23	593.58	266.20	2.230		
		zono	prev_pgd	0.5	1	3	23	571.64	395.88	1.444		
					1	4	23	357.66	368.63	0.970		
					1	5	23	3618.14	3614.10	1.001		
					1	6	23	3619.31	3614.10	1.001		
					1	7	23	497.84	532.92	0.934		
					1	8	23	367.76	524.34	0.701		
					1	9	23	599.02	707.85	0.846		
		zono	prev_pgd	0.0	1	0	0	636.41	791.94	0.804	<i>0.889</i>	<i>0.06</i>
					1	2	0	593.58	502.81	1.181		
					1	3	0	571.64	574.91	0.994		
					1	4	0	357.66	594.03	0.602		
					1	5	0	3618.14	3616.97	1.000		
					1	6	0	3619.31	3621.74	0.999		
					1	7	0	497.84	865.41	0.575		
					1	8	0	367.76	608.59	0.604		
					1	9	0	599.02	482.87	1.241		

Table 58: Architecture **3x50** — perturbation **translation(3,3)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$; continuation of Table 39 (4 combos, 22 cell rows). Column meanings: see Table 39.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
linf	0.1	zono	prev_pgd	0.0+sg	0	1	0	3616.22	3614.03	1.001	1.000	0.55
					0	2	0	3616.40	3614.68	1.000		
					0	3	0	3613.34	3611.84	1.000		
					0	4	0	3614.18	3611.45	1.001		
					0	5	0	3614.08	3611.72	1.001		
					0	6	0	3614.68	3611.97	1.001		
					0	7	0	3614.31	3617.20	0.999		
					0	8	0	3613.60	3616.40	0.999		
					0	9	0	3613.23	3616.64	0.999		
		zono	prev_pgd	0.5	0	1	21	3616.22	3616.75	1.000	1.000	0.46
					0	2	21	3616.40	3613.78	1.001		
					0	3	21	3613.34	3613.74	1.000		
					0	4	21	3614.18	3618.04	0.999		
					0	5	21	3614.08	3617.88	0.999		
					0	6	21	3614.68	3618.06	0.999		
					0	7	21	3614.31	3616.35	0.999		
					0	8	21	3613.60	3614.56	1.000		
					0	9	21	3613.23	3613.42	1.000		
		interval	prev_pgd	0.5+sg	0	1	18	3616.22	3620.12	0.999	0.999	0.80
					0	2	18	3616.40	3615.63	1.000		
					0	3	18	3613.34	3618.16	0.999		
					0	4	18	3614.18	3618.74	0.999		
					0	5	18	3614.08	3615.98	0.999		
					0	6	18	3614.68	3618.12	0.999		
					0	7	18	3614.31	3615.38	1.000		
					0	8	18	3613.60	3617.23	0.999		
					0	9	18	3613.23	3617.23	0.999		
		zono	prev_pgd	0.5+sg	0	1	21	3616.22	3621.67	0.998	0.999	0.71
					0	2	21	3616.40	3617.40	1.000		
					0	3	21	3613.34	3617.91	0.999		
					0	4	21	3614.18	3614.02	1.000		
					0	5	21	3614.08	3618.35	0.999		
					0	6	21	3614.68	3619.22	0.999		
					0	7	21	3614.31	3619.18	0.999		
					0	8	21	3613.60	3616.09	0.999		
					0	9	21	3613.23	3618.80	0.998		
		zono	prev_pgd	0.0	0	1	0	3616.22	3618.82	0.999	0.999	0.50
					0	2	0	3616.40	3614.84	1.000		
					0	3	0	3613.34	3622.19	0.998		
					0	4	0	3614.18	3621.24	0.998		
					0	5	0	3614.08	3620.32	0.998		
					0	6	0	3614.68	3621.28	0.998		
					0	7	0	3614.31	3619.55	0.999		
					0	8	0	3613.60	3618.87	0.999		
					0	9	0	3613.23	3613.78	1.000		
		interval	prev_pgd	0.0+sg	0	1	0	3616.22	3613.64	1.001	0.998	0.53
					0	2	0	3616.40	3622.53	0.998		
					0	3	0	3613.34	3623.38	0.997		
					0	4	0	3614.18	3623.85	0.997		
					0	5	0	3614.08	3622.39	0.998		
					0	6	0	3614.68	3622.20	0.998		
					0	7	0	3614.31	3618.60	0.999		
					0	8	0	3613.60	3620.57	0.998		
					0	9	0	3613.23	3616.04	0.999		

Table 59: Architecture **3x50** — perturbation **linf(0.1)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$; continuation of Table 39 (6 combos, 53 cell rows). Column meanings: see Table 39.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \overline{\delta}_{std} - \overline{\delta}_{adv} _{c_src}$
linf	0.1	zono	prev_pgd	0.0	1	0	0	3615.27	3616.96	1.000	1.000	0.33
					1	2	0	3618.23	3615.99	1.001		
					1	3	0	3613.90	3615.39	1.000		
					1	4	0	3613.86	3616.25	0.999		
					1	6	0	3613.51	3616.48	0.999		
					1	7	0	3611.58	3615.70	0.999		
					1	8	0	3613.61	3614.42	1.000		
					1	9	0	3613.16	3614.48	1.000		
		zono	prev_pgd	0.5	1	0	21	3615.27	3617.87	0.999	0.999	1.36
					1	2	21	3618.23	3619.99	1.000		
					1	3	21	3613.90	3617.14	0.999		
					1	4	21	3613.86	3614.71	1.000		
					1	5	21	3614.68	3615.00	1.000		
					1	6	21	3613.51	3615.49	0.999		
					1	7	21	3611.58	3618.77	0.998		
					1	8	21	3613.61	3614.24	1.000		
					1	9	21	3613.16	3614.20	1.000		

Table 60: Architecture **3x50** — perturbation **linf(0.1)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$; continuation of Table 39 (2 combos, 17 cell rows). Column meanings: see Table 39.

p_type	p_size	bound_tight	varHint	τ	c_src	n_timeout	n_advstd_tighter	avg_gap_std	avg_gap_adv	$ \text{gap}_{\text{std}} - \text{gap}_{\text{adv}} $
patch	1,14,14,3	interval	prev_pgd	0.0+sg	0	9	9	4.2376	3.3668	0.8709
		interval	prev_pgd	0.5+sg	0	9	9	4.2376	3.0433	1.1944
		zono	prev_pgd	0.0+sg	0	8	8	3.8954	3.1973	0.6980
		zono	prev_pgd	0.5+sg	0	9	8	4.2376	3.8012	0.4364
		interval	prev_pgd	0.0+sg	1	8	8	3.0814	2.2257	0.8558
		zono	prev_pgd	0.0	2	9	8	5.7273	5.1289	0.5984
		zono	prev_pgd	0.5	2	8	8	5.6432	4.8723	0.7709
		zono	prev_pgd	0.0+sg	1	8	7	3.0814	2.6551	0.4263
		interval	prev_pgd	0.5+sg	1	8	7	3.0814	2.6517	0.4297
		zono	prev_pgd	0.5+sg	1	8	7	3.0814	2.5464	0.5350
		zono	prev_pgd	0.5	0	9	6	4.2376	4.0335	0.2042
		zono	prev_pgd	0.0	1	8	5	3.0814	3.0952	0.0137
		zono	prev_pgd	0.5	1	7	5	3.3334	3.1975	0.1360
		zono	prev_pgd	0.0	0	8	3	4.1406	4.4030	0.2623
		occ	prev_pgd	0.5+sg	0	9	9	0.1882	0.1151	0.0732
		zono	prev_pgd	0.5+sg	0	9	9	0.1882	0.1097	0.0786
occ	3,3,5	zono	prev_pgd	0.5	1	9	9	0.1732	0.0958	0.0774
		zono	prev_pgd	0.5	0	9	8	0.1882	0.1385	0.0497
		zono	prev_pgd	0.0	0	9	7	0.1882	0.1680	0.0203
		zono	prev_pgd	0.0+sg	0	9	6	0.1882	0.1786	0.0097
		interval	prev_pgd	0.0+sg	0	9	5	0.1882	0.1821	0.0061
		zono	prev_pgd	0.0	1	9	4	0.1732	0.1712	0.0020
		occ	prev_pgd	0.5+sg	0	9	9	1.3688	1.0441	0.3247
		zono	prev_pgd	0.5	1	9	9	1.4283	0.9056	0.5226
		interval	prev_pgd	0.5+sg	0	9	8	1.3688	1.0684	0.3004
		zono	prev_pgd	0.5	0	9	8	1.3688	1.1494	0.2193
		interval	prev_pgd	0.0+sg	0	9	7	1.3688	1.2542	0.1146
		zono	prev_pgd	0.0	1	8	7	1.4783	1.3223	0.1560
		zono	prev_pgd	0.0+sg	0	8	5	1.3856	1.2628	0.1227
		zono	prev_pgd	0.0	0	9	4	1.3688	1.3953	0.0265
		translation	prev_pgd	0.5+sg	0	9	9	1.9030	1.2254	0.6776
		interval	prev_pgd	0.0+sg	0	9	9	1.9030	1.4188	0.4842
translation	1,1	interval	prev_pgd	0.5+sg	0	9	9	1.9030	1.1729	0.7301
		zono	prev_pgd	0.0+sg	0	9	9	1.9030	1.4731	0.4299
		zono	prev_pgd	0.5	0	9	9	1.9030	1.4212	0.4819
		zono	prev_pgd	0.0	0	8	6	1.9992	1.6741	0.3251
		zono	prev_pgd	0.0+sg	1	4	4	0.2851	0.2513	0.0337
		zono	prev_pgd	0.5+sg	1	3	3	0.3312	0.2455	0.0857
		zono	prev_pgd	0.0	1	4	3	0.2851	0.2631	0.0219
		interval	prev_pgd	0.0+sg	1	3	3	0.3312	0.2648	0.0664
		interval	prev_pgd	0.5+sg	1	4	3	0.2851	0.2710	0.0141
		zono	prev_pgd	0.5	1	3	1	0.2520	0.2654	0.0134
		translation	prev_pgd	0.5+sg	0	1	1	1.8819	1.1274	0.7545
		zono	prev_pgd	0.0+sg	0	1	1	1.8819	1.1351	0.7468
		zono	prev_pgd	0.5+sg	0	1	1	1.8819	1.2142	0.6676
		zono	prev_pgd	0.5	0	1	1	1.8819	0.8027	1.0792
		zono	prev_pgd	0.0	0	1	1	1.8819	0.8896	0.9922
		interval	prev_pgd	0.0+sg	0	1	1	1.8819	1.0667	0.8152
translation	3,3	zono	prev_pgd	0.0	0	2	2	0.2853	0.2624	0.0229
		zono	prev_pgd	0.5	1	2	2	0.7587	0.1981	0.5606
		zono	prev_pgd	0.0	1	2	1	0.7587	0.3189	0.4399
		linf	prev_pgd	0.5+sg	0	9	9	9.5874	8.1594	1.4280
linf	0.1	zono	prev_pgd	0.5	0	9	9	9.5874	8.6600	0.9274
		interval	prev_pgd	0.5+sg	0	8	8	9.7615	8.1628	1.5987
		zono	prev_pgd	0.0+sg	0	9	8	9.5874	8.5966	0.9907
		interval	prev_pgd	0.0+sg	0	9	7	9.5874	8.6065	0.9808
		zono	prev_pgd	0.0	0	9	7	9.5874	8.8719	0.7155
		zono	prev_pgd	0.0	1	8	6	4.0697	3.6442	0.4255
		zono	prev_pgd	0.5	1	9	5	4.0109	5.6863	1.6755

Table 61: Architecture **3x50** — gap comparison on cells where both modes hit TIME_LIMIT. One row per (combo, c_{src}), aggregated over all tested c_{tgt} . $n_{\text{advstd_tighter}}$: count where advstd’s $u - l$ gap < standard’s. avg_gap_std : mean $u - l$ per method. $\text{avg_gap_adv} < \text{avg_gap_std}$ in **bold**. Includes every combo with at least one such cell (may include combos absent from Table 39).

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
patch	1,14,14,3	interval	prev_pgd	0.5	0	3	0	14.75	12.51	1.179	1.145	0.01
					0	4	0	15.24	12.50	1.219		
					0	5	0	16.13	12.50	1.290		
					0	6	0	13.07	12.83	1.019		
					0	7	0	13.37	12.01	1.113		
					0	8	0	13.05	12.45	1.048		
	zono	no	no	0.5	0	3	0	14.75	12.02	1.227	1.134	0.01
					0	4	0	15.24	13.78	1.106		
					0	5	0	16.13	12.36	1.305		
					0	6	0	13.07	13.36	0.978		
					0	7	0	13.37	12.50	1.070		
					0	8	0	13.05	11.70	1.115		
	interval	direct_pgd	direct_pgd	0.5	0	3	0	14.75	12.90	1.143	1.119	0.01
					0	4	0	15.24	12.55	1.214		
					0	5	0	16.13	12.82	1.258		
					0	6	0	13.07	13.27	0.985		
					0	7	0	13.37	12.52	1.068		
					0	8	0	13.05	12.49	1.045		
	interval	no	no	0.5	0	3	0	14.75	12.87	1.146	1.113	0.01
					0	4	0	15.24	12.77	1.193		
					0	5	0	16.13	12.87	1.253		
					0	6	0	13.07	12.95	1.009		
					0	7	0	13.37	11.02	1.213		
					0	8	0	13.05	15.13	0.863		
	interval	direct_pgd	direct_pgd	0.0	0	3	0	14.75	12.51	1.179	1.103	0.01
					0	4	0	15.24	12.85	1.186		
					0	5	0	16.13	13.00	1.241		
					0	6	0	13.07	13.27	0.985		
					0	7	0	13.37	13.26	1.008		
					0	8	0	13.05	12.80	1.020		
	zono	direct_pgd	direct_pgd	0.5	0	3	0	14.75	12.64	1.167	1.098	0.01
					0	4	0	15.24	12.68	1.202		
					0	5	0	16.13	12.63	1.277		
					0	6	0	13.07	13.68	0.955		
					0	7	0	13.37	14.30	0.935		
					0	8	0	13.05	12.40	1.052		

Table 62: Architecture **3x10** — every recorded advstd combination at all seeds, all τ values, perturbation **patch**(1,14,14,3), $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 1/3, combos ranked 1–6 of 16 by mean **sp**) (6 combos, 36 cell rows; see Tables labelled **tab:safe_3x10_*** for the other perturbations / c_{src} / τ -buckets of this arch). Rows are tinted by c_{src} (one distinct color per source class); **sp** > 1 (advstd faster) in **bold**. Combos sorted by descending mean **sp**; cells within a combo by (c_{src}, c_{tgt}). Total 1224 cell rows across this architecture.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_src}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_src}$
patch	1,14,14,3	zono	no	0.0	0	3	0	14.75	12.33	1.196	1.072	0.01
					0	4	0	15.24	14.17	1.076		
					0	5	0	16.13	13.72	1.176		
					0	6	0	13.07	12.88	1.015		
					0	7	0	13.37	13.88	0.963		
					0	8	0	13.05	12.93	1.009		
		zono	prev_pgd	0.0	0	3	0	14.75	13.67	1.079	1.072	0.01
					0	4	0	15.24	13.98	1.090		
					0	5	0	16.13	13.34	1.209		
					0	6	0	13.07	13.51	0.967		
					0	7	0	13.37	12.40	1.078		
					0	8	0	13.05	12.92	1.010		
		zono	prev_pgd	0.5	0	3	0	14.75	13.68	1.078	1.069	0.01
					0	4	0	15.24	14.87	1.025		
					0	5	0	16.13	13.04	1.237		
					0	6	0	13.07	12.64	1.034		
					0	7	0	13.37	12.20	1.096		
					0	8	0	13.05	13.82	0.944		
		zono	direct_pgd	0.0	0	3	0	14.75	14.53	1.015	1.020	0.01
					0	4	0	15.24	16.43	0.928		
					0	5	0	16.13	12.98	1.243		
					0	6	0	13.07	13.51	0.967		
					0	7	0	13.37	13.82	0.967		
					0	8	0	13.05	13.02	1.002		
		interval	no	0.0	0	3	0	14.75	14.00	1.054	1.020	0.01
					0	4	0	15.24	13.15	1.159		
					0	5	0	16.13	15.18	1.063		
					0	6	0	13.07	14.08	0.928		
					0	7	0	13.37	12.98	1.030		
					0	8	0	13.05	14.68	0.889		
		interval	prev_pgd	0.0+sg	0	1	0	17.14	19.18	0.894	0.997	0.01
					0	2	0	21.33	12.14	1.757		
					0	3	0	14.75	12.84	1.149		
					0	4	0	15.24	17.57	0.867		
					0	5	0	16.13	17.30	0.932		
					0	6	0	13.07	18.22	0.717		
					0	7	0	13.37	19.20	0.696		
					0	8	0	13.05	19.91	0.655		
					0	9	0	20.34	15.60	1.304		

Table 63: Architecture **3x10** — perturbation **patch(1,14,14,3)**, $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5\}$ (part 2/3, combos ranked 7–12 of 16 by mean **sp**); continuation of Table 5 (6 combos, 39 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
patch	1,14,14,3	interval	prev_pgd	0.0	0	3	0	14.75	20.83	0.708	0.981	0.01
					0	4	0	15.24	17.97	0.848		
					0	5	0	16.13	12.34	1.307		
					0	6	0	13.07	12.80	1.021		
					0	7	0	13.37	11.89	1.124		
					0	8	0	13.05	14.86	0.878		
	interval	prev_pgd	0.5+sg	0	1	0	17.14	21.88	0.783	0.924	0.01	
				0	2	0	21.33	18.53	1.151			
				0	3	0	14.75	19.36	0.762			
				0	4	0	15.24	16.68	0.914			
				0	5	0	16.13	15.63	1.032			
				0	6	0	13.07	15.86	0.824			
				0	7	0	13.37	14.14	0.946			
				0	8	0	13.05	16.95	0.770			
	zono	prev_pgd	0.5+sg	0	9	0	20.34	17.93	1.134	0.868	0.01	
				0	1	0	17.14	20.79	0.824			
				0	2	0	21.33	19.61	1.088			
				0	3	0	14.75	19.82	0.744			
				0	4	0	15.24	18.42	0.827			
				0	5	0	16.13	19.14	0.843			
				0	6	0	13.07	20.07	0.651			
				0	7	0	13.37	19.97	0.670			
	zono	prev_pgd	0.0+sg	0	8	0	13.05	15.65	0.834	0.861	0.01	
				0	9	0	20.34	15.28	1.331			
				0	1	0	17.14	19.51	0.879			
				0	2	0	21.33	18.72	1.139			
				0	3	0	14.75	19.93	0.740			
				0	4	0	15.24	19.34	0.788			
				0	5	0	16.13	17.98	0.897			
				0	6	0	13.07	22.83	0.572			
	zono	prev_pgd	0.0+sg	0	7	0	13.37	20.00	0.668	0.861	0.01	
				0	8	0	13.05	19.93	0.655			
				0	9	0	20.34	14.39	1.413			

Table 64: Architecture **3x10** — perturbation **patch(1,14,14,3)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 3/3, combos ranked 13–16 of 16 by mean **sp**); continuation of Table 5 (4 combos, 33 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_src}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_src}$
patch	1,14,14,3	interval	prev_pgd	0.0+sg	1	0	0	19.11	16.50	1.158	1.509	<i>0.02</i>
					1	2	0	12.95	11.51	1.125		
					1	3	0	22.55	13.76	1.639		
					1	4	0	23.72	16.32	1.453		
					1	5	0	26.84	14.25	1.884		
					1	6	0	25.81	13.62	1.895		
					1	7	0	26.70	13.31	2.006		
					1	8	0	22.57	15.31	1.474		
					1	9	0	14.65	15.48	0.946		
		interval	prev_pgd	0.0	1	3	0	22.55	15.39	1.465	1.504	<i>0.02</i>
					1	4	0	23.72	16.10	1.473		
					1	5	0	26.84	17.94	1.496		
					1	6	0	25.81	15.82	1.631		
					1	7	0	26.70	16.51	1.617		
					1	8	0	22.57	16.85	1.339		
		interval	prev_pgd	0.5	1	3	0	22.55	15.48	1.457	1.499	<i>0.02</i>
					1	4	0	23.72	16.64	1.425		
					1	5	0	26.84	17.57	1.528		
					1	6	0	25.81	16.05	1.608		
					1	7	0	26.70	16.37	1.631		
					1	8	0	22.57	16.80	1.343		
		interval	prev_pgd	0.5+sg	1	0	0	19.11	18.76	1.019	1.494	<i>0.02</i>
					1	2	0	12.95	17.20	0.753		
					1	3	0	22.55	13.19	1.710		
					1	4	0	23.72	13.35	1.777		
					1	5	0	26.84	12.90	2.081		
					1	6	0	25.81	15.85	1.628		
					1	7	0	26.70	15.58	1.714		
					1	8	0	22.57	12.72	1.774		
					1	9	0	14.65	14.82	0.989		
		zono	direct_pgd	0.0	1	3	0	22.55	15.40	1.464	1.484	<i>0.01</i>
					1	4	0	23.72	16.28	1.457		
					1	5	0	26.84	18.67	1.438		
					1	6	0	25.81	16.28	1.585		
					1	7	0	26.70	16.47	1.621		
					1	8	0	22.57	16.83	1.341		
		zono	prev_pgd	0.0	1	3	0	22.55	17.33	1.301	1.476	<i>0.02</i>
					1	4	0	23.72	16.52	1.436		
					1	5	0	26.84	17.18	1.562		
					1	6	0	25.81	16.15	1.598		
					1	7	0	26.70	16.30	1.638		
					1	8	0	22.57	17.12	1.318		

Table 65: Architecture **3x10** — perturbation **patch(1,14,14,3)**, $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5\}$ (part 1/3, combos ranked 1–6 of 16 by mean sp); continuation of Table 5 (6 combos, 42 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_src}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_src}$
patch	1,14,14,3	interval	direct_pgd	0.0	1	3	0	22.55	16.11	1.400	1.471	0.01
					1	4	0	23.72	16.96	1.399		
					1	5	0	26.84	19.21	1.397		
					1	6	0	25.81	15.59	1.656		
					1	7	0	26.70	15.80	1.690		
					1	8	0	22.57	17.53	1.288		
					1	3	0	22.55	16.41	1.374	1.464	0.02
					1	4	0	23.72	16.22	1.462		
					1	5	0	26.84	18.40	1.459		
					1	6	0	25.81	16.24	1.589		
		zono	prev_pgd	0.5	1	7	0	26.70	16.28	1.640		
					1	8	0	22.57	17.92	1.259		
		zono	direct_pgd	0.5	1	3	0	22.55	15.98	1.411	1.461	0.01
					1	4	0	23.72	15.96	1.486		
					1	5	0	26.84	18.85	1.424		
					1	6	0	25.81	15.97	1.616		
					1	7	0	26.70	16.23	1.645		
					1	8	0	22.57	19.03	1.186		
		interval	direct_pgd	0.5	1	3	0	22.55	16.00	1.409	1.447	0.01
					1	4	0	23.72	16.80	1.412		
					1	5	0	26.84	18.52	1.449		
					1	6	0	25.81	16.12	1.601		
					1	7	0	26.70	17.76	1.503		
					1	8	0	22.57	17.27	1.307		
		interval	no	0.5	1	3	0	22.55	17.04	1.323	1.430	0.02
					1	4	0	23.72	17.04	1.392		
					1	5	0	26.84	19.35	1.387		
					1	6	0	25.81	16.77	1.539		
					1	7	0	26.70	15.49	1.724		
					1	8	0	22.57	18.57	1.215		
		interval	no	0.0	1	3	0	22.55	17.29	1.304	1.412	0.02
					1	4	0	23.72	16.20	1.464		
					1	5	0	26.84	19.44	1.381		
					1	6	0	25.81	16.93	1.525		
					1	7	0	26.70	17.17	1.555		
					1	8	0	22.57	18.12	1.246		

Table 66: Architecture **3x10** — perturbation **patch(1,14,14,3)**, $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5\}$ (part 2/3, combos ranked 7–12 of 16 by mean **sp**); continuation of Table 5 (6 combos, 36 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
patch	1,14,14,3	zono	no	0.0	1	3	0	22.55	16.54	1.363	1.411	0.02
					1	4	0	23.72	16.41	1.445		
					1	5	0	26.84	19.87	1.351		
					1	6	0	25.81	16.61	1.554		
					1	7	0	26.70	17.87	1.494		
					1	8	0	22.57	17.93	1.259		
		zono	no	0.5	1	3	0	22.55	20.87	1.080	1.360	0.02
					1	4	0	23.72	16.98	1.397		
					1	5	0	26.84	18.94	1.417		
					1	6	0	25.81	18.42	1.401		
					1	7	0	26.70	15.91	1.678		
					1	8	0	22.57	19.03	1.186		
		zono	prev_pgd	0.5+sg	1	0	0	19.11	19.84	0.963	1.305	0.02
					1	2	0	12.95	19.76	0.655		
					1	3	0	22.55	17.43	1.294		
					1	4	0	23.72	17.73	1.338		
					1	5	0	26.84	18.19	1.476		
					1	6	0	25.81	15.11	1.708		
					1	7	0	26.70	14.57	1.833		
					1	8	0	22.57	15.67	1.440		
					1	9	0	14.65	14.06	1.042		
		zono	prev_pgd	0.0+sg	1	0	0	19.11	18.16	1.052	1.227	0.02
					1	2	0	12.95	21.00	0.617		
					1	3	0	22.55	17.81	1.266		
					1	4	0	23.72	17.99	1.319		
					1	5	0	26.84	18.87	1.422		
					1	6	0	25.81	18.22	1.417		
					1	7	0	26.70	20.32	1.314		
					1	8	0	22.57	14.29	1.579		
					1	9	0	14.65	13.89	1.055		

Table 67: Architecture **3x10** — perturbation **patch(1,14,14,3)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$ (part 3/3, combos ranked 13–16 of 16 by mean sp); continuation of Table 5 (4 combos, 30 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
occ	3,3,5	interval	prev_pgd	0.5+sg	0	1	0	21.45	17.68	1.213	1.022	0.00
					0	2	0	20.52	19.92	1.030		
					0	3	0	14.59	17.78	0.821		
					0	4	0	16.49	16.41	1.005		
					0	5	0	14.70	15.91	0.924		
					0	6	0	13.77	15.90	0.866		
					0	7	0	15.26	14.12	1.081		
					0	8	0	13.88	14.20	0.977		
					0	9	0	18.70	14.58	1.283		
		zono	prev_pgd	0.5+sg	0	1	0	21.45	17.45	1.229	1.012	0.00
					0	2	0	20.52	18.15	1.131		
					0	3	0	14.59	16.80	0.868		
					0	4	0	16.49	15.77	1.046		
					0	5	0	14.70	14.71	0.999		
					0	6	0	13.77	16.08	0.856		
					0	7	0	15.26	16.52	0.924		
					0	8	0	13.88	17.49	0.794		
					0	9	0	18.70	14.86	1.258		
		zono	prev_pgd	0.0+sg	0	1	0	21.45	18.96	1.131	0.996	0.00
					0	2	0	20.52	21.32	0.962		
					0	3	0	14.59	16.93	0.862		
					0	4	0	16.49	17.14	0.962		
					0	5	0	14.70	17.29	0.850		
					0	6	0	13.77	15.66	0.879		
					0	7	0	15.26	13.28	1.149		
					0	8	0	13.88	15.10	0.919		
					0	9	0	18.70	14.99	1.247		
		interval	prev_pgd	0.0+sg	0	1	0	21.45	18.31	1.171	0.943	0.00
					0	2	0	20.52	19.78	1.037		
					0	3	0	14.59	18.39	0.793		
					0	4	0	16.49	17.21	0.958		
					0	5	0	14.70	19.68	0.747		
					0	6	0	13.77	18.82	0.732		
					0	7	0	15.26	17.37	0.879		
					0	8	0	13.88	16.39	0.847		
					0	9	0	18.70	14.14	1.322		

Table 68: Architecture **3x10** — perturbation **occ(3,3,5)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$; continuation of Table 5 (4 combos, 36 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
occ	3,3,5	interval	prev_pgd	0.0+sg	1	0	0	20.57	11.19	1.838	1.722	0.02
					1	2	0	19.27	12.28	1.569		
					1	3	0	26.38	12.68	2.080		
					1	4	0	22.64	11.99	1.888		
					1	5	0	26.24	11.84	2.216		
					1	6	0	24.21	11.55	2.096		
					1	7	0	14.84	15.69	0.946		
					1	8	0	24.15	15.94	1.515		
					1	9	0	16.72	12.36	1.353		
		zono	prev_pgd	0.5+sg	1	0	0	20.57	11.60	1.773	1.695	0.02
					1	2	0	19.27	14.71	1.310		
					1	3	0	26.38	12.51	2.109		
					1	4	0	22.64	11.30	2.004		
					1	5	0	26.24	12.79	2.052		
					1	6	0	24.21	13.52	1.791		
					1	7	0	14.84	13.25	1.120		
					1	8	0	24.15	12.33	1.959		
					1	9	0	16.72	14.72	1.136		
		interval	prev_pgd	0.5+sg	1	0	0	20.57	12.76	1.612	1.520	0.02
					1	2	0	19.27	15.64	1.232		
					1	3	0	26.38	19.26	1.370		
					1	4	0	22.64	18.34	1.234		
					1	5	0	26.24	13.14	1.997		
					1	6	0	24.21	13.26	1.826		
					1	7	0	14.84	12.40	1.197		
					1	8	0	24.15	11.80	2.047		
					1	9	0	16.72	14.30	1.169		
		zono	prev_pgd	0.0+sg	1	0	0	20.57	15.99	1.286	1.464	0.02
					1	2	0	19.27	17.27	1.116		
					1	3	0	26.38	18.67	1.413		
					1	4	0	22.64	14.33	1.580		
					1	5	0	26.24	14.05	1.868		
					1	6	0	24.21	13.41	1.805		
					1	7	0	14.84	13.50	1.099		
					1	8	0	24.15	12.81	1.885		
					1	9	0	16.72	14.85	1.126		

Table 69: Architecture **3x10** — perturbation **occ(3,3,5)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$; continuation of Table 5 (4 combos, 36 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_{\text{src}}}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_{\text{src}}}$
occ	5,5,5	zono	no	0.5	0	3	0	21.76	13.58	1.602	1.353	0.01
					0	4	0	15.18	11.43	1.328		
					0	5	0	15.59	11.55	1.350		
					0	6	0	14.17	11.33	1.251		
					0	7	0	17.19	11.29	1.523		
					0	8	0	14.13	13.29	1.063		
		zono	no	0.0	0	3	0	21.76	14.30	1.522	1.169	0.01
					0	4	0	15.18	14.98	1.013		
					0	5	0	15.59	13.52	1.153		
					0	6	0	14.17	12.85	1.103		
					0	7	0	17.19	13.16	1.306		
					0	8	0	14.13	15.39	0.918		
		interval	no	0.5	0	3	0	21.76	14.62	1.488	1.115	0.01
					0	4	0	15.18	14.46	1.050		
					0	5	0	15.59	16.53	0.943		
					0	6	0	14.17	14.13	1.003		
					0	7	0	17.19	14.17	1.213		
					0	8	0	14.13	14.21	0.994		
		interval	no	0.0	0	3	0	21.76	17.07	1.275	1.106	0.01
					0	4	0	15.18	13.35	1.137		
					0	5	0	15.59	14.81	1.053		
					0	6	0	14.17	13.87	1.022		
					0	7	0	17.19	14.10	1.219		
					0	8	0	14.13	15.15	0.933		
		zono	prev_pgd	0.5+sg	0	1	0	23.57	17.23	1.368	1.101	0.01
					0	2	0	25.54	18.15	1.407		
					0	3	0	21.76	20.26	1.074		
					0	4	0	15.18	17.71	0.857		
					0	5	0	15.59	17.86	0.873		
					0	6	0	14.17	15.65	0.905		
					0	7	0	17.19	16.09	1.068		
					0	8	0	14.13	17.96	0.787		
					0	9	0	22.61	14.39	1.571		
		interval	prev_pgd	0.0+sg	0	1	0	23.57	17.34	1.359	1.063	0.01
					0	2	0	25.54	16.43	1.554		
					0	3	0	21.76	21.73	1.001		
					0	4	0	15.18	18.23	0.833		
					0	5	0	15.59	17.04	0.915		
					0	6	0	14.17	17.56	0.807		
					0	7	0	17.19	18.34	0.937		
					0	8	0	14.13	18.91	0.747		
					0	9	0	22.61	15.97	1.416		

Table 70: Architecture **3x10** — perturbation **occ(5,5,5)**, $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5\}$ (part 1/3, combos ranked 1–6 of 16 by mean **sp**); continuation of Table 5 (6 combos, 42 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
occ	5,5,5	zono	prev_pgd	0.0+sg	0	1	0	23.57	17.22	1.369	1.013	0.01
					0	2	0	25.54	20.15	1.267		
					0	3	0	21.76	21.41	1.016		
					0	4	0	15.18	18.98	0.800		
					0	5	0	15.59	20.59	0.757		
					0	6	0	14.17	19.04	0.744		
					0	7	0	17.19	17.34	0.991		
					0	8	0	14.13	18.36	0.770		
					0	9	0	22.61	16.16	1.399		
		zono	direct_pgd	0.0	0	3	0	21.76	18.52	1.175	0.988	0.01
					0	4	0	15.18	14.80	1.026		
					0	5	0	15.59	15.59	1.000		
					0	6	0	14.17	14.84	0.955		
					0	7	0	17.19	15.37	1.118		
		interval	prev_pgd	0.5+sg	0	8	0	14.13	21.58	0.655		0.01
					0	1	0	23.57	18.21	1.294	0.985	
					0	2	0	25.54	21.91	1.166		
					0	3	0	21.76	24.27	0.897		
					0	4	0	15.18	18.45	0.823		
					0	5	0	15.59	17.04	0.915		
					0	6	0	14.17	16.27	0.871		
					0	7	0	17.19	17.96	0.957		
					0	8	0	14.13	20.40	0.693		
					0	9	0	22.61	18.13	1.247		
		interval	direct_pgd	0.0	0	3	0	21.76	17.22	1.264	0.980	0.01
					0	4	0	15.18	16.92	0.897		
					0	5	0	15.59	15.95	0.977		
					0	6	0	14.17	15.86	0.893		
					0	7	0	17.19	16.75	1.026		
		zono	direct_pgd	0.5	0	8	0	14.13	17.22	0.821		0.01
					0	3	0	21.76	17.12	1.271	0.967	
					0	4	0	15.18	15.65	0.970		
					0	5	0	15.59	17.05	0.914		
					0	6	0	14.17	16.81	0.843		
		interval	direct_pgd	0.5	0	7	0	17.19	17.00	1.011		0.01
					0	8	0	14.13	17.89	0.790		
					0	3	0	21.76	18.26	1.192	0.962	
					0	4	0	15.18	16.25	0.934		
					0	5	0	15.59	15.49	1.006		
					0	6	0	14.17	18.06	0.785		
					0	7	0	17.19	15.84	1.085		
					0	8	0	14.13	18.41	0.768		

Table 71: Architecture **3x10** — perturbation **occ(5,5,5)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 2/3, combos ranked 7–12 of 16 by mean sp); continuation of Table 5 (6 combos, 42 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_{\text{src}}}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_{\text{src}}}$
occ	5,5,5	zono	prev_pgd	0.5	0	3	0	21.76	19.24	1.131	0.959	0.01
					0	4	0	15.18	15.97	0.951		
					0	5	0	15.59	17.28	0.902		
					0	6	0	14.17	17.21	0.823		
					0	7	0	17.19	15.05	1.142		
					0	8	0	14.13	17.60	0.803		
		interval	prev_pgd	0.0	0	3	0	21.76	18.82	1.156	0.957	0.01
					0	4	0	15.18	16.69	0.910		
					0	5	0	15.59	16.90	0.922		
					0	6	0	14.17	16.08	0.881		
					0	7	0	17.19	15.51	1.108		
					0	8	0	14.13	18.50	0.764		
		interval	prev_pgd	0.5	0	3	0	21.76	19.61	1.110	0.957	0.01
					0	4	0	15.18	15.77	0.963		
					0	5	0	15.59	15.64	0.997		
					0	6	0	14.17	16.19	0.875		
					0	7	0	17.19	16.38	1.049		
					0	8	0	14.13	18.92	0.747		
		zono	prev_pgd	0.0	0	3	0	21.76	20.00	1.088	0.938	0.01
					0	4	0	15.18	15.91	0.954		
					0	5	0	15.59	17.34	0.899		
					0	6	0	14.17	17.47	0.811		
					0	7	0	17.19	15.20	1.131		
					0	8	0	14.13	19.02	0.743		

Table 72: Architecture **3x10** — perturbation **occ(5,5,5)**, $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5\}$ (part 3/3, combos ranked 13–16 of 16 by mean **sp**); continuation of Table 5 (4 combos, 24 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
occ	5,5,5	zono	prev_pgd	0.5+sg	1	0	0	19.80	13.61	1.455	1.222	0.02
					1	2	0	12.88	11.73	1.098		
					1	3	0	16.02	12.64	1.267		
					1	4	0	14.21	11.16	1.273		
					1	5	0	16.39	16.01	1.024		
					1	6	0	17.03	11.81	1.442		
					1	7	0	16.60	14.63	1.135		
					1	8	0	15.69	12.98	1.209		
					1	9	0	13.80	12.56	1.099		
		interval	prev_pgd	0.5+sg	1	0	0	19.80	16.54	1.197	1.130	0.02
					1	2	0	12.88	16.99	0.758		
					1	3	0	16.02	14.81	1.082		
					1	4	0	14.21	13.00	1.093		
					1	5	0	16.39	14.94	1.097		
					1	6	0	17.03	13.11	1.299		
					1	7	0	16.60	12.75	1.302		
					1	8	0	15.69	12.93	1.213		
					1	9	0	13.80	12.18	1.133		
		interval	prev_pgd	0.0+sg	1	0	0	19.80	13.45	1.472	1.092	0.02
					1	2	0	12.88	17.07	0.755		
					1	3	0	16.02	14.50	1.105		
					1	4	0	14.21	16.10	0.883		
					1	5	0	16.39	16.22	1.010		
					1	6	0	17.03	13.78	1.236		
					1	7	0	16.60	14.29	1.162		
					1	8	0	15.69	13.98	1.122		
					1	9	0	13.80	12.76	1.082		
		zono	prev_pgd	0.0+sg	1	0	0	19.80	15.06	1.315	1.051	0.02
					1	2	0	12.88	18.57	0.694		
					1	3	0	16.02	17.63	0.909		
					1	4	0	14.21	14.19	1.001		
					1	5	0	16.39	15.27	1.073		
					1	6	0	17.03	13.61	1.251		
					1	7	0	16.60	14.59	1.138		
					1	8	0	15.69	15.26	1.028		
					1	9	0	13.80	13.19	1.046		
		interval	direct_pgd	0.0	1	3	0	16.02	16.08	0.996	1.029	0.02
					1	4	0	14.21	14.16	1.004		
					1	5	0	16.39	17.36	0.944		
					1	6	0	17.03	14.97	1.138		
					1	7	0	16.60	14.52	1.143		
		zono	no	0.5	1	8	0	15.69	16.55	0.948		
					1	3	0	16.02	16.62	0.964	1.022	0.02
					1	4	0	14.21	14.00	1.015		
					1	5	0	16.39	16.13	1.016		
					1	6	0	17.03	14.91	1.142		
					1	7	0	16.60	15.32	1.084		
					1	8	0	15.69	17.22	0.911		

Table 73: Architecture **3x10** — perturbation **occ(5,5,5)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$ (part 1/3, combos ranked 1–6 of 16 by mean **sp**); continuation of Table 5 (6 combos, 48 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
occ	5,5,5	zono	prev_pgd	0.5	1	3	0	16.02	17.47	0.917	1.008	0.02
					1	4	0	14.21	15.18	0.936		
					1	5	0	16.39	14.93	1.098		
					1	6	0	17.03	15.51	1.098		
					1	7	0	16.60	15.13	1.097		
1	8	0	15.69	17.38	0.903							
	interval	prev_pgd	0.0	1	3	0	16.02	17.60	0.910	0.997	0.02	
				1	4	0	14.21	14.45	0.983			
				1	5	0	16.39	14.92	1.099			
				1	6	0	17.03	15.46	1.102			
				1	7	0	16.60	15.19	1.093			
1	8	0	15.69	19.73	0.795							
	zono	prev_pgd	0.0	1	3	0	16.02	17.07	0.938	0.994	0.02	
				1	4	0	14.21	14.68	0.968			
				1	5	0	16.39	14.86	1.103			
				1	6	0	17.03	16.00	1.064			
				1	7	0	16.60	15.66	1.060			
1	8	0	15.69	18.91	0.830							
	interval	prev_pgd	0.5	1	3	0	16.02	16.45	0.974	0.981	0.02	
				1	4	0	14.21	14.78	0.961			
				1	5	0	16.39	18.19	0.901			
				1	6	0	17.03	15.26	1.116			
				1	7	0	16.60	15.25	1.089			
1	8	0	15.69	18.62	0.843							
	interval	direct_pgd	0.5	1	3	0	16.02	17.68	0.906	0.960	0.02	
				1	4	0	14.21	15.18	0.936			
				1	5	0	16.39	18.06	0.908			
				1	6	0	17.03	17.00	1.002			
				1	7	0	16.60	15.04	1.104			
1	8	0	15.69	17.32	0.906							
	zono	no	0.0	1	3	0	16.02	17.94	0.893	0.957	0.02	
				1	4	0	14.21	15.81	0.899			
				1	5	0	16.39	17.45	0.939			
				1	6	0	17.03	16.24	1.049			
				1	7	0	16.60	15.49	1.072			
1	8	0	15.69	17.66	0.888							

Table 74: Architecture **3x10** — perturbation **occ(5,5,5)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$ (part 2/3, combos ranked 7–12 of 16 by mean **sp**); continuation of Table 5 (6 combos, 36 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
occ	5,5,5	zono	direct_pgd	0.0	1	3	0	16.02	19.43	0.824	<i>0.938</i>	<i>0.02</i>
					1	4	0	14.21	16.59	0.857		
					1	5	0	16.39	19.81	0.827		
					1	6	0	17.03	14.80	1.151		
					1	7	0	16.60	15.55	1.068		
					1	8	0	15.69	17.40	0.902		
		interval	no	0.0	1	3	0	16.02	17.38	0.922	<i>0.928</i>	<i>0.02</i>
					1	4	0	14.21	16.17	0.879		
					1	5	0	16.39	17.56	0.933		
					1	6	0	17.03	15.99	1.065		
					1	7	0	16.60	19.11	0.869		
					1	8	0	15.69	17.47	0.898		
		zono	direct_pgd	0.5	1	3	0	16.02	19.55	0.819	<i>0.915</i>	<i>0.02</i>
					1	4	0	14.21	15.98	0.889		
					1	5	0	16.39	18.93	0.866		
					1	6	0	17.03	17.46	0.975		
					1	7	0	16.60	15.51	1.070		
					1	8	0	15.69	18.08	0.868		
		interval	no	0.5	1	3	0	16.02	18.00	0.890	<i>0.910</i>	<i>0.02</i>
					1	4	0	14.21	14.99	0.948		
					1	5	0	16.39	17.98	0.912		
					1	6	0	17.03	17.40	0.979		
					1	7	0	16.60	17.60	0.943		
					1	8	0	15.69	19.83	0.791		

Table 75: Architecture **3x10** — perturbation **occ(5,5,5)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$ (part 3/3, combos ranked 13–16 of 16 by mean **sp**); continuation of Table 5 (4 combos, 24 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	1,1	interval	direct_pgd	0.5	0	3	0	13.93	11.07	1.258	1.483	0.03
					0	4	0	15.18	11.23	1.352		
					0	5	0	15.40	11.05	1.394		
					0	6	0	18.67	10.79	1.730		
					0	7	0	20.37	11.21	1.817		
					0	8	0	16.12	11.95	1.349		
		zono	prev_pgd	0.0	0	3	0	13.93	11.17	1.247	1.442	0.00
					0	4	0	15.18	11.08	1.370		
					0	5	0	15.40	11.17	1.379		
					0	6	0	18.67	10.99	1.699		
					0	7	0	20.37	13.76	1.480		
					0	8	0	16.12	10.92	1.476		
		zono	direct_pgd	0.0	0	3	0	13.93	11.19	1.245	1.434	0.03
					0	4	0	15.18	12.59	1.206		
					0	5	0	15.40	11.46	1.344		
					0	6	0	18.67	11.72	1.593		
					0	7	0	20.37	11.48	1.774		
					0	8	0	16.12	11.16	1.444		
		interval	prev_pgd	0.5	0	3	0	13.93	11.95	1.166	1.409	0.00
					0	4	0	15.18	11.87	1.279		
					0	5	0	15.40	11.94	1.290		
					0	6	0	18.67	12.22	1.528		
					0	7	0	20.37	11.70	1.741		
					0	8	0	16.12	11.12	1.450		
		zono	direct_pgd	0.5	0	3	0	13.93	11.76	1.185	1.362	0.03
					0	4	0	15.18	12.23	1.241		
					0	5	0	15.40	11.69	1.317		
					0	6	0	18.67	11.87	1.573		
					0	7	0	20.37	13.77	1.479		
					0	8	0	16.12	11.73	1.374		
		interval	direct_pgd	0.0	0	3	0	13.93	13.33	1.045	1.333	0.03
					0	4	0	15.18	12.11	1.254		
					0	5	0	15.40	11.69	1.317		
					0	6	0	18.67	12.62	1.479		
					0	7	0	20.37	12.18	1.672		
					0	8	0	16.12	13.08	1.232		

Table 76: Architecture **3x10** — perturbation **translation(1,1)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 1/3, combos ranked 1–6 of 16 by mean **sp**); continuation of Table 5 (6 combos, 36 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	1,1	interval	prev_pgd	0.0	0	3	0	13.93	12.84	1.085	1.315	<i>0.00</i>
					0	4	0	15.18	13.57	1.119		
					0	5	0	15.40	12.37	1.245		
					0	6	0	18.67	12.50	1.494		
					0	7	0	20.37	12.46	1.635		
					0	8	0	16.12	12.28	1.313		
	zono	prev_pgd	0.5	0	3	0	13.93	14.06	0.991	1.297		<i>0.00</i>
				0	4	0	15.18	13.20	1.150			
				0	5	0	15.40	12.45	1.237			
				0	6	0	18.67	13.54	1.379			
				0	7	0	20.37	12.73	1.600			
				0	8	0	16.12	11.30	1.427			
	zono	prev_pgd	0.5+sg	0	1	0	20.98	13.33	1.574	1.270		<i>0.01</i>
				0	2	0	21.15	13.29	1.591			
				0	3	0	13.93	13.97	0.997			
				0	4	0	15.18	13.74	1.105			
				0	5	0	15.40	13.50	1.141			
				0	6	0	18.67	13.87	1.346			
				0	7	0	20.37	14.18	1.437			
				0	8	0	16.12	14.36	1.123			
				0	9	0	18.52	16.60	1.116			
	zono	prev_pgd	0.0+sg	0	1	0	20.98	13.11	1.600	1.266		<i>0.01</i>
				0	2	0	21.15	13.16	1.607			
				0	3	0	13.93	13.29	1.048			
				0	4	0	15.18	14.16	1.072			
				0	5	0	15.40	13.80	1.116			
				0	6	0	18.67	13.43	1.390			
				0	7	0	20.37	14.52	1.403			
				0	8	0	16.12	15.46	1.043			
				0	9	0	18.52	16.67	1.111			
	interval	prev_pgd	0.5+sg	0	1	0	20.98	13.09	1.603	1.255		<i>0.01</i>
				0	2	0	21.15	13.83	1.529			
				0	3	0	13.93	13.79	1.010			
				0	4	0	15.18	13.65	1.112			
				0	5	0	15.40	13.58	1.134			
				0	6	0	18.67	14.20	1.315			
				0	7	0	20.37	14.90	1.367			
				0	8	0	16.12	15.07	1.070			
				0	9	0	18.52	16.03	1.155			
	zono	no	0.0	0	3	0	13.93	14.10	0.988	1.176		<i>0.00</i>
				0	4	0	15.18	13.74	1.105			
				0	5	0	15.40	14.85	1.037			
				0	6	0	18.67	13.58	1.375			
				0	7	0	20.37	14.49	1.406			
				0	8	0	16.12	14.08	1.145			

Table 77: Architecture **3x10** — perturbation **translation(1,1)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 2/3, combos ranked 7–12 of 16 by mean **sp**); continuation of Table 5 (6 combos, 45 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_src}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_src}$
translation	1,1	interval	no	0.5	0	3	0	13.93	13.64	1.021	1.148	0.00
					0	4	0	15.18	14.05	1.080		
					0	5	0	15.40	13.72	1.122		
					0	6	0	18.67	13.77	1.356		
					0	7	0	20.37	14.09	1.446		
					0	8	0	16.12	18.68	0.863		
		interval	prev-pgd	0.0+sg	0	1	0	20.98	22.40	0.937	1.087	0.01
					0	2	0	21.15	17.73	1.193		
					0	3	0	13.93	16.68	0.835		
					0	4	0	15.18	13.02	1.166		
					0	5	0	15.40	13.56	1.136		
					0	6	0	18.67	15.49	1.205		
					0	7	0	20.37	18.79	1.084		
					0	8	0	16.12	18.32	0.880		
					0	9	0	18.52	13.74	1.348		
		interval	no	0.0	0	3	0	13.93	14.13	0.986	1.076	0.00
					0	4	0	15.18	14.58	1.041		
					0	5	0	15.40	20.79	0.741		
					0	6	0	18.67	14.74	1.267		
					0	7	0	20.37	15.44	1.319		
					0	8	0	16.12	14.66	1.100		
		zono	no	0.5	0	3	0	13.93	23.82	0.585	0.731	0.00
					0	4	0	15.18	27.68	0.548		
					0	5	0	15.40	15.65	0.984		
					0	6	0	18.67	23.79	0.785		
					0	7	0	20.37	33.73	0.604		
					0	8	0	16.12	18.28	0.882		

Table 78: Architecture **3x10** — perturbation **translation(1,1)**, $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5\}$ (part 3/3, combos ranked 13–16 of 16 by mean **sp**); continuation of Table 5 (4 combos, 27 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	1,1	zono	prev_pgd	0.0+sg	1	0	0	20.62	9.85	2.093	1.730	0.01
					1	2	0	16.58	10.44	1.588		
					1	3	0	22.44	10.24	2.191		
					1	4	0	22.97	10.99	2.090		
					1	5	0	21.27	12.26	1.735		
					1	6	0	21.10	13.34	1.582		
					1	7	0	22.94	14.09	1.628		
					1	8	0	22.62	14.35	1.576		
					1	9	0	15.86	14.62	1.085		
		zono	prev_pgd	0.5+sg	1	0	0	20.62	10.36	1.990	1.713	0.01
					1	2	0	16.58	11.25	1.474		
					1	3	0	22.44	11.55	1.943		
					1	4	0	22.97	13.60	1.689		
					1	5	0	21.27	12.34	1.724		
					1	6	0	21.10	12.59	1.676		
					1	7	0	22.94	14.13	1.623		
					1	8	0	22.62	12.11	1.868		
					1	9	0	15.86	11.07	1.433		
		interval	prev_pgd	0.5+sg	1	0	0	20.62	11.27	1.830	1.676	0.01
					1	2	0	16.58	11.23	1.476		
					1	3	0	22.44	11.20	2.004		
					1	4	0	22.97	12.00	1.914		
					1	5	0	21.27	13.38	1.590		
					1	6	0	21.10	13.97	1.510		
					1	7	0	22.94	14.50	1.582		
					1	8	0	22.62	13.22	1.711		
					1	9	0	15.86	10.84	1.463		
		zono	prev_pgd	0.0	1	3	0	22.44	14.42	1.556	1.580	0.01
					1	4	0	22.97	14.79	1.553		
					1	5	0	21.27	13.40	1.587		
					1	6	0	21.10	13.65	1.546		
					1	7	0	22.94	13.68	1.677		
					1	8	0	22.62	14.49	1.561		
		zono	no	0.0	1	3	0	22.44	13.62	1.648	1.579	0.00
					1	4	0	22.97	14.62	1.571		
					1	5	0	21.27	14.13	1.505		
					1	6	0	21.10	13.54	1.558		
					1	7	0	22.94	13.95	1.644		
					1	8	0	22.62	14.60	1.549		
		interval	prev_pgd	0.0	1	3	0	22.44	14.60	1.537	1.576	0.01
					1	4	0	22.97	14.22	1.615		
					1	5	0	21.27	14.62	1.455		
					1	6	0	21.10	13.62	1.549		
					1	7	0	22.94	14.19	1.617		
					1	8	0	22.62	13.42	1.686		

Table 79: Architecture **3x10** — perturbation **translation(1,1)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$ (part 1/3, combos ranked 1–6 of 16 by mean **sp**); continuation of Table 5 (6 combos, 45 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	1,1	interval	direct_pgd	0.5	1	3	0	22.44	13.76	1.631	1.575	0.01
					1	4	0	22.97	13.93	1.649		
					1	5	0	21.27	14.81	1.436		
					1	6	0	21.10	14.10	1.496		
					1	7	0	22.94	14.01	1.637		
					1	8	0	22.62	14.15	1.599		
		interval	direct_pgd	0.0	1	3	0	22.44	13.35	1.681	1.570	0.01
					1	4	0	22.97	14.17	1.621		
					1	5	0	21.27	14.26	1.492		
					1	6	0	21.10	13.42	1.572		
					1	7	0	22.94	13.97	1.642		
					1	8	0	22.62	15.99	1.415		
		zono	prev_pgd	0.5	1	3	0	22.44	14.15	1.586	1.569	0.01
					1	4	0	22.97	14.13	1.626		
					1	5	0	21.27	13.71	1.551		
					1	6	0	21.10	15.23	1.385		
					1	7	0	22.94	14.56	1.576		
					1	8	0	22.62	13.40	1.688		
		zono	direct_pgd	0.0	1	3	0	22.44	13.49	1.663	1.566	0.01
					1	4	0	22.97	14.08	1.631		
					1	5	0	21.27	14.67	1.450		
					1	6	0	21.10	13.56	1.556		
					1	7	0	22.94	13.54	1.694		
					1	8	0	22.62	16.12	1.403		
		interval	prev_pgd	0.5	1	3	0	22.44	13.53	1.659	1.566	0.01
					1	4	0	22.97	14.04	1.636		
					1	5	0	21.27	14.17	1.501		
					1	6	0	21.10	13.69	1.541		
					1	7	0	22.94	15.74	1.457		
					1	8	0	22.62	14.13	1.601		
		zono	no	0.5	1	3	0	22.44	13.80	1.626	1.562	0.00
					1	4	0	22.97	14.52	1.582		
					1	5	0	21.27	14.54	1.463		
					1	6	0	21.10	14.37	1.468		
					1	7	0	22.94	14.04	1.634		
					1	8	0	22.62	14.13	1.601		

Table 80: Architecture **3x10** — perturbation **translation(1,1)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$ (part 2/3, combos ranked 7–12 of 16 by mean **sp**); continuation of Table 5 (6 combos, 36 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_src}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_src}$
translation	1,1	interval	no	0.0	1	3	0	22.44	13.97	1.606	1.560	<i>0.00</i>
					1	4	0	22.97	14.25	1.612		
					1	5	0	21.27	14.36	1.481		
					1	6	0	21.10	13.63	1.548		
					1	7	0	22.94	14.80	1.550		
					1	8	0	22.62	14.48	1.562		
		zono	direct.pgd	0.5	1	3	0	22.44	13.60	1.650	1.556	<i>0.01</i>
					1	4	0	22.97	14.39	1.596		
					1	5	0	21.27	15.79	1.347		
					1	6	0	21.10	13.77	1.532		
					1	7	0	22.94	14.35	1.599		
					1	8	0	22.62	14.04	1.611		
		interval	no	0.5	1	3	0	22.44	12.91	1.738	1.542	<i>0.00</i>
					1	4	0	22.97	15.83	1.451		
					1	5	0	21.27	15.43	1.378		
					1	6	0	21.10	14.78	1.428		
					1	7	0	22.94	14.15	1.621		
					1	8	0	22.62	13.83	1.636		
		interval	prev.pgd	0.0+sg	1	0	0	20.62	19.25	1.071	1.539	<i>0.01</i>
					1	2	0	16.58	14.36	1.155		
					1	3	0	22.44	11.14	2.014		
					1	4	0	22.97	11.40	2.015		
					1	5	0	21.27	16.20	1.313		
					1	6	0	21.10	13.53	1.559		
					1	7	0	22.94	11.63	1.972		
					1	8	0	22.62	13.00	1.740		
					1	9	0	15.86	15.70	1.010		

Table 81: Architecture **3x10** — perturbation **translation(1,1)**, $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5\}$ (part 3/3, combos ranked 13–16 of 16 by mean **sp**); continuation of Table 5 (4 combos, 27 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_src}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_src}$
translation	3,1	interval	prev_pgd	0.5+sg	0	1	0	21.16	14.12	1.499	1.071	0.01
					0	2	0	19.76	14.06	1.405		
					0	3	0	13.40	14.04	0.954		
					0	4	0	14.38	13.54	1.062		
					0	5	0	13.93	14.05	0.991		
					0	6	0	11.80	15.58	0.757		
					0	7	0	11.82	15.08	0.784		
					0	8	0	12.12	13.83	0.876		
					0	9	0	19.29	14.69	1.313		
		interval	prev_pgd	0.0+sg	0	1	0	21.16	13.61	1.555	1.061	0.01
					0	2	0	19.76	13.61	1.452		
					0	3	0	13.40	13.18	1.017		
					0	4	0	14.38	13.87	1.037		
					0	5	0	13.93	14.89	0.936		
					0	6	0	11.80	15.68	0.753		
					0	7	0	11.82	16.07	0.736		
					0	8	0	12.12	16.68	0.727		
					0	9	0	19.29	14.42	1.338		
		zono	prev_pgd	0.0+sg	0	1	0	21.16	13.95	1.517	1.058	0.01
					0	2	0	19.76	14.33	1.379		
					0	3	0	13.40	14.38	0.932		
					0	4	0	14.38	14.28	1.007		
					0	5	0	13.93	14.30	0.974		
					0	6	0	11.80	15.13	0.780		
					0	7	0	11.82	15.35	0.770		
					0	8	0	12.12	14.68	0.826		
					0	9	0	19.29	14.45	1.335		
		zono	prev_pgd	0.5+sg	0	1	0	21.16	14.00	1.511	1.047	0.01
					0	2	0	19.76	14.14	1.397		
					0	3	0	13.40	15.97	0.839		
					0	4	0	14.38	15.74	0.914		
					0	5	0	13.93	13.77	1.012		
					0	6	0	11.80	15.45	0.764		
					0	7	0	11.82	14.50	0.815		
					0	8	0	12.12	13.89	0.873		
					0	9	0	19.29	14.83	1.301		
		zono	prev_pgd	0.5	0	3	0	13.40	13.21	1.014	0.972	0.02
					0	4	0	14.38	13.06	1.101		
					0	5	0	13.93	12.93	1.077		
					0	6	0	11.80	13.07	0.903		
					0	7	0	11.82	14.03	0.842		
					0	8	0	12.12	13.59	0.892		
		zono	direct_pgd	0.0	0	3	0	13.40	13.14	1.020	0.965	0.04
					0	4	0	14.38	13.30	1.081		
					0	5	0	13.93	12.99	1.072		
					0	6	0	11.80	13.60	0.868		
					0	7	0	11.82	13.91	0.850		
					0	8	0	12.12	13.51	0.897		

Table 82: Architecture **3x10** — perturbation **translation(3,1)**, $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5\}$ (part 1/3, combos ranked 1–6 of 16 by mean **sp**); continuation of Table 5 (6 combos, 48 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_src}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_src}$
translation	3,1	zono	direct_pgd	0.5	0	3	0	13.40	13.55	0.989	0.953	0.04
					0	4	0	14.38	13.57	1.060		
					0	5	0	13.93	13.22	1.054		
					0	6	0	11.80	14.15	0.834		
					0	7	0	11.82	13.54	0.873		
					0	8	0	12.12	13.32	0.910		
		interval	direct_pgd	0.0	0	3	0	13.40	13.04	1.028	0.941	0.04
					0	4	0	14.38	13.43	1.071		
					0	5	0	13.93	13.21	1.055		
					0	6	0	11.80	13.58	0.869		
					0	7	0	11.82	16.05	0.736		
					0	8	0	12.12	13.63	0.889		
		interval	prev_pgd	0.5	0	3	0	13.40	13.08	1.024	0.925	0.02
					0	4	0	14.38	13.33	1.079		
					0	5	0	13.93	14.92	0.934		
					0	6	0	11.80	13.07	0.903		
					0	7	0	11.82	14.08	0.839		
					0	8	0	12.12	15.67	0.773		
		zono	prev_pgd	0.0	0	3	0	13.40	14.11	0.950	0.914	0.02
					0	4	0	14.38	13.60	1.057		
					0	5	0	13.93	14.55	0.957		
					0	6	0	11.80	13.51	0.873		
					0	7	0	11.82	13.90	0.850		
					0	8	0	12.12	15.26	0.794		
		interval	direct_pgd	0.5	0	3	0	13.40	14.81	0.905	0.905	0.04
					0	4	0	14.38	14.57	0.987		
					0	5	0	13.93	14.02	0.994		
					0	6	0	11.80	13.73	0.859		
					0	7	0	11.82	13.96	0.847		
					0	8	0	12.12	14.47	0.838		
		interval	prev_pgd	0.0	0	3	0	13.40	15.33	0.874	0.901	0.02
					0	4	0	14.38	13.35	1.077		
					0	5	0	13.93	14.79	0.942		
					0	6	0	11.80	14.34	0.823		
					0	7	0	11.82	14.08	0.839		
					0	8	0	12.12	14.27	0.849		

Table 83: Architecture **3x10** — perturbation **translation(3,1)**, $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5\}$ (part 2/3, combos ranked 7–12 of 16 by mean **sp**); continuation of Table 5 (6 combos, 36 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	3,1	zono	no	0.5	0	3	0	13.40	35.84	0.374	0.594	0.02
					0	4	0	14.38	30.59	0.470		
					0	5	0	13.93	14.22	0.980		
					0	6	0	11.80	16.32	0.723		
					0	7	0	11.82	16.88	0.700		
					0	8	0	12.12	38.26	0.317		
		zono	no	0.0	0	3	0	13.40	36.02	0.372	0.500	0.02
					0	4	0	14.38	38.00	0.378		
					0	5	0	13.93	36.07	0.386		
					0	6	0	11.80	19.66	0.600		
					0	7	0	11.82	13.66	0.865		
					0	8	0	12.12	30.43	0.398		
		interval	no	0.5	0	3	0	13.40	14.20	0.944	0.469	0.02
					0	4	0	14.38	36.57	0.393		
					0	5	0	13.93	38.16	0.365		
					0	6	0	11.80	36.55	0.323		
					0	7	0	11.82	26.00	0.455		
					0	8	0	12.12	36.04	0.336		
		interval	no	0.0	0	3	0	13.40	35.54	0.377	0.404	0.02
					0	4	0	14.38	23.88	0.602		
					0	5	0	13.93	33.35	0.418		
					0	6	0	11.80	38.52	0.306		
					0	7	0	11.82	29.95	0.395		
					0	8	0	12.12	37.10	0.327		

Table 84: Architecture **3x10** — perturbation **translation(3,1)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 3/3, combos ranked 13–16 of 16 by mean **sp**); continuation of Table 5 (4 combos, 24 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_{\text{src}}}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_{\text{src}}}$
translation	3,1	interval	prev_pgd	0.5+sg	1	0	0	18.86	9.59	1.967	1.719	0.04
					1	2	0	16.73	11.73	1.426		
					1	3	0	23.91	10.16	2.353		
					1	4	0	22.60	10.49	2.154		
					1	5	0	15.45	12.39	1.247		
					1	6	0	23.56	14.40	1.636		
					1	7	0	24.44	15.18	1.610		
					1	8	0	19.60	11.28	1.738		
					1	9	0	16.56	12.38	1.338		
		interval	prev_pgd	0.0+sg	1	0	0	18.86	11.89	1.586	1.687	0.04
					1	2	0	16.73	12.08	1.385		
					1	3	0	23.91	14.04	1.703		
					1	4	0	22.60	11.72	1.928		
					1	5	0	15.45	12.37	1.249		
					1	6	0	23.56	11.81	1.995		
					1	7	0	24.44	11.89	2.056		
					1	8	0	19.60	10.96	1.788		
					1	9	0	16.56	11.10	1.492		
		zono	prev_pgd	0.5+sg	1	0	0	18.86	9.83	1.919	1.682	0.04
					1	2	0	16.73	10.59	1.580		
					1	3	0	23.91	11.28	2.120		
					1	4	0	22.60	13.70	1.650		
					1	5	0	15.45	13.65	1.132		
					1	6	0	23.56	13.73	1.716		
					1	7	0	24.44	11.33	2.157		
					1	8	0	19.60	12.10	1.620		
					1	9	0	16.56	13.34	1.241		
		zono	prev_pgd	0.0+sg	1	0	0	18.86	9.78	1.928	1.664	0.04
					1	2	0	16.73	11.74	1.425		
					1	3	0	23.91	10.76	2.222		
					1	4	0	22.60	11.37	1.988		
					1	5	0	15.45	11.35	1.361		
					1	6	0	23.56	13.02	1.810		
					1	7	0	24.44	14.61	1.673		
					1	8	0	19.60	14.00	1.400		
					1	9	0	16.56	14.17	1.169		
		interval	no	0.5	1	3	0	23.91	13.71	1.744	1.567	0.03
					1	4	0	22.60	13.55	1.668		
					1	5	0	15.45	13.55	1.140		
					1	6	0	23.56	13.89	1.696		
					1	7	0	24.44	14.03	1.742		
					1	8	0	19.60	13.86	1.414		
		interval	direct_pgd	0.5	1	3	0	23.91	13.47	1.775	1.554	0.04
					1	4	0	22.60	14.26	1.585		
					1	5	0	15.45	13.57	1.139		
					1	6	0	23.56	13.69	1.721		
					1	7	0	24.44	13.90	1.758		
					1	8	0	19.60	14.59	1.343		

Table 85: Architecture **3x10** — perturbation **translation(3,1)**, $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5\}$ (part 1/3, combos ranked 1–6 of 16 by mean **sp**); continuation of Table 5 (6 combos, 48 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_{\text{src}}}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_{\text{src}}}$
translation	3,1	zono	direct_pgd	0.0	1	3	0	23.91	13.48	1.774	1.550	0.04
					1	4	0	22.60	13.84	1.633		
					1	5	0	15.45	13.93	1.109		
					1	6	0	23.56	14.03	1.679		
					1	7	0	24.44	13.63	1.793		
					1	8	0	19.60	14.97	1.309		
		zono	prev_pgd	0.0	1	3	0	23.91	14.01	1.707	1.527	0.04
					1	4	0	22.60	13.58	1.664		
					1	5	0	15.45	13.50	1.144		
					1	6	0	23.56	13.72	1.717		
					1	7	0	24.44	15.31	1.596		
					1	8	0	19.60	14.67	1.336		
		interval	prev_pgd	0.0	1	3	0	23.91	14.07	1.699	1.524	0.04
					1	4	0	22.60	13.66	1.654		
					1	5	0	15.45	13.76	1.123		
					1	6	0	23.56	13.58	1.735		
					1	7	0	24.44	14.38	1.700		
					1	8	0	19.60	15.87	1.235		
		zono	direct_pgd	0.5	1	3	0	23.91	14.09	1.697	1.519	0.04
					1	4	0	22.60	14.43	1.566		
					1	5	0	15.45	13.12	1.178		
					1	6	0	23.56	14.46	1.629		
					1	7	0	24.44	14.76	1.656		
					1	8	0	19.60	14.14	1.386		
		interval	prev_pgd	0.5	1	3	0	23.91	13.58	1.761	1.506	0.04
					1	4	0	22.60	13.81	1.636		
					1	5	0	15.45	13.87	1.114		
					1	6	0	23.56	15.36	1.534		
					1	7	0	24.44	14.46	1.690		
					1	8	0	19.60	15.09	1.299		
		zono	prev_pgd	0.5	1	3	0	23.91	15.70	1.523	1.500	0.04
					1	4	0	22.60	13.75	1.644		
					1	5	0	15.45	14.66	1.054		
					1	6	0	23.56	14.66	1.607		
					1	7	0	24.44	14.04	1.741		
					1	8	0	19.60	13.68	1.433		

Table 86: Architecture **3x10** — perturbation **translation(3,1)**, $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5\}$ (part 2/3, combos ranked 7–12 of 16 by mean **sp**); continuation of Table 5 (6 combos, 36 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	3,1	interval	direct_pgd	0.0	1	3	0	23.91	14.19	1.685	1.495	0.04
					1	4	0	22.60	13.77	1.641		
					1	5	0	15.45	13.57	1.139		
					1	6	0	23.56	13.88	1.697		
					1	7	0	24.44	15.10	1.619		
					1	8	0	19.60	16.51	1.187		
					1	3	0	23.91	13.76	1.738	1.492	0.03
					1	4	0	22.60	15.60	1.449		
					1	5	0	15.45	14.25	1.084		
					1	6	0	23.56	14.22	1.657		
					1	7	0	24.44	14.93	1.637		
					1	8	0	19.60	14.13	1.387		
				0.5	1	3	0	23.91	15.48	1.545	1.423	0.03
					1	4	0	22.60	14.83	1.524		
					1	5	0	15.45	14.82	1.043		
					1	6	0	23.56	14.63	1.610		
					1	7	0	24.44	15.20	1.608		
					1	8	0	19.60	16.22	1.208		
					1	3	0	23.91	14.50	1.649	1.419	0.03
					1	4	0	22.60	15.32	1.475		
					1	5	0	15.45	17.55	0.880		
					1	6	0	23.56	14.32	1.645		
					1	7	0	24.44	15.16	1.612		
					1	8	0	19.60	15.66	1.252		

Table 87: Architecture **3x10** — perturbation **translation(3,1)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$ (part 3/3, combos ranked 13–16 of 16 by mean **sp**); continuation of Table 5 (4 combos, 24 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	3,3	zono	no	0.0	0	3	0	12.78	10.75	1.189	1.130	<i>0.03</i>
					0	4	0	14.57	11.46	1.271		
					0	5	0	14.10	11.20	1.259		
					0	6	0	10.86	11.53	0.942		
					0	7	0	11.99	12.22	0.981		
					0	8	0	12.37	10.88	1.137		
	interval	no	no	0.5	0	3	0	12.78	10.70	1.194	1.129	<i>0.03</i>
					0	4	0	14.57	11.48	1.269		
					0	5	0	14.10	10.66	1.323		
					0	6	0	10.86	11.24	0.966		
					0	7	0	11.99	13.40	0.895		
					0	8	0	12.37	10.99	1.126		
	zono	no	no	0.5	0	3	0	12.78	11.18	1.143	1.102	<i>0.03</i>
					0	4	0	14.57	11.47	1.270		
					0	5	0	14.10	12.17	1.159		
					0	6	0	10.86	10.40	1.044		
					0	7	0	11.99	11.33	1.058		
					0	8	0	12.37	13.17	0.939		
	interval	prev_pgd	0.5+sg		0	1	0	20.13	14.22	1.416	1.078	<i>0.02</i>
					0	2	0	21.34	15.53	1.374		
					0	3	0	12.78	14.75	0.866		
					0	4	0	14.57	14.59	0.999		
					0	5	0	14.10	13.47	1.047		
					0	6	0	10.86	13.63	0.797		
					0	7	0	11.99	13.90	0.863		
					0	8	0	12.37	13.78	0.898		
					0	9	0	19.27	13.34	1.445		
	zono	prev_pgd	0.0+sg		0	1	0	20.13	12.83	1.569	1.072	<i>0.02</i>
					0	2	0	21.34	13.83	1.543		
					0	3	0	12.78	12.96	0.986		
					0	4	0	14.57	15.84	0.920		
					0	5	0	14.10	13.66	1.032		
					0	6	0	10.86	14.51	0.748		
					0	7	0	11.99	16.35	0.733		
					0	8	0	12.37	14.70	0.841		
					0	9	0	19.27	15.17	1.270		
	interval	prev_pgd	0.0+sg		0	1	0	20.13	14.39	1.399	1.038	<i>0.02</i>
					0	2	0	21.34	13.06	1.634		
					0	3	0	12.78	15.18	0.842		
					0	4	0	14.57	16.51	0.882		
					0	5	0	14.10	14.81	0.952		
					0	6	0	10.86	15.15	0.717		
					0	7	0	11.99	16.18	0.741		
					0	8	0	12.37	15.82	0.782		
					0	9	0	19.27	13.85	1.391		

Table 88: Architecture **3x10** — perturbation **translation(3,3)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 1/3, combos ranked 1–6 of 16 by mean **sp**); continuation of Table 5 (6 combos, 45 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	3,3	zono	prev_pgd	0.5+sg	0	1	0	20.13	13.33	1.510	1.003	0.02
					0	2	0	21.34	13.09	1.630		
					0	3	0	12.78	13.89	0.920		
					0	4	0	14.57	15.73	0.926		
					0	5	0	14.10	17.22	0.819		
					0	6	0	10.86	16.49	0.659		
					0	7	0	11.99	16.10	0.745		
					0	8	0	12.37	16.89	0.732		
					0	9	0	19.27	17.70	1.089		
		zono	prev_pgd	0.5	0	3	0	12.78	13.13	0.973	0.962	0.03
					0	4	0	14.57	13.28	1.097		
					0	5	0	14.10	13.30	1.060		
					0	6	0	10.86	13.16	0.825		
					0	7	0	11.99	13.40	0.895		
					0	8	0	12.37	13.39	0.924		
		interval	prev_pgd	0.0	0	3	0	12.78	13.25	0.965	0.953	0.03
					0	4	0	14.57	13.40	1.087		
					0	5	0	14.10	13.28	1.062		
					0	6	0	10.86	13.13	0.827		
					0	7	0	11.99	13.22	0.907		
					0	8	0	12.37	14.26	0.867		
		interval	direct_pgd	0.5	0	3	0	12.78	13.00	0.983	0.952	0.05
					0	4	0	14.57	13.67	1.066		
					0	5	0	14.10	13.38	1.054		
					0	6	0	10.86	13.22	0.821		
					0	7	0	11.99	14.31	0.838		
					0	8	0	12.37	13.01	0.951		
		zono	direct_pgd	0.5	0	3	0	12.78	13.00	0.983	0.936	0.05
					0	4	0	14.57	13.55	1.075		
					0	5	0	14.10	13.17	1.071		
					0	6	0	10.86	13.26	0.819		
					0	7	0	11.99	15.40	0.779		
					0	8	0	12.37	13.91	0.889		
		zono	direct_pgd	0.0	0	3	0	12.78	13.12	0.974	0.935	0.05
					0	4	0	14.57	13.69	1.064		
					0	5	0	14.10	13.58	1.038		
					0	6	0	10.86	13.30	0.817		
					0	7	0	11.99	14.53	0.825		
					0	8	0	12.37	13.89	0.891		

Table 89: Architecture **3x10** — perturbation **translation(3,3)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 2/3, combos ranked 7–12 of 16 by mean **sp**); continuation of Table 5 (6 combos, 39 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	3,3	interval	prev_pgd	0.5	0	3	0	12.78	13.37	0.956	<i>0.934</i>	<i>0.03</i>
					0	4	0	14.57	14.16	1.029		
					0	5	0	14.10	13.35	1.056		
					0	6	0	10.86	14.22	0.764		
					0	7	0	11.99	12.44	0.964		
					0	8	0	12.37	14.84	0.834		
	zono		prev_pgd	0.0	0	3	0	12.78	14.89	0.858	<i>0.923</i>	<i>0.03</i>
					0	4	0	14.57	13.62	1.070		
					0	5	0	14.10	13.36	1.055		
					0	6	0	10.86	13.69	0.793		
					0	7	0	11.99	13.68	0.876		
					0	8	0	12.37	13.98	0.885		
	interval		direct_pgd	0.0	0	3	0	12.78	13.64	0.937	<i>0.908</i>	<i>0.05</i>
					0	4	0	14.57	14.46	1.008		
					0	5	0	14.10	13.14	1.073		
					0	6	0	10.86	13.53	0.803		
					0	7	0	11.99	16.07	0.746		
					0	8	0	12.37	14.00	0.884		
	interval		no	0.0	0	3	0	12.78	35.34	0.362	<i>0.670</i>	<i>0.03</i>
					0	4	0	14.57	20.81	0.700		
					0	5	0	14.10	12.69	1.111		
					0	6	0	10.86	15.43	0.704		
					0	7	0	11.99	15.06	0.796		
					0	8	0	12.37	35.55	0.348		

Table 90: Architecture **3x10** — perturbation **translation(3,3)**, $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$ (part 3/3, combos ranked 13–16 of 16 by mean **sp**); continuation of Table 5 (4 combos, 24 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
translation	3,3	zono	prev_pgd	0.0+sg	1	0	0	19.27	9.84	1.958	1.312	0.02
					1	2	0	17.32	10.02	1.729		
					1	3	0	14.78	10.34	1.429		
					1	4	0	13.53	10.42	1.298		
					1	5	0	13.74	11.07	1.241		
					1	6	0	12.78	16.24	0.787		
					1	7	0	14.15	14.02	1.009		
					1	8	0	13.03	11.45	1.138		
					1	9	0	16.29	13.33	1.222		
		zono	prev_pgd	0.5+sg	1	0	0	19.27	11.77	1.637	1.214	0.02
					1	2	0	17.32	13.25	1.307		
					1	3	0	14.78	10.92	1.353		
					1	4	0	13.53	11.03	1.227		
					1	5	0	13.74	11.97	1.148		
					1	6	0	12.78	16.08	0.795		
					1	7	0	14.15	12.64	1.119		
					1	8	0	13.03	13.39	0.973		
					1	9	0	16.29	11.89	1.370		
		interval	prev_pgd	0.5+sg	1	0	0	19.27	11.16	1.727	1.210	0.02
					1	2	0	17.32	13.53	1.280		
					1	3	0	14.78	13.49	1.096		
					1	4	0	13.53	12.93	1.046		
					1	5	0	13.74	12.08	1.137		
					1	6	0	12.78	13.02	0.982		
					1	7	0	14.15	13.80	1.025		
					1	8	0	13.03	11.17	1.167		
					1	9	0	16.29	11.42	1.426		
		interval	prev_pgd	0.0+sg	1	0	0	19.27	12.44	1.549	1.143	0.02
					1	2	0	17.32	11.51	1.505		
					1	3	0	14.78	11.85	1.247		
					1	4	0	13.53	13.63	0.993		
					1	5	0	13.74	15.28	0.899		
					1	6	0	12.78	14.96	0.854		
					1	7	0	14.15	14.40	0.983		
					1	8	0	13.03	13.47	0.967		
					1	9	0	16.29	12.64	1.289		
		interval	prev_pgd	0.0	1	3	0	14.78	13.77	1.073	0.986	0.03
					1	4	0	13.53	14.15	0.956		
					1	5	0	13.74	13.71	1.002		
					1	6	0	12.78	14.29	0.894		
					1	7	0	14.15	13.74	1.030		
		interval	prev_pgd	0.5	1	8	0	13.03	13.61	0.957		0.03
					1	3	0	14.78	13.66	1.082	0.962	
					1	4	0	13.53	14.16	0.956		
					1	5	0	13.74	13.85	0.992		
					1	6	0	12.78	16.02	0.798		
					1	7	0	14.15	14.40	0.983		
					1	8	0	13.03	13.54	0.962		

Table 91: Architecture **3x10** — perturbation **translation(3,3)**, $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$ (part 1/3, combos ranked 1–6 of 16 by mean **sp**); continuation of Table 5 (6 combos, 48 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_src}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_src}$
translation	3,3	interval	direct_pgd	0.0	1	3	0	14.78	13.14	1.125	0.959	0.03
					1	4	0	13.53	13.71	0.987		
					1	5	0	13.74	13.83	0.993		
					1	6	0	12.78	15.33	0.834		
					1	7	0	14.15	14.26	0.992		
					1	8	0	13.03	15.81	0.824		
					1	3	0	14.78	15.25	0.969		
					1	4	0	13.53	13.32	1.016		
					1	5	0	13.74	14.23	0.966		
					1	6	0	12.78	13.87	0.921		
					1	7	0	14.15	13.77	1.028	0.953	0.03
					1	8	0	13.03	15.87	0.821		
					1	3	0	14.78	13.81	1.070		
					1	4	0	13.53	13.76	0.983		
					1	5	0	13.74	13.66	1.006		
					1	6	0	12.78	14.24	0.897	0.949	0.03
					1	7	0	14.15	16.97	0.834		
					1	8	0	13.03	14.47	0.900		
					1	3	0	14.78	14.28	1.035		
					1	4	0	13.53	13.84	0.978		
					1	5	0	13.74	15.91	0.864	0.931	0.03
					1	6	0	12.78	14.22	0.899		
					1	7	0	14.15	15.15	0.934		
					1	8	0	13.03	14.82	0.879		
					1	3	0	14.78	14.18	1.042		
					1	4	0	13.53	14.24	0.950	0.930	0.03
					1	5	0	13.74	14.06	0.977		
					1	6	0	12.78	14.13	0.904		
					1	7	0	14.15	14.99	0.944		
					1	8	0	13.03	17.04	0.765		
					1	3	0	14.78	13.64	1.084	0.928	0.03
					1	4	0	13.53	13.75	0.984		
					1	5	0	13.74	14.25	0.964		
					1	6	0	12.78	14.33	0.892		
					1	7	0	14.15	17.38	0.814		
					1	8	0	13.03	15.66	0.832		

Table 92: Architecture **3x10** — perturbation **translation(3,3)**, $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5\}$ (part 2/3, combos ranked 7–12 of 16 by mean **sp**); continuation of Table 5 (6 combos, 36 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c_src}$	$ \delta_{\text{std}} - \delta_{\text{adv}} _{c_src}$
translation	3,3	zono	no	0.0	1	3	0	14.78	14.21	1.040	<i>0.921</i>	<i>0.03</i>
					1	4	0	13.53	15.10	0.896		
					1	5	0	13.74	16.91	0.813		
					1	6	0	12.78	14.84	0.861		
					1	7	0	14.15	13.96	1.014		
					1	8	0	13.03	14.45	0.902		
		zono	direct_pgd	0.5	1	3	0	14.78	15.24	0.970	<i>0.885</i>	<i>0.03</i>
					1	4	0	13.53	16.23	0.834		
					1	5	0	13.74	16.17	0.850		
					1	6	0	12.78	16.16	0.791		
					1	7	0	14.15	14.58	0.971		
					1	8	0	13.03	14.59	0.893		
		zono	prev_pgd	0.5	1	3	0	14.78	15.72	0.940	<i>0.879</i>	<i>0.03</i>
					1	4	0	13.53	15.91	0.850		
					1	5	0	13.74	16.46	0.835		
					1	6	0	12.78	17.32	0.738		
					1	7	0	14.15	14.49	0.977		
					1	8	0	13.03	13.91	0.937		
		zono	no	0.5	1	3	0	14.78	16.24	0.910	<i>0.845</i>	<i>0.03</i>
					1	4	0	13.53	18.49	0.732		
					1	5	0	13.74	16.85	0.815		
					1	6	0	12.78	14.31	0.893		
					1	7	0	14.15	16.64	0.850		
					1	8	0	13.03	14.99	0.869		

Table 93: Architecture **3x10** — perturbation **translation(3,3)**, $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5\}$ (part 3/3, combos ranked 13–16 of 16 by mean **sp**); continuation of Table 5 (4 combos, 24 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
linf	0.1	interval	prev_pgd	0.0+sg	0	1	0	24.23	15.86	1.528	1.251	<i>0.04</i>
					0	2	0	23.54	18.44	1.277		
					0	3	0	21.68	15.83	1.370		
					0	4	0	21.14	15.92	1.328		
					0	5	0	28.15	16.77	1.679		
					0	6	0	19.07	15.84	1.204		
					0	7	0	16.40	16.94	0.968		
					0	8	0	17.09	19.92	0.858		
					0	9	0	18.54	17.64	1.051		
		interval	prev_pgd	0.5+sg	0	1	0	24.23	16.39	1.478	1.223	<i>0.04</i>
					0	2	0	23.54	19.27	1.222		
					0	3	0	21.68	17.45	1.242		
					0	4	0	21.14	17.96	1.177		
					0	5	0	28.15	17.66	1.594		
					0	6	0	19.07	17.13	1.113		
					0	7	0	16.40	15.39	1.066		
					0	8	0	17.09	18.11	0.944		
					0	9	0	18.54	15.84	1.170		
		zono	prev_pgd	0.5+sg	0	1	0	24.23	17.45	1.389	1.207	<i>0.04</i>
					0	2	0	23.54	19.98	1.178		
					0	3	0	21.68	20.55	1.055		
					0	4	0	21.14	18.69	1.131		
					0	5	0	28.15	16.90	1.666		
					0	6	0	19.07	16.04	1.189		
					0	7	0	16.40	15.09	1.087		
					0	8	0	17.09	17.00	1.005		
					0	9	0	18.54	15.98	1.160		
		zono	prev_pgd	0.0+sg	0	1	0	24.23	15.82	1.532	1.186	<i>0.04</i>
					0	2	0	23.54	17.91	1.314		
					0	3	0	21.68	17.82	1.217		
					0	4	0	21.14	17.92	1.180		
					0	5	0	28.15	19.90	1.415		
					0	6	0	19.07	18.87	1.011		
					0	7	0	16.40	16.75	0.979		
					0	8	0	17.09	17.43	0.980		
					0	9	0	18.54	17.68	1.049		

Table 94: Architecture **3x10** — perturbation **linf**(0.1), $c_{src} = 0$, $\tau \in \{0.0, 0.5\}$; continuation of Table 5 (4 combos, 36 cell rows). Column meanings: see Table 5.

p_type	p_size	bound_tight	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{sp}_{c_src}$	$ \delta_{std} - \delta_{adv} _{c_src}$
linf	0.1	interval	prev_pgd	0.0+sg	1	0	0	22.43	13.79	1.627	1.076	0.05
					1	2	0	20.13	15.68	1.284		
					1	3	0	19.97	17.33	1.152		
					1	4	0	13.25	14.88	0.890		
					1	5	0	14.60	15.10	0.967		
					1	6	0	12.79	14.09	0.908		
					1	7	0	14.48	13.85	1.045		
					1	8	0	13.58	15.07	0.901		
					1	9	0	12.66	13.92	0.909		
		interval	prev_pgd	0.5+sg	1	0	0	22.43	15.03	1.492	1.065	0.05
					1	2	0	20.13	14.30	1.408		
					1	3	0	19.97	16.80	1.189		
					1	4	0	13.25	15.46	0.857		
					1	5	0	14.60	15.90	0.918		
					1	6	0	12.79	15.37	0.832		
					1	7	0	14.48	14.80	0.978		
					1	8	0	13.58	14.94	0.909		
					1	9	0	12.66	12.66	1.000		
		zono	prev_pgd	0.0+sg	1	0	0	22.43	16.08	1.395	1.011	0.05
					1	2	0	20.13	16.82	1.197		
					1	3	0	19.97	19.09	1.046		
					1	4	0	13.25	17.06	0.777		
					1	5	0	14.60	15.80	0.924		
					1	6	0	12.79	13.99	0.914		
					1	7	0	14.48	13.48	1.074		
					1	8	0	13.58	15.70	0.865		
					1	9	0	12.66	13.90	0.911		
		zono	prev_pgd	0.5+sg	1	0	0	22.43	18.57	1.208	0.950	0.05
					1	2	0	20.13	19.32	1.042		
					1	3	0	19.97	20.61	0.969		
					1	4	0	13.25	14.82	0.894		
					1	5	0	14.60	15.64	0.934		
					1	6	0	12.79	14.77	0.866		
					1	7	0	14.48	16.21	0.893		
					1	8	0	13.58	15.85	0.857		
					1	9	0	12.66	14.32	0.884		

Table 95: Architecture **3x10** — perturbation **linf**(0.1), $c_{src} = 1$, $\tau \in \{0.0, 0.5\}$; continuation of Table 5 (4 combos, 36 cell rows). Column meanings: see Table 5.